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Medge functions

Synopsis

The functions prefixed with Medge make up the Edge Finder module. The Edge Finder module provides powerful operations that allow you to extract different types of edges in images, such as object contours or thin curvilinear shapes. The module offers many settings including image processing controls, feature controls, and annotations controls. It also provides analysis capabilities allowing for the advanced manipulations of results. The Edge Finder module can also be used in conjunction with the Aurora Imaging Library Geometric Model Finder module; that is, you can define models from the result of an edge extraction, or you can find occurrences in the result of an edge extraction. The Edge Finder module offers complete support for camera calibration. Extraction of edges can be performed in the calibrated real-world such that, even without physically correcting your images, results are in real-world units.

Functions

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MedgeAlloc

Synopsis

Allocate an Edge Finder context.

Syntax

```
AIL_ID MedgeAlloc(
    AIL_ID SysId,             //in
    AIL_INT64 EdgeFinderType, //in
    AIL_INT64 ControlFlag,    //in
    AIL_ID *ContextIdPtr      //out
)
```

Description

This function allocates an Edge Finder context on the specified system. An Edge Finder context contains all the information necessary to perform an MedgeCalculate() operation, including global processing settings, and edge features to calculate.

Edge Finder context settings can be adjusted using MedgeControl().

When the Edge Finder context is no longer required, release it using MedgeFree() unless M_UNIQUE_ID was specified during allocation; if M_UNIQUE_ID was specified, the smart identifier manages the Edge Finder context's lifetime and you must not manually free it.

Parameters

SysId INQ

Specifies the system on which to allocate the Edge Finder context.

This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the system																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

EdgeFinderType INQ

Specifies the type of Edge Finder context. This parameter must be set to one of the values below.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For the Edge Finder context

Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_CONTOUR	Specifies a contour context type, which is used to find object contours in images.	c		e	f	g	a		i		j	k	l	m	n	b
M_CREST	Specifies a crest context type, which is used to find thin line crests in images.	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

ContextIdPtr

Specifies the address of the variable in which to write the Edge Finder context identifier or specifies the data type that the function should use to return the Edge Finder context identifier.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving the identifier or specifying how to return it																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_NULL	Specifies that you will use this function's return value to obtain the identifier of the allocated Edge Finder context ; in this case, a standard Aurora Imaging Library identifier of type *AIL_ID* is returned.	c		e	f	g	a		i		j	k	l	m	n	b

M_UNIQUE_ID	Specifies that you will use this function's return value to obtain the identifier of the allocated Edge Finder context ; in this case, an Aurora Imaging Library smart identifier of type *AIL_UNIQUE_EDGE_ID* is returned instead of a standard Aurora Imaging Library identifier. > This setting is only available when using C++11 (or later). An Aurora Imaging Library smart identifier manages the lifespan of the Aurora Imaging Library object it owns (similar to a *std::unique_ptr*). Note, you can use an Aurora Imaging Library smart identifier as though it were a standard Aurora Imaging Library identifier, except that you cannot use it to manually free the Edge Finder context (it is freed automatically). For more information, see the Aurora Imaging Library smart identifiers subsection of the Aurora Imaging Library custom data types, void pointers, extensions, and portability functions section of Chapter 2: Key information and building an application.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the contour Edge Finder context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated contour Edge Finder context. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the crest Edge Finder context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated crest Edge Finder context. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the Edge Finder context identifier either as a standard identifier (*AIL_ID*) or a smart

identifier (*AIL_UNIQUE_EDGE_ID*). If allocation fails, **M_NULL** is returned (or *nullptr* if M_UNIQUE_ID was specified).

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailedge11.lib.
DLL	Requires ail11.dll; ailedge11.dll.

MedgeAllocResult

Synopsis

Allocate an Edge Finder result buffer.

Syntax

```
AIL_ID MedgeAllocResult(
    AIL_ID SysId,           //in
    AIL_INT64 ControlFlag,  //in
    AIL_ID *EdgeResultIdPtr //out
)
```

Description

This function allocates an Edge Finder result buffer, on the specified system, to store results obtained from an MedgeCalculate() operation.

When the Edge Finder result buffer is no longer required, release it using MedgeFree() unless M_UNIQUE_ID was specified during allocation; if M_UNIQUE_ID was specified, the smart identifier manages the Edge Finder result buffer's lifetime and you must not manually free it.

Parameters

SysId INQ

Specifies the system on which to allocate the Edge Finder result buffer.

This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the system																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

EdgeResultIdPtr

Specifies the address of the variable in which to write the Edge Finder result buffer identifier or specifies the data type that the function should use to return the Edge Finder result buffer identifier.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving the identifier or specifying how to return it																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_NULL	Specifies that you will use this function's return value to obtain the identifier of the allocated Edge Finder result buffer ; in this case, a standard Aurora Imaging Library identifier of type *AIL_ID* is returned.	c		e	f	g	a		i		j	k	l	m	n	b
M_UNIQUE_ID	Specifies that you will use this function's return value to obtain the identifier of the allocated Edge Finder result buffer ; in this case, an Aurora Imaging Library smart identifier of type *AIL_UNIQUE_EDGE_ID* is returned instead of a standard Aurora Imaging Library identifier. This setting is only available when using C++11 (or later). An Aurora Imaging Library smart identifier manages the lifespan of the Aurora Imaging Library object it owns (similar to a *std::unique_ptr*). Note, you can use an Aurora Imaging Library smart identifier as though it were a standard Aurora Imaging Library identifier, except that you cannot use it to manually free the Edge Finder result buffer (it is freed automatically). For more information, see the Aurora Imaging Library smart identifiers subsection of the Aurora Imaging Library custom data types, void pointers, extensions, and portability functions section of Chapter 2: Key information and building an application.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the allocated Edge Finder result buffer. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the Edge Finder result buffer identifier either as a standard identifier (*AIL_ID*) or a smart identifier (*AIL_UNIQUE_EDGE_ID*). If allocation fails, **M_NULL** is returned (or *nullptr* if M_UNIQUE_ID was specified).

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailedge11.lib.
DLL	Requires ail11.dll; ailedge11.dll.

MedgeCalculate

Synopsis

Perform edge extraction and feature calculations.

Syntax

```
void MedgeCalculate(
    AIL_ID ContextId,      //in
    AIL_ID SourceImageId,  //in
    AIL_ID SourceDeriv1Id, //in
    AIL_ID SourceDeriv2Id, //in
    AIL_ID SourceDeriv3Id, //in
    AIL_ID EdgeResultId,   //out
    AIL_INT64 ControlFlag  //in
)
```

Description

This function extracts edges and calculates edge features in the source image buffer or in the derivative image buffers. You can also use this function to perform post-calculations on included edges of an Edge Finder result buffer.

Results are stored in the specified Edge Finder result buffer. Select which edge features to calculate with MedgeControl(). Include, exclude, or delete edges that meet a specified criterion with MedgeSelect(). Note that to calculate new features and/or select features of a new set of edges, **MedgeCalculate**() must be called after MedgeControl() and/or MedgeSelect().

If the source image or derivative images are calibrated, results are calculated in the world coordinate system, otherwise they are calculated in the pixel coordinate system.

If the source image or derivative images are calibrated, you can retrieve results in real-world or in pixel units. However, in the presence of distortion, some results are meaningless when converted from real-world to pixel units (for example, Feret angles). For example, if an edge appears warped in the source image, but the camera calibration context of the source image compensates for this during the extraction, the resulting Feret angles are meaningful in the real-world coordinate system, and meaningless in the pixel coordinate system.

Calculations are typically done on a source image buffer (SourceImageId), where **MedgeCalculate**() calculates the required derivative image buffers that are used to extract edges. In this case, the derivative image buffers (SourceDeriv1Id, SourceDeriv2Id, and SourceDeriv3Id) must be set to **M_NULL**. However, if you specify your own derivative image buffers, the edge extraction process is done with them instead. In this case, SourceImageId must be set to **M_NULL**. Note that, when specifying your own derivative image buffers, the majority of processing control settings remain in effect, except for filter settings (for example, M_FILTER_SMOOTHNESS).

When calculated by Edge Finder, derivative buffers are always consistent (for example, they have the same scaling factor) and are always normalized to 10-bit images such that the maximum output value is 1024 for 8-bit source images. When you provide the derivative buffers, they should respect these limitations for best performance.

To avoid repeatedly calculating time-consuming edge features for all edges in a source image, **MedgeCalculate**() can also perform post-calculations on included edges of an Edge Finder result buffer.

That is, once the results of a calculation have been stored in an Edge Finder result buffer, you can select a subset of edges from that Edge Finder result buffer (with MedgeSelect()) and/or add new features for calculation to that Edge Finder result buffer (with MedgeControl()), and then perform the post-calculation with **MedgeCalculate**(). In this case, processing time greatly decreases since calculations are not unnecessarily repeated. The operation of selecting edges and adding new features can be repeated until the required result is calculated. To perform post-calculations on the included edges of an Edge Finder result buffer, set SourceImageId, SourceDeriv1Id, SourceDeriv2Id, and SourceDeriv3Id to **M_NULL**.

Note that there are some restrictions when post-calculating edges; for more information, see the Post-calculation subsection of the Calculating and retrieving results section of Chapter 10: Edge Finder.

Parameters

ContextId

Specifies the Edge Finder context to use for the extraction. The Edge Finder context must have been previously allocated on the required system using MedgeAlloc().

SourceImageId

Specifies the source image buffer from which to extract edges. Typically, the source image buffer should be 1-band 8-bit unsigned. Other buffer depths and types are generally accepted, but can slightly decrease performance. Note that 3-band source image buffers are only supported when extracting edge contours. When the source image buffer is 32-bit float, Edge Finder uses floating-point precision calculations.

This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.

If you specify your own derivative image buffers, or if you are performing post-calculations on the included edges of an Edge Finder result buffer, this parameter must be set to **M_NULL**.

> Note that when MedgeControl() with M_MODEL_FINDER_COMPATIBLE or M_AGM_COMPATIBLE is enabled, the maximum image buffer size is 32768x32768 pixels. This is the target size limitation for both the Model Finder and AGM modules.

SourceDeriv1Id

Specifies the X-source derivative image buffer used to extract edges. When the Edge Finder context type is M_CONTOUR, SourceDeriv1Id specifies the first derivative of the image in the X-direction. When the Edge Finder context type is M_CREST, SourceDeriv1Id specifies the second derivative of the image in the X-direction.

Derivative image buffers must be 1-band 16-bit signed or 1-band 32-bit float. In addition, all derivative image buffers must be consistent (for example, they must have the same scaling factor), and they must be of the same type and size. When the derivative image buffers are 32-bit float, Edge Finder uses floating-point precision calculations.

This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.

If you specify a source image buffer, or if you are performing post-calculations on the included edges of an Edge Finder result buffer, this parameter must be set to **M_NULL**.

> Note that when MedgeControl() with M_MODEL_FINDER_COMPATIBLE or M_AGM_COMPATIBLE is enabled, the maximum image buffer size is 32768x32768 pixels. This is the target size limitation for

> both the Model Finder and AGM modules.

SourceDeriv2Id

Specifies the Y-source derivative image buffer used to extract edges. When the Edge Finder context type is M_CONTOUR, SourceDeriv2Id specifies the first derivative of the image in the Y-direction. When the Edge Finder context type is M_CREST, SourceDeriv2Id specifies the second derivative of the image in the Y-direction.

Derivative image buffers must be 1-band 16-bit signed or 1-band 32-bit float. In addition, all derivative image buffers must be consistent (for example, they must have the same scaling factor), and they must be of the same type and size. When the derivative image buffers are 32-bit float, Edge Finder uses floating-point precision calculations.

This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.

If you specify a source image buffer, or if you are performing post-calculations on the included edges of an Edge Finder result buffer, this parameter must be set to **M_NULL**.

> Note that when MedgeControl() with M_MODEL_FINDER_COMPATIBLE or M_AGM_COMPATIBLE is enabled, the maximum image buffer size is 32768x32768 pixels. This is the target size limitation for both the Model Finder and AGM modules.

SourceDeriv3Id

Specifies the source cross-derivative image buffer used to extract edges. The cross-derivative can only be specified if the Edge Finder context type is M_CREST. If the Edge Finder context type is M_CONTOUR, set SourceDeriv3Id to **M_NULL**.

Derivative image buffers must be 1-band 16-bit signed or 1-band 32-bit float. In addition, all derivative image buffers must be consistent (for example, they must have the same scaling factor), and they must be of the same type and size. When the derivative image buffers are 32-bit float, Edge Finder uses floating-point precision calculations.

This image buffer must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.

If you specify a source image buffer, or if you are performing post-calculations on the included edges of an Edge Finder result buffer, this parameter must be set to **M_NULL**.

> Note that when MedgeControl() with M_MODEL_FINDER_COMPATIBLE or M_AGM_COMPATIBLE is enabled, the maximum image buffer size is 32768x32768 pixels. This is the target size limitation for both the Model Finder and AGM modules.

EdgeResultId

Specifies the Edge Finder result buffer in which to write the results of the extraction. The Edge Finder result buffer must have been previously allocated on the required system using MedgeAllocResult().

ControlFlag

Specifies whether optimal performance during Edge Finder calculations is ensured. This parameter can only be used when providing derivative image buffers. When providing a source image buffer, optimal

performance is always ensured; therefore, this parameter must be set to M_DEFAULT.

When providing derivative image buffers, this parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For the derivative image buffers																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies that optimal performance is not ensured.	c		e	f	g	a		i		j	k	l	m	n	b
M_NO_CHECK	Specifies that optimal performance is ensured. M_NO_CHECK should only be used if each derivative image buffer has pixel values that do not go beyond the range of positive or negative 1024 (that is, 10-bit signed values). This ensures optimal performance since Edge Finder does not have to verify unnecessary values.	c		e	f	g	a		i		j	k	l	m	n	b

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailedge11.lib.
DLL	Requires ail11.dll; ailedge11.dll.

MedgeControl

Synopsis

Control an Edge Finder context or an Edge Finder result buffer setting.

Syntax

```
void MedgeControl(
    AIL_ID ContextOrResultId, //out
    AIL_INT64 ControlType,    //in
    AIL_DOUBLE ControlValue   //in
)
```

Description

This function sets the specified control for either an Edge Finder context or an Edge Finder result buffer. For Edge Finder contexts, these settings control the execution of MedgeCalculate() operations and select which edge features MedgeCalculate() should calculate. For Edge Finder result buffers, these settings control the post manipulation of results. For example, to draw a zoomed region of the source image that was used to calculate results, the drawing control values must be appropriately set. Similarly, to select edges based on the proximity of an edge or edges to a specified point, M_NEAREST_NEIGHBOR_RADIUS must be appropriately set. For more information, see MedgeDraw() or MedgeSelect().

All control settings can typically be inquired with MedgeInquire().

For new context settings to take effect, you must calculate the settings, using MedgeCalculate().

Note that some control settings have post-calculation restrictions. For more information, see the Post-calculation subsection of the Calculating and retrieving results section of Chapter 10: Edge Finder.

> Note that when using a C compiler (not a C++ or other compiler) in 64-bit mode, **MedgeControl()** internally calls the *AIL_DOUBLE* version of this function (MedgeControlDouble()). If you need to pass large integer values (including pointers), call the *AIL_INT64* version of this function (MedgeControlInt64()).

Parameters

ContextOrResultId

Specifies either the Edge Finder context or the Edge Finder result buffer whose settings you want to modify. The Edge Finder context or the Edge Finder result buffer must have been previously allocated on the required system using MedgeAlloc() or MedgeAllocResult(), respectively.

ControlType

Specifies the setting to change.

See the Parameter associations section for possible values that can be specified.

ControlValue

Specifies the setting's new value.

See the Parameter associations section for possible values that can be specified.

Parameter associations

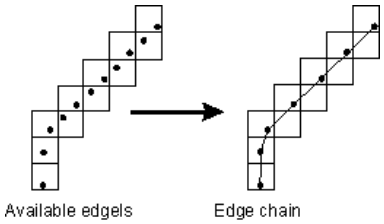
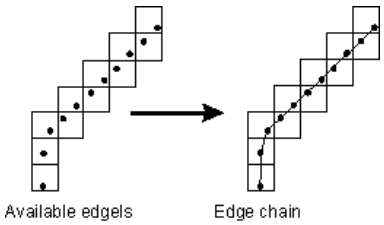
The tables below list possible values for the **ControlType** and **ControlValue** parameters.

- For operation settings for object contours and line crests
- For operation settings for line crests
- For saving internal buffers in the Edge Finder result buffer
- For calculating edge features
- For calculating Feret values
- For selecting groups of edge features
- For performing post-calculations on extracted edges
- For a result buffer
- For setting closest-edgel constraints

The following ControlType and corresponding ControlValue parameter settings are used to control the Edge Finder context operation settings and can be specified for both M_CONTOUR and M_CREST Edge Finder contexts, unless otherwise specified.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For operation settings for object contours and line crests																	
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																
M_ACCURACY		Sets the edgel accuracy of the edge extraction. Accuracy depends on the image's dynamic range, sharpness, and noise. The best accuracy is achieved in well-contrasted noise-free images. INQ															
	M_DEFAULT	Same as M_HIGH.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that edgel accuracy will be disabled. Edgels will be calculated with pixel accuracy.	c		e	f	g	a		i		j	k	l	m	n	b
	M_HIGH	Specifies high accuracy. Edgels will be calculated with subpixel accuracy.	c		e	f	g	a		i		j	k	l	m	n	b
	M_VERY_HIGH	Specifies very high accuracy. Edgels will be calculated with very precise subpixel accuracy. M_VERY_HIGH uses a classical camera model to compensate for any pixel-distortion aberration.	c		e	f	g	a		i		j	k	l	m	n	b

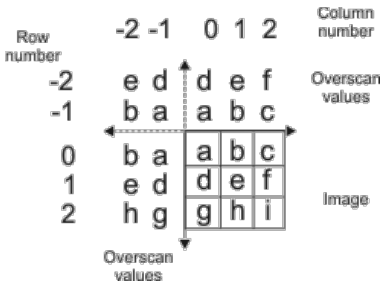
M_ANGLE_ACCURACY		Sets the precision with which to estimate edgel angles when extracting edges. M_ANGLE_ACCURACY can only be used with M_CONTOUR Edge Finder contexts. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_HIGH.	c		e	f	g	a		i		j	k	l	m	n	b
	M_HIGH	Specifies high precision. The angle estimation is performed in increments of 360/256 degrees.	c		e	f	g	a		i		j	k	l	m	n	b
	M_LOW	Specifies low precision. The angle estimation is performed in increments of 45 degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_CHAIN_ALL_NEIGHBORS		Sets how edge chains are built. Edge chains are built using an 8-connected lattice. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that edge chains are built with the least amount of edgel information possible.  	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that edge chains are built with as much edgel information as possible. Enabling M_CHAIN_ALL_NEIGHBORS can result in slightly longer calculations, however, the edge chain will contain more edgel information.  	c		e	f	g	a		i		j	k	l	m	n	b

M_DETAIL_LEVEL		Sets the level of details to extract from the image. The detail level determines what is considered an edge/background. A higher detail level will include more edges than a lower detail level. Essentially, M_DETAIL_LEVEL sets the threshold mode of the Edge Finder context. Note that an M_DETAIL_LEVEL setting overrides an M_THRESHOLD_MODE setting. Typically, M_DETAIL_LEVEL is used when interfacing with the Aurora Imaging Library Geometric Model Finder module. Otherwise, M_THRESHOLD_MODE should be used instead.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_MEDIUM.	c		e	f	g	a		i		j	k	l	m	n	b
	M_HIGH	Sets the detail level to high.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MEDIUM	Sets the detail level to medium.	c		e	f	g	a		i		j	k	l	m	n	b
	M_VERY_HIGH	Sets the detail level to very high.	c		e	f	g	a		i		j	k	l	m	n	b
M_EXTRACTION_SCALE		Sets the scale of the image at which to do the edge extraction. Once the extraction is complete, the results are scaled to the original scale of the image. An extraction scale less than one speeds up the calculation or the search but can result in a less reliable result, including, the loss of important details and/or a reduction in the accuracy of the search results. M_EXTRACTION_SCALE is for advanced users of the EdgeFinder module. The default setting usually provides the most accurate search results. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0	Specifies the extraction scale.	c		e	f	g	a		i		j	k	l	m	n	b

M_FILTER_SMOOTHNESS	Sets the degree of smoothness (strength of denoising) applied by the filter during the neighborhood operation. M_FILTER_SMOOTHNESS only has an effect if **MedgeControl**() with M_FILTER_TYPE is set to M_DERICHE or M_SHEN. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 50.0.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 100.0	Specifies the smoothness value. A value of 100.0 results in a strong noise reduction effect, while a value of 0.0 has almost no noise reduction effect.	c		e	f	g	a		i		j	k	l	m	n	b
M_FILTER_TYPE	Sets the type of filter used when performing the neighborhood operation used to extract edges. The type of filter determines the distribution of the neighborhoods' influence. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_SHEN.	c		e	f	g	a		i		j	k	l	m	n	b
M_DERICHE	Specifies a Deriche infinite support filter. This is an exponential weighting function, of the general form: $k(a\ \lvert n\rvert + 1)e^{-a\lvert n\rvert}$ Deriche is an Infinite Impulse Response (IIR) filter. For the Deriche filter, the neighborhoods' influence decreases much slower as the distance from the central pixel increases, compared to the Shen-Castan filter (M_SHEN). M_DERICHE can be used with both M_CONTOUR and M_CREST Edge Finder contexts. Typically, M_DERICHE is used for unusually thick crests. M_DERICHE allows you to use Edge Finder's smoothing capabilities. To do so, use **MedgeControl**() with M_FILTER_SMOOTHNESS.	c		e	f	g	a		i		j	k	l	m	n	b

	M_FREI_CHEN	Specifies a Frei Chen filter. This is a Finite Impulse Response (FIR) filter that can be represented with the following convolution kernels: $\begin{bmatrix} -1 & -\sqrt{2} & -1 \\ 0 & 0 & 0 \\ 1 & \sqrt{2} & 1 \end{bmatrix}$ and $\begin{bmatrix} -1 & 0 & 1 \\ -\sqrt{2} & 0 & \sqrt{2} \\ -1 & 0 & 1 \end{bmatrix}$ M_FREI_CHEN can only be used with M_CONTOUR Edge Finder contexts. Also, when using M_FREI_CHEN, you cannot smooth images using **MedgeControl**() with M_FILTER_SMOOTHNESS.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PREWITT	Specifies a Prewitt filter. This is a Finite Impulse Response (FIR) filter that can be represented with the following convolution kernels: $\begin{bmatrix} -1 & -1 & -1 \\ 0 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$ and $\begin{bmatrix} -1 & 0 & 1 \\ -1 & 0 & 1 \\ -1 & 0 & 1 \end{bmatrix}$ M_PREWITT can only be used with M_CONTOUR Edge Finder contexts. Also, when using M_PREWITT, you cannot smooth images using **MedgeControl**() with M_FILTER_SMOOTHNESS.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SHEN	Specifies a Shen-Castan infinite support exponential filter. This is an exponential weighting function, of the general form: $Ke^{-\beta\lvert n\rvert}$ Shen-Castan is an Infinite Impulse Response (IIR) filter. For the Shen-Castan filter, the neighborhoods' influence decreases much faster as the distance from the central pixel increases, compared to the Deriche filter (M_DERICHE). M_SHEN can be used with both M_CONTOUR and M_CREST Edge Finder contexts. Typically, M_SHEN performs an excellent edge extraction on most images; however, if you are extracting unusually thick crests that yield inappropriate results, you should use M_DERICHE. M_SHEN allows you to use Edge Finder's smoothing capabilities. To do so, use **MedgeControl**() with M_FILTER_SMOOTHNESS.	c		e	f	g	a		i		j	k	l	m	n	b

	M_SOBEL	Specifies a Sobel filter. This is a Finite Impulse Response (FIR) filter that can be represented with the following convolution kernels: $\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$ and $\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$ M_SOBEL can only be used with M_CONTOUR Edge Finder contexts. Also, when using M_SOBEL, you cannot smooth images using **MedgeControl**() with M_FILTER_SMOOTHNESS.	c		e	f	g	a		i		j	k	l	m	n	b
M_FLOAT_MODE		Sets whether to force all the processing of edge extraction operations to be performed using floating-point precision calculations. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that all edge extractions are not forced to be performed using floating-point precision calculations.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that all edge extractions are forced to be performed using floating-point precision calculations.	c		e	f	g	a		i		j	k	l	m	n	b
M_MAGNITUDE_TYPE		Sets how to calculate the magnitude of the edge at each edgel position. Note that for M_CONTOUR Edge Finder contexts, the magnitude is the norm of the gradient vector at the edgel position. For M_CREST Edge Finder contexts, the magnitude is equal to the maximum eigenvalue of the Hessian matrix at the edgel position. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value. For M_CONTOUR Edge Finder contexts, the default is M_SQR_NORM. For M_CREST Edge Finder contexts, the default is M_NORM.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NORM	Specifies that the magnitude will be used.	c		e	f	g	a		i		j	k	l	m	n	b

	M_SQR_NORM	Specifies that the square of the magnitude will be used. This value optimizes the edge extraction operation while still preserving a very good edgel position accuracy. M_SQR_NORM is calculated faster than M_NORM, though it is less accurate.	c		e	f	g	a		i		j	k	l	m	n	b
M_OVERSCAN		Sets the type of overscan used to handle the source image's bordering pixels. Note that M_OVERSCAN is ignored if using an IIR filter (e.g. Shen and Deriche filter types). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_MIRROR.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that no overscan will be used, unless processing the border pixels is faster than ignoring them; in the latter case, Aurora Imaging Library automatically selects the overscan to optimize speed according to the specified operation and the target system.	c		e	f	g	a		i		j	k	l	m	n	b
	M_MIRROR	Specifies that the border pixels of a source image are processed using overscan pixel values that mirror the source buffer pixel values. That is, the overscan pixel values will be a mirror copy of the source buffer's borders. For example:  	c		e	f	g	a		i		j	k	l	m	n	b
	M_REPLACE	Specifies that the border pixels of a source image are processed using overscan pixel values set to the overscan replacement value (**MedgeControl**() with M_OVERSCAN_REPLACE_VALUE).	c		e	f	g	a		i		j	k	l	m	n	b

	M_TRANSPARENT	Specifies that the border pixels of a source image are processed using transparent overscan pixel values. That is, the overscan pixel values will be those of the parent buffer. If they are not available, a mirror type overscan is used instead.	c		e	f	g	a		i		j	k	l	m	n	b
M_OVERSCAN_REPLACE_VALUE		Sets a replacement value for the overscan pixel values. Note that M_OVERSCAN_REPLACE_VALUE is ignored unless M_OVERSCAN with **MedgeControl**() is set to M_REPLACE. In addition, M_OVERSCAN_REPLACE_VALUE is ignored if using an IIR filter (e.g. Shen and Deriche filter types). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.	c		e	f	g	a		i		j	k	l	m	n	b
	M_REPLACE_MAX	Specifies that the overscan neighborhood pixel values will be set to the maximum value of the source image buffer.	c		e	f	g	a		i		j	k	l	m	n	b
	M_REPLACE_MIN	Specifies that the overscan neighborhood pixel values will be set to the minimum value of the source image buffer.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the value of the overscan neighborhood pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_THRESHOLD_HIGH		Sets the upper bound of the hysteresis threshold value. M_THRESHOLD_HIGH is ignored unless **MedgeControl**() with M_THRESHOLD_MODE is set to M_USER_DEFINED. Note that lower threshold values result in a more sensitive edgel detection; that is, a higher (or equal) number of edgels are detected. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the upper bound of the hysteresis threshold.	c		e	f	g	a		i		j	k	l	m	n	b

M_THRESHOLD_LOW	Sets the lower bound of the hysteresis threshold value. M_THRESHOLD_LOW is ignored unless **MedgeControl**() with M_THRESHOLD_MODE is set to M_USER_DEFINED. Note that lower threshold values result in a more sensitive edgel detection; that is, a higher (or equal) number of edgels are detected. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0.0	Specifies the lower bound of the hysteresis threshold.	c		e	f	g	a		i		j	k	l	m	n	b
M_THRESHOLD_MODE	Sets the threshold of the edge extraction. The Edge Finder module uses a classical hysteresis threshold to extract relevant edges in the image. Threshold values can either be set manually or determined automatically by the Edge Finder module. Note that lower threshold values result in a more sensitive edgel detection; that is, a higher (or equal) number of edgels are detected. When interfacing with the Aurora Imaging Library Geometric Model Finder module, you should use M_DETAIL_LEVEL to set the threshold mode. Note that an M_DETAIL_LEVEL setting overrides an M_THRESHOLD_MODE setting. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_HIGH.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies no threshold. All edges are extracted.	c		e	f	g	a		i		j	k	l	m	n	b
M_HIGH	Specifies a high threshold. M_HIGH should be used for images with some contrast variations, noise, and non-uniform illumination. M_HIGH always results in a lower (or equal) number of edgels than M_MEDIUM.	c		e	f	g	a		i		j	k	l	m	n	b
M_LOW	Specifies a low threshold. All edges over a minimum noise-based estimated threshold are extracted. M_LOW always results in a lower (or equal) number of edgels than M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b

	M_MEDIUM	Specifies a medium threshold. M_MEDIUM should be used for multi-contrast images, or for images with a lot of noise or non-uniform illumination. M_MEDIUM always results in a lower (or equal) number of edgels than M_LOW.	c		e	f	g	a		i		j	k	l	m	n	b
	M_USER_DEFINED	Specifies that the threshold values will be user-defined. Set the threshold values with M_THRESHOLD_LOW and M_THRESHOLD_HIGH.	c		e	f	g	a		i		j	k	l	m	n	b
	M_VERY_HIGH	Specifies a very high threshold. Only the strongest edges in the image are extracted. M_VERY_HIGH always results in a lower (or equal) number of edgels than M_HIGH.	c		e	f	g	a		i		j	k	l	m	n	b
M_THRESHOLD_TYPE		Sets the type of hysteresis threshold used when performing the edge extraction. Edge chains are built such that the magnitude values of all connected edgels are stronger than a lower bound threshold value and such that at least one of the edgel's magnitude in each edge chain is stronger than a upper bound threshold value. Note that M_THRESHOLD_TYPE sets how to use the lower and upper threshold bounds, while M_THRESHOLD_MODE defines what the bounds are. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_HYSTERESIS.	c		e	f	g	a		i		j	k	l	m	n	b
	M_FULL_HYSTERESIS	Specifies that the lower bound threshold value is 0.	c		e	f	g	a		i		j	k	l	m	n	b
	M_HYSTERESIS	Specifies that both the lower bound threshold value and the upper bound threshold value will be used.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NO_HYSTERESIS	Specifies that the lower bound threshold value is equal to the upper bound threshold value.	c		e	f	g	a		i		j	k	l	m	n	b

M_TIMEOUT		Sets the maximum edge extraction and calculation time for MedgeCalculate(), in msec. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies an infinite amount of edge extraction and calculation time.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0	Specifies the maximum edge extraction and calculation time, in msec.	c		e	f	g	a		i		j	k	l	m	n	b

The following ControlType and corresponding ControlValue settings are used to control the Edge Finder context operation settings and can only be specified for M_CREST Edge Finder contexts.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For operation settings for line crests																	
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																
M_FOREGROUND_VALUE		Sets the color of the line crests to extract from the image. INQ															
	M_DEFAULT	Same as M_FOREGROUND_BLACK.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ANY	Specifies that the line crests are both lighter and darker than the image's background color. This corresponds to both valley-like and ridge-like lines extracted from the image.	c		e	f	g	a		i		j	k	l	m	n	b
	M_FOREGROUND_BLACK	Specifies that the line crests are darker than the image's background color. This corresponds to valley-like lines extracted from the image (a valley-like Gaussian profile).	c		e	f	g	a		i		j	k	l	m	n	b
	M_FOREGROUND_WHITE	Specifies that the line crests are lighter than the image's background color. This corresponds to ridge-like lines extracted from the image (a ridge-like Gaussian profile).	c		e	f	g	a		i		j	k	l	m	n	b

The following ControlType and corresponding ControlValue parameter settings are used to save internal buffers in the Edge Finder result buffer and can be specified for both M_CONTOUR and M_CREST Edge Finder contexts.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For saving internal buffers in the Edge Finder result buffer																	
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																
M_SAVE_ANGLE		Sets whether the internal angle buffer, used when extracting edges, is saved in the Edge Finder result buffer. Note that the angle value is not meaningful in the whole image; it is only meaningful for edges whose magnitude value is above the lower bound threshold value. Note that the angle is measured between the horizontal axis and the perpendicular direction of the edge chain at each edgel location. For more information on angle convention in Aurora Imaging Library, see the Internal processing buffers subsection of the Calculating and retrieving results section of Chapter 10: Edge Finder. INQ															
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the angle buffer will not be saved.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the angle buffer will be saved.	c		e	f	g	a		i		j	k	l	m	n	b
M_SAVE_CHAIN_ANGLE		Sets whether the angle value of the edge at each edgel position is saved in the Edge Finder result buffer. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the angle values will not be saved.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the angle values will be saved.	c		e	f	g	a		i		j	k	l	m	n	b
M_SAVE_CHAIN_MAGNITUDE		Sets whether the magnitude value of the edge at each edgel position is saved in the Edge Finder result buffer. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies that the magnitude values will not be saved.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the magnitude values will be saved.	c		e	f	g	a		i		j	k	l	m	n	b
M_SAVE_DERIVATIVES		Sets whether the internal derivative buffers used when extracting edges are saved in the Edge Finder result buffer. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the derivative buffers will not be saved.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the derivative buffers will be saved.	c		e	f	g	a		i		j	k	l	m	n	b
M_SAVE_IMAGE		Sets whether the source image is saved in the Edge Finder result buffer. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the image will not be saved.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the image will be saved.	c		e	f	g	a		i		j	k	l	m	n	b
M_SAVE_MAGNITUDE		Sets whether the internal magnitude buffer, used when extracting edges, is saved in the Edge Finder result buffer. For M_CONTOUR Edge Finder contexts, the magnitude is the norm of the gradient vector at the edgel position. For M_CREST Edge Finder contexts, the magnitude is equal to the maximum eigenvalue of the Hessian matrix at the edgel position. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the magnitude buffer will not be saved.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the magnitude buffer will be saved.	c		e	f	g	a		i		j	k	l	m	n	b

M_SAVE_MASK		Sets whether the mask buffer is saved in the Edge Finder result buffer. For more information, see MedgeMask(). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the mask will not be saved.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the mask will be saved.	c		e	f	g	a		i		j	k	l	m	n	b

The following ControlType and corresponding ControlValue parameter settings are used to control the edge features calculated for each edge and can be specified for both M_CONTOUR and M_CREST Edge Finder contexts.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For calculating edge features																	
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																
M_AVERAGE_STRENGTH +		Sets whether to calculate the average strength of each edge. This is the average energy of each edge, which is defined as the energy of the edge (M_STRENGTH) divided by its number of edgels. INQ															
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the average strength will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the average strength will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_X_MAX +		Sets whether to calculate the extreme right edgel coordinate of each edge. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the extreme right edgel coordinate will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b

	M_ENABLE	Specifies that the extreme right edgel coordinate will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_X_MIN +		Sets whether to calculate the extreme left edgel coordinate of each edge. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the extreme left edgel coordinate will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the extreme left edgel coordinate will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_Y_MAX +		Sets whether to calculate the extreme bottom edgel coordinate of each edge. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the extreme bottom edgel coordinate will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the extreme bottom edgel coordinate will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_Y_MIN +		Sets whether to calculate the extreme top edgel coordinate of each edge. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the extreme top edgel coordinate will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the extreme top edgel coordinate will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b

M_CENTER_OF_GRAVITY_X +		Sets whether to calculate the X-position of each edge's center of gravity. This is equal to the average position of edgel positions in X. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the X-position of the center of gravity will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the X-position of the center of gravity will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_OF_GRAVITY_Y +		Sets whether to calculate the Y-position of each edge's center of gravity. This is equal to the average position of edgel positions in Y. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the Y-position of the center of gravity will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the Y-position of the center of gravity will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_CIRCLE_FIT_CENTER_X +		Sets whether to calculate the X-coordinate of the center of the circle that is the best fit for each edge. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the X-coordinate will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the X-coordinate will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b

M_CIRCLE_FIT_CENTER_Y +		Sets whether to calculate the Y-coordinate of the center of the circle that is the best fit for each edge. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the Y-coordinate will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the Y-coordinate will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_CIRCLE_FIT_COVERAGE +		Sets whether to calculate the coverage of the circle that is the best fit for each edge. The circle fit coverage indicates what angular fraction of the fitted circle is subtended by the radii going to the endpoints of the edge. If the edge is closed, the coverage will be 1. For a quarter circle the coverage would be 0.25. A perfectly straight line has a coverage of 0, it doesn't subtend much of an infinitely large circle. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the coverage of the circle will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the coverage of the circle will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_CIRCLE_FIT_ERROR +		Sets whether to calculate the fit error of the circle that is the best fit for each edge. This is calculated as the average quadratic error. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the fit error of the circle will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b

	M_ENABLE	Specifies that the fit error of the circle will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_CIRCLE_FIT_RADIUS +		Sets whether to calculate the radius of the circle that is the best fit for each edge. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the radius will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the radius will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_CLOSURE +		Sets whether to calculate the closure state of each edge. The closure state identifies whether the edge forms a closed chain. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the edge's closure state will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the edge's closure state will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_CONVEX_PERIMETER +		Sets whether to calculate the convex elongation of each edge. This is an approximation of the perimeter of the convex hull of an edge. It is derived from several Feret diameters; therefore, a larger number of Ferets gives a more accurate result (see M_NUMBER_OF_FERETS). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the convex elongation will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the convex elongation will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b

M_ELLIPSE_FIT_ANGLE +		Sets whether to calculate the angle of the ellipse that is the best fit for each edge. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the angle will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the angle will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_ELLIPSE_FIT_CENTER_X +		Sets whether to calculate the X-coordinate of the center of the ellipse that is the best fit for each edge. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the X-coordinate will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the X-coordinate will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_ELLIPSE_FIT_CENTER_Y +		Sets whether to calculate the Y-coordinate of the center of the ellipse that is the best fit for each edge. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the Y-coordinate will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the Y-coordinate will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b

M_ELLIPSE_FIT_COVERAGE +		Sets whether to calculate the coverage of the ellipse that is the best fit for each edge. The coverage describes the portion of the ellipse covered by the edge. The value returned is between 0.0 and 1.0, inclusive, where 0.0 equals no coverage, 0.5 equals 50 percent coverage, and 1.0 equals 100 percent coverage. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the coverage of the ellipse will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the coverage of the ellipse will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_ELLIPSE_FIT_ERROR +		Sets whether to calculate the fit error of the ellipse that is the best fit for each edge. This is calculated as the average quadratic error. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the fit error of the ellipse will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the fit error of the ellipse will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_ELLIPSE_FIT_MAJOR_AXIS +		Sets whether to calculate the major axis of the ellipse that is the best fit for each edge. The major axis is the line passing through the foci, center, and vertices of an ellipse. It is also the principal axis of symmetry. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the major axis will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b

	M_ENABLE	Specifies that the major axis will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_ELLIPSE_FIT_MINOR_AXIS +		Sets whether to calculate the minor axis of the ellipse that is the best fit for each edge. The minor axis is the line through the center of an ellipse that is perpendicular to the major axis. The minor axis is an axis of symmetry. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the minor axis will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the minor axis will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_FAST_LENGTH +		Sets whether to calculate a coarse approximation for the length of each edge. M_FAST_LENGTH gives a less accurate but faster approximation of the edge's length than M_LENGTH. M_FAST_LENGTH is equal to: INQ $(\sum horizontal_edge_pairs) + (\sum vertical_edge_pairs) + \sqrt{2}\,(\sum diagonal_edge_pairs)$	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that a fast length will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that a fast length will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_ELONGATION +		Sets whether to calculate the Feret elongation of each edge. This is a measure of the shape of each edge. It is equal to M_FERET_MAX_DIAMETER / M_FERET_MIN_DIAMETER. It is accurate for reasonably compact objects, but becomes less accurate for very elongated objects (because M_FERET_MIN_DIAMETER becomes less accurate). INQ	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the Feret elongation will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the Feret elongation will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_GENERAL +		Sets whether to calculate the general Feret of each edge. This is the Feret diameter calculated at M_FERET_GENERAL_ANGLE. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the general Feret will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the general Feret will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_MAX_ANGLE +		Sets whether to calculate the maximum Feret angle of each edge. This is the angle at which the maximum Feret diameter is found. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the maximum Feret angle will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the maximum Feret angle will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_MAX_DIAMETER +		Sets whether to calculate the maximum Feret diameter of each edge. This is the largest Feret diameter found after checking a certain number of angles (see M_NUMBER_OF_FERETS). More angles will give a more accurate result, but will take longer to calculate. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies that the maximum Feret diameter will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the maximum Feret diameter will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_MEAN_DIAMETER +		Sets whether to calculate the average Feret diameter at all the angles checked. See M_NUMBER_OF_FERETS for more information. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the average Feret diameter will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the average Feret diameter will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_MIN_ANGLE +		Sets whether to calculate the minimum Feret angle of each edge. This is the angle at which the minimum Feret diameter is found. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the minimum Feret angle will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the minimum Feret angle will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_MIN_DIAMETER +		Sets whether to calculate the minimum Feret diameter of each edge. This is the smallest Feret diameter found after checking a certain number of angles (see M_NUMBER_OF_FERETS). More angles will give a more accurate result, but will take longer to calculate. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies that the minimum Feret diameter will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the minimum Feret diameter will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_X +		Sets whether to calculate the X-Feret value of each edge. This is the dimension of the minimum bounding box of an edge in the horizontal direction; that is, M_BOX_X_MAX - M_BOX_X_MIN. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the X-Feret value will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the X-Feret value will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_Y +		Sets whether to calculate the Y-Feret value of each edge. This is the dimension of the minimum bounding box of an edge in the vertical direction; that is, M_BOX_Y_MAX - M_BOX_Y_MIN. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the Y-Feret value will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the Y-Feret value will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIRST_POINT_X +		Sets whether to calculate the X-coordinate of each edge's first point (starting point). Together with M_FIRST_POINT_Y, these values define a unique point for each edge, which is always the first point of each edge chain. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies that the first point's X-coordinate will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the first point's X-coordinate will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIRST_POINT_Y +		Sets whether to calculate the Y-coordinate of each edge's first point (starting point). Together with M_FIRST_POINT_X, these values define a unique point for each edge, which is always the first point of each edge chain. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the first point's Y-coordinate will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the first point's Y-coordinate will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_LABEL_VALUE +		Sets whether to calculate the label value of each edge in an image. The label value is a positive integer greater or equal to one; each edge in an image has a unique label value. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the label value will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the label value will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_LENGTH +		Sets whether to calculate the length of each edge. M_LENGTH gives a more accurate but slower approximation of the edge's length than M_FAST_LENGTH. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies that the length of each edge will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the length of each edge will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_LINE_FIT_A +		Sets whether to calculate the coefficient *A* of the line that is the best fit for each edge. The line fit approximation is based on the following equation: $A x + B y + C = 0$. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the coefficient *A* will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the coefficient *A* will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_LINE_FIT_B +		Sets whether to calculate the coefficient *B* of the line that is the best fit for each edge. The line fit approximation is based on the following equation: $A x + B y + C = 0$. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the coefficient *B* will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the coefficient *B* will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_LINE_FIT_C +		Sets whether to calculate the coefficient *C* of the line that is the best fit for each edge. The line fit approximation is based on the following equation: $A x + B y + C = 0$. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the coefficient *C* will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b

	M_ENABLE	Specifies that the coefficient *C* will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_LINE_FIT_ERROR +		Sets whether to calculate the fit error of the line that is the best fit for each edge. This is calculated as the average quadratic error. The line fit approximation is based on the following equation: $A\,x + B\,y + C = 0$. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the fit error will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the fit error will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOMENT_ELONGATION +		Sets whether to calculate the moment elongation of each edge. M_MOMENT_ELONGATION specifies the ratio of the principal values of the edge's inertial matrix, which corresponds to the principal directions of the edge shape. This can be approximately defined as the ratio between the edge's minimum and maximum moment. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the moment elongation will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the moment elongation will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOMENT_ELONGATION_ANGLE +		Sets whether to calculate the angle of the principal axis along each edge's moment elongation (M_MOMENT_ELONGATION). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies that the angle of the moment elongation will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the angle of the moment elongation will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION_X +		Sets whether to calculate the X-position of each edge. The position of the edge is defined by the middle edgel of the edge chain. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the X-position will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the X-position will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION_Y +		Sets whether to calculate the Y-position of each edge. The position of the edge is defined by the middle edgel of the edge chain. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the Y-position will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the Y-position will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE +		Sets whether to calculate the number of edgels of each edge. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the number of edgels will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b

	M_ENABLE	Specifies that the number of edgels will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_STRENGTH +		Sets whether to calculate the strength of each edge. This is the energy of each edge, which is defined as the sum of the square of the edgels' magnitude values. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the strength will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the strength will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_TORTUOSITY +		Sets whether to calculate the tortuosity measure of each edge. The tortuosity measure is equal to the diagonal length of the edge's bounding box (M_BOX), divided by the length of the edge (M_LENGTH). Therefore, a non-tortuous edge (a straight line) will have a tortuosity of 1.0 while a tortuous edge will have its tortuosity decreasing toward zero. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the tortuosity measure will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the tortuosity measure will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_X_MAX_AT_Y_MAX +		Sets whether to calculate the maximum X-coordinate at the maximum Y-coordinate of each edge. Together with M_BOX_Y_MAX, this is one of four contact points on the convex perimeter of the edge. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies that the X-maximum at Y-maximum will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the X-maximum at Y-maximum will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_X_MIN_AT_Y_MIN +		Sets whether to calculate the minimum X-coordinate at the minimum Y-coordinate of each edge. Together with M_BOX_Y_MIN, this is one of four contact points on the convex perimeter of the edge. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the X-minimum at Y-minimum will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the X-minimum at Y-minimum will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_Y_MAX_AT_X_MIN +		Sets whether to calculate the maximum Y-coordinate at the minimum X-coordinate of each edge. Together with M_BOX_X_MIN, this is one of four contact points on the convex perimeter of the edge. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the Y-maximum at X-minimum will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the Y-maximum at X-minimum will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_Y_MIN_AT_X_MAX +		Sets whether to calculate the minimum Y-coordinate at the maximum X-coordinate of each edge. Together with M_BOX_X_MAX, this is one of four contact points on the convex perimeter of the edge. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies that the Y-minimum at X-maximum will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the Y-minimum at X-maximum will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for the values listed in For calculating edge features.

You can add one of the following values to the above-mentioned values to specify a sorting key for result retrieval.

Note that only one edge feature can be selected as the first, second, or third sorting key.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying a sorting key																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_NO_SORT	Removes the specified sorting key.	c		e	f	g	a		i		j	k	l	m	n	b
M_SORT*n*_DOWN	Specifies the feature as the *n* th sorting key (in descending order), where *n* stands for an integer between 1 and 3.	c		e	f	g	a		i		j	k	l	m	n	b
M_SORT*n*_UP	Specifies the feature as the *n* th sorting key (in ascending order), where *n* stands for an integer between 1 and 3.	c		e	f	g	a		i		j	k	l	m	n	b

The following ControlType and corresponding ControlValue parameter settings are used to calculate Feret values and can be specified for both M_CONTOUR and M_CREST Edge Finder contexts.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For calculating Feret values																	
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																

M_FERET_ANGLE_SEARCH_END	Sets the end of the angular range at which to search for Feret diameters. The angular range is used to calculate the following Feret features: M_FERET_MAX_DIAMETER, M_FERET_MIN_DIAMETER, M_FERET_MAX_ANGLE, and M_FERET_MIN_ANGLE. The search is done in the counter-clockwise direction in steps of (M_FERET_ANGLE_SEARCH_END - M_FERET_ANGLE_SEARCH_START) / M_NUMBER_OF_FERETS. INQ															
M_DEFAULT	Specifies the default value; the default value is 360.0°.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 360.0	Specifies the end of the angular region.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_ANGLE_SEARCH_START	Sets the start of the angular range at which to search for Feret diameters. The angular range is used to calculate the following Feret features: M_FERET_MAX_DIAMETER, M_FERET_MIN_DIAMETER, M_FERET_MAX_ANGLE, and M_FERET_MIN_ANGLE. The search is done in the counter-clockwise direction in steps of (M_FERET_ANGLE_SEARCH_END - M_FERET_ANGLE_SEARCH_START) / M_NUMBER_OF_FERETS. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 0.0°.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 360.0	Specifies the start of the angular region.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_GENERAL_ANGLE	Sets the angle at which to calculate the M_FERET_GENERAL value. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 0.0°.	c		e	f	g	a		i		j	k	l	m	n	b

	M_MOMENT_ELONGATION_ANGLE + *n*	Specifies to use the moment elongation angle plus the offset *n*. The moment elongation angle is the angle of the principal axis for an edge's moment elongation (M_MOMENT_ELONGATION). Adding an offset is optional. For example, if you want to get the general Feret measure to be perpendicular to the moment elongation angle, set the offset to 90.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 360.0	Specifies the angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_FERETS		Sets the number of Feret angles to use to calculate M_FERET_MAX_DIAMETER, M_FERET_MIN_DIAMETER, M_FERET_MAX_ANGLE, M_FERET_MIN_ANGLE, M_FERET_MEAN_DIAMETER, M_FERET_ELONGATION, and M_CONVEX_PERIMETER. The first Feret angle used is established using M_FERET_ANGLE_SEARCH_START, and the difference between successive angles is (M_FERET_ANGLE_SEARCH_END - M_FERET_ANGLE_SEARCH_START) / M_NUMBER_OF_FERETS. Note that increasing the number of Ferets increases the accuracy of the results; however, it also increases the processing time. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 8.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the number of Ferets.	c		e	f	g	a		i		j	k	l	m	n	b

The following values allow you to select groups of edge features for calculation in a single call.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For selecting groups of edge features																		
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																	

M_ALL_FEATURES		Sets whether to calculate all edge features.														
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that all edge features will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that all edge features will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX		Sets whether to calculate the four extreme edgel coordinates of each edge (M_BOX_X_MIN, M_BOX_X_MAX, M_BOX_Y_MIN, and M_BOX_Y_MAX).	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that all the extreme edgel coordinates will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that all the extreme edgel coordinates will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_OF_GRAVITY		Sets whether to calculate the X- and Y-coordinates of the center of gravity of each edge (M_CENTER_OF_GRAVITY_X and M_CENTER_OF_GRAVITY_Y).	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that both the X- and Y-coordinates of the center of gravity will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that both the X- and Y-coordinates of the center of gravity will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_CIRCLE_FIT		Sets whether to calculate the circle fit values of each edge (M_CIRCLE_FIT_CENTER_X, M_CIRCLE_FIT_CENTER_Y, M_CIRCLE_FIT_RADIUS, M_CIRCLE_FIT_ERROR, and M_CIRCLE_FIT_COVERAGE).	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies that all the circle fit values will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that all the circle fit values will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_CONTACT_POINTS		Sets whether to calculate the minimum/maximum X-coordinate at the minimum/maximum Y-coordinate of each edge (M_X_MIN_AT_Y_MIN, M_X_MAX_AT_Y_MAX, M_Y_MAX_AT_X_MIN, and M_Y_MIN_AT_X_MAX). Together with its corresponding extreme edgel coordinate, these points represent four contact points on the convex perimeter of the edge. For example, M_X_MIN_AT_Y_MIN and M_BOX_Y_MIN is one of the four contact points.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the four contact points will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the four contact points will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_ELLIPSE_FIT		Sets whether to calculate the ellipse fit values of each edge (M_ELLIPSE_FIT_ANGLE, M_ELLIPSE_FIT_CENTER_X, M_ELLIPSE_FIT_CENTER_Y, M_ELLIPSE_FIT_COVERAGE, M_ELLIPSE_FIT_ERROR, M_ELLIPSE_FIT_MINOR_AXIS, and M_ELLIPSE_FIT_MAJOR_AXIS).	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that all the ellipse fit values will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that all the ellipse fit values will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_BOX		Sets whether to calculate the X- and Y-Feret values of each edge (M_FERET_X and M_FERET_Y).	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the X- and Y-Feret values will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the X- and Y-Feret values will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIRST_POINT		Sets whether to calculate the X- and Y-coordinate of each edge's first point (M_FIRST_POINT_X, and M_FIRST_POINT_Y).	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the X- and Y-coordinate of each edge's first point will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the X- and Y-coordinate of each edge's first point will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_LINE_FIT		Sets whether to calculate the line fit values of each edge (M_LINE_FIT_A, M_LINE_FIT_B, M_LINE_FIT_C, and M_LINE_FIT_ERROR).	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that all the line fit values will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that all the line fit values will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION		Sets whether to calculate both the X- and Y-position of each edge (M_POSITION_X and M_POSITION_Y).	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that both the X- and Y-position will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that both the X- and Y-position will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b

The following ControlType and corresponding ControlValue parameter settings are used when performing

post-calculations on extracted edges and can typically be specified for both M_CONTOUR and M_CREST Edge Finder contexts.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For performing post-calculations on extracted edges																	
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																
M_APPROXIMATION_TOLERANCE		Sets the resolution of edge approximation. INQ															
	M_DEFAULT	Specifies the default value; the default value is 50.0.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the resolution. When set to 0.0, a very fine edge approximation is performed. When set to 100.0, a coarse edge approximation is performed.	c		e	f	g	a		i		j	k	l	m	n	b
M_CHAIN_APPROXIMATION		Sets the simple geometric feature used when performing the edge approximation. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the edge approximation will not be performed. In this case, only the edge map is calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_LINE	Specifies that the edge approximation will be performed using a polygonal segmentation of each edge, in pixel units. In this case, both the edge map and the edge approximation are calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_WORLD_LINE	Specifies that the edge approximation will be performed using a polygonal segmentation of each edge, in world units. In this case, both the edge map and the edge approximation are calculated. If M_CHAIN_APPROXIMATION is set to M_WORLD_LINE, calling MedgeCalculate() to perform the calculation generates an error if the operation is not performed on a calibrated image.	c		e	f	g	a		i		j	k	l	m	n	b

M_FILL_GAP_ANGLE	Sets the aperture angle, starting at the line's tangent at the point of interest, where Edge Finder searches for edge extremity candidates when filling edge gaps. That is, M_FILL_GAP_ANGLE, along with M_FILL_GAP_DISTANCE, define a region where two chain extremities can be linked. Note that two chain extremities can only be linked if both are included in the search region of the other. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 360.0°.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 360.0	Specifies the aperture angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_FILL_GAP_CANDIDATE	Sets whether to join an edge extremity with the other extremity of the same edge, or with an extremity of any edge. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_ANY.	c		e	f	g	a		i		j	k	l	m	n	b
M_ANY	Specifies that the extremity of an edge can be connected with the extremity of any edge.	c		e	f	g	a		i		j	k	l	m	n	b
M_SAME	Specifies that the extremity of an edge can only be connected with the other extremity of the same edge.	c		e	f	g	a		i		j	k	l	m	n	b
M_FILL_GAP_CONTINUITY	Sets the continuity constraint used when performing edge gap filling, when more than one edge extremity candidate is present. Among the whole set of extremity candidate edgels found, the one which fulfills the continuity criteria best is chosen to fill the gap. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 50.0.	c		e	f	g	a		i		j	k	l	m	n	b

	0.0 <= *Value* <= 100.0	Specifies the continuity constraint. When set to 0.0, the closest candidate is chosen to link edge segments together. When set to 100.0, the candidate that gives the most continuous result (minimum curvature) is chosen.	c		e	f	g	a		i		j	k	l	m	n	b
M_FILL_GAP_DISTANCE		Sets the maximum distance radius where Edge Finder searches for edge extremity candidates when filling edge gaps. That is, M_FILL_GAP_DISTANCE, along with M_FILL_GAP_ANGLE, define a region where two chain extremities can be linked. Note that two chain extremities can only be linked if both are included in the search region of the other. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0 pixels.	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies an infinite maximum distance radius.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the maximum distance radius, in pixels. Note that generally, large values should not be set. Typically, you would set M_FILL_GAP_DISTANCE to a value in the range of 2 to 5 pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_FILL_GAP_POLARITY		Sets the use of the edge polarity to perform the filling of edge gaps. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_ANY.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ANY	Specifies that gaps between edges will be filled, regardless of their polarity.	c		e	f	g	a		i		j	k	l	m	n	b
	M_REVERSE	Specifies that gaps between edges that have reverse polarity will be filled. Note that M_REVERSE is not available for M_CREST Edge Finder contexts.	c		e	f	g	a		i		j	k	l	m	n	b

	M_SAME	Specifies that gaps between edges that have the same polarity will be filled. Note that M_SAME is not available for M_CREST Edge Finder contexts.	c		e	f	g	a		i		j	k	l	m	n	b

The following ControlType and corresponding ControlValue parameter settings can be specified for an Edge Finder result buffer.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For a result buffer																	
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																
M_AGM_COMPATIBLE		Sets whether the Edge Finder result buffer can be used with an AGM context. When used together, you can define models from the result of an edge extraction, or you can find model occurrences in the result of an edge extraction. Note that you must call MedgeCalculate() for this setting to take effect. INQ															
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the Edge Finder result buffer cannot be used with an AGM context.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the Edge Finder result buffer can be used with an AGM context.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_CROSS_SIZE		Sets the size of the cross used to identify certain features when drawing with MedgeDraw(). This cross is used to draw the edge's edgels (M_DRAW_EDGELS), mid-point (M_DRAW_POSITION), and center of gravity (M_DRAW_CENTER_OF_GRAVITY). For more information, see MedgeDraw(). INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 4.0 pixels.	c		e	f	g	a		i		j	k	l	m	n	b

Value	Specifies the size of the cross, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_MODEL_FINDER_COMPATIBLE	Sets whether the Edge Finder result buffer can be used with a Model Finder context. When used together, you can define models from the result of an edge extraction, or you can find model occurrences in the result of an edge extraction. Note that you must call MedgeCalculate() for this setting to take effect. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that the Edge Finder result buffer cannot be used with a Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies that the Edge Finder result buffer can be used with a Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_NEIGHBOR_RADIUS	Sets the radius distance used to select the closest edge or edges from a point. For more information, see the MedgeSelect() Condition parameter values M_NEAREST_NEIGHBOR and M_ALL_NEAREST_NEIGHBORS. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 2.0 pixels.	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the radius distance, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_RESULT_OUTPUT_UNITS	Sets whether to return results in pixels or world units. This essentially sets the output coordinate system to use. The setting of this control type will only affect functions within this module which return positional results. This control type can be changed at any time to return results in the required output units. INQ	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Same as M_ACCORDING_TO_CALIBRATION.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ACCORDING_TO_CALIBRATION	Specifies that results are returned in world units if the result was calculated on an image associated with a camera calibration context; otherwise, specifies that results are returned in pixel units.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PIXEL	Specifies that results are returned in pixel units, with respect to the pixel coordinate system.	c		e	f	g	a		i		j	k	l	m	n	b
	M_WORLD	Specifies that results are returned in world units, with respect to the relative coordinate system. If world units are specifed, calling MedgeGetResult() or MedgeGetNeighbors() generates an error if the result was not calculated on a calibrated image.	c		e	f	g	a		i		j	k	l	m	n	b

The following ControlType and corresponding ControlValue parameter settings are used to set closest-edgel constraints for MedgeGetNeighbors(). This function retrieves the coordinates of edgels, from an Edge Finder result buffer, that meet the constraints set below, and correspond to the closest neighbors from a list of user-specified source points.

The following values can only be specified for an Edge Finder result buffer.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For setting closest-edgel constraints																	
ControlType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	ControlValue																

M_NEIGHBOR_ANGLE		Sets the gradient angle that an edgel must have, before being considered a candidate, when finding the closest edgels to a list of points. M_NEIGHBOR_ANGLE is based on a source angle, which you must provide for each source point, with MedgeGetNeighbors(). Note that you can set a tolerance for this angle, using **MedgeControl**() with M_NEIGHBOR_ANGLE_TOLERANCE. M_NEIGHBOR_ANGLE can only be used if the internal angle buffer (**MedgeControl**() with M_SAVE_ANGLE) was initially saved. INQ															
	M_DEFAULT	Same as M_ANY.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ANY	Specifies that the edgel candidate can have any gradient angle. Note that M_ANY is equivalent to setting M_NEIGHBOR_ANGLE_TOLERANCE to 360.0°. In this case, you need not provide the source angle(s) in MedgeGetNeighbors().	c		e	f	g	a		i		j	k	l	m	n	b
	M_REVERSE	Specifies that the gradient angle of the edgel candidate must have the reverse angle (+ 180°) of the source point.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SAME	Specifies that the gradient angle of the edgel candidate must have the same angle as the source point.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SAME_OR_REVERSE	Specifies that the gradient angle of the edgel candidate must either have the same, or the reverse angle of the source point.	c		e	f	g	a		i		j	k	l	m	n	b

M_NEIGHBOR_ANGLE_TOLERANCE		Sets the angular tolerance to use for the angle constraint (M_NEIGHBOR_ANGLE), when finding the closest edgels to a list of points. Only edgels that have a gradient angle that falls within this angular range can be returned. Note that M_NEIGHBOR_ANGLE_TOLERANCE is ignored if M_NEIGHBOR_ANGLE is set to M_ANY. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 180.0°.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 360.0	Specifies the angular tolerance, in degrees. This value is applied to the angle constraint. Note that setting M_NEIGHBOR_ANGLE_TOLERANCE to 360.0° is equivalent to setting M_NEIGHBOR_ANGLE to M_ANY.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEIGHBOR_MAXIMUM_NUMBER		Sets the maximum number of edgels that can be returned (for each point), when finding the closest edgels to a list of points. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the maximum number of edgels.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEIGHBOR_MINIMUM_SPACING		Sets the minimum distance separating closest edgel candidates within the same edge, when finding the closest edgels to a list of points. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_INFINITE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies no minimum distance separating two edgel candidates.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 1	Specifies the minimum distance, in edgels.	c		e	f	g	a		i		j	k	l	m	n	b

M_SEARCH_ANGLE		Sets the search angle constraint (applied the Edge Finder result buffer), when finding the closest edgels to a list of points. Only edgels that are located along this angle can be returned. M_SEARCH_ANGLE is relative to the source angle set with MedgeGetNeighbors(). Note that you can set the angular tolerance and the angular orientation for M_SEARCH_ANGLE, using **MedgeControl**() with M_SEARCH_ANGLE_TOLERANCE and M_SEARCH_ANGLE_SIGN, respectively. M_SEARCH_ANGLE, M_SEARCH_ANGLE_TOLERANCE, and M_SEARCH_ANGLE_SIGN limit the search region for potential edgel candidates to an angular sector within the Edge Finder result buffer. Only edgels that are located within this region can be returned. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0°.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 360.0	Specifies the angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b

M_SEARCH_ANGLE_SIGN		Sets the orientation to use for the search angle constraint, when finding the closest edgels to a list of points. The orientation is relative to the source angle set with MedgeGetNeighbors(). For example, if the source angle is 45°, then the reverse orientation would be 225°. Note that you can set the search angle and the angular tolerance for M_SEARCH_ANGLE_SIGN, using **MedgeControl**() with M_SEARCH_ANGLE and M_SEARCH_ANGLE_TOLERANCE, respectively. M_SEARCH_ANGLE, M_SEARCH_ANGLE_TOLERANCE, and M_SEARCH_ANGLE_SIGN limit the search region for potential edgel candidates to an angular sector within the Edge Finder result buffer. Only edgels that are located within this region can be returned. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_SAME.	c		e	f	g	a		i		j	k	l	m	n	b
	M_REVERSE	Specifies that the orientation of the angle must be the reverse (+ 180°) of the source angle. That is, the angle used will look flipped when compared to the source angle.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SAME	Specifies that the orientation of the angle must be the same as the source angle.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SAME_OR_REVERSE	Specifies that the orientation of the angle can be the same as, or the reverse of, the source angle.	c		e	f	g	a		i		j	k	l	m	n	b

M_SEARCH_ANGLE_TOLERANCE	Sets the angular tolerance to use for the search angle constraint, when finding the closest edgels to a list of points. Only edgels that are located within this angular range (in the Edge Finder result buffer) can be returned. Note that you can set the search angle and the angular orientation for M_SEARCH_ANGLE_TOLERANCE, using **MedgeControl**() with M_SEARCH_ANGLE and M_SEARCH_ANGLE_SIGN, respectively. M_SEARCH_ANGLE, M_SEARCH_ANGLE_TOLERANCE, and M_SEARCH_ANGLE_SIGN limit the search region for potential edgel candidates to an angular sector within the Edge Finder result buffer. Only edgels that are located within this region can be returned. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 360.0°.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 360.0	Specifies the angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_SEARCH_RADIUS_MAX	Sets the maximum radius distance in the Edge Finder result buffer that will be searched for the closest edgels. You must use MedgeGetNeighbors() to specify the source point from which the radius is measured. Used together, M_SEARCH_RADIUS_MAX and M_SEARCH_RADIUS_MIN define a ring-like region, in the Edge Finder result buffer, that will be used to find the closest edgels. All edgels that fall outside this ring will be ignored. INQ	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_INFINITE.	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies a maximum distance radius that spans all edgels in the Edge Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b

	Value	Specifies the maximum distance radius, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_SEARCH_RADIUS_MIN		Sets the minimum radius distance in the Edge Finder result buffer that will be searched for the closest edgels. You must use MedgeGetNeighbors() to specify the source point from which the radius is measured. Used together, M_SEARCH_RADIUS_MAX and M_SEARCH_RADIUS_MIN define a ring-like region, in the Edge Finder result buffer, that will be used to find the closest edgels. All edgels that fall outside this ring will be ignored. INQ	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0 pixels.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the minimum distance radius, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b

Type-specific versions of the function when using a C compiler under 64-bit

```
void MedgeControlInt64 (AIL_ID ContextOrResultId, AIL_INT64 ControlType, AIL_INT64 ControlValue)
```

Details

Parameters

ContextOrResultId

See ContextOrResultId of the main function for a description.

ControlType

See ControlType of the main function for a description.

ControlValue

See ControlValue of the main function for a description.

```
void MedgeControlDouble (AIL_ID ContextOrResultId, AIL_INT64 ControlType, AIL_DOUBLE ControlValue)
```

Details

Parameters

ContextOrResultId

See ContextOrResultId of the main function for a description.

ControlType

See ControlType of the main function for a description.

ControlValue

See ControlValue of the main function for a description.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailedge11.lib.
DLL	Requires ail11.dll; ailedge11.dll.

MedgeDraw

Synopsis

Draw specific edge results in the destination image buffer or 2D graphics list.

Syntax

```
void MedgeDraw(
    AIL_ID ContextGraId,           //in
    AIL_ID ResultEdgeId,           //in
    AIL_ID DstImageBufOrListGraId, //out
    AIL_INT64 Operation,           //in
    AIL_INT IndexOrLabel,          //in
    AIL_INT64 ControlFlag          //in
)
```

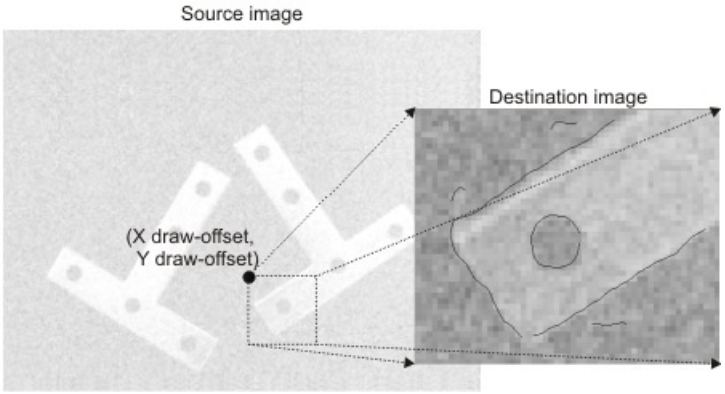
Description

This function draws specific edge results in the destination image buffer or 2D graphics list.

To draw edge results, the required information must be available for the specified edge or group of edges, in the Edge Finder result buffer. To verify availability, call MedgeGetResult() with the required edge result combined with M_AVAILABLE.

You can draw results and settings, obtained relative to an offset, at the top-left corner of the destination image, using MgraControl() with M_DRAW_OFFSET_X and M_DRAW_OFFSET_Y and zoom them using MgraControl() with M_DRAW_ZOOM_X and M_DRAW_ZOOM_Y. For example, you can draw a zoomed section of the edge map found in the target image at the top-left corner of the destination image. For more information, see the Drawing graphics with offset and zoom subsection of the Drawing graphics section of Chapter 26: Generating graphics.

Drawing a zoomed region of the edge map



Note that if the edges are calculated using a calibrated source image, Edge Finder takes the camera calibration into account; that is, drawings might be distorted, according to the camera calibration. For example, a straight line (in the world) might be drawn as a curve.

Parameters

ContextGraId

Specifies the identifier of the 2D graphics context to use when drawing. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the 2D graphics context																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Specifies that the default 2D graphics context of the current Aurora Imaging Library application is used. > Note that there is a different default 2D graphics context for each thread.	c		e	f	g	a		i		j	k	l	m	n	b
2D graphics context identifier	Specifies a valid 2D graphics context identifier, which you have allocated using MgraAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

ResultEdgeId

Specifies the identifier of the Edge Finder result buffer from which to extract the results to draw. The Edge Finder result buffer must have been previously allocated on the required system using MedgeAllocResult().

DstImageBufOrListGraId

Specifies the identifier of the destination image buffer or 2D graphics list in which to draw. The buffer can be any valid 1- or 3-band image buffer allocated using MbufAlloc...(). The 2D graphics list must be previously allocated using MgraAllocList(). By drawing into a display's overlay buffer or associating the 2D graphics list with the display, you can also annotate an image non-destructively.

If specifying an image buffer, it must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.

When using **MedgeDraw**() to draw results extracted from a calibrated source image, the destination drawing buffer need not be calibrated to the same world. **MedgeDraw**() draws all the requested results according to the calibrated source image.

Operation

Specifies the type of operation to perform. The possible Operation parameter values in the table below can be added together to draw multiple feature results at once.

The following operations can draw in a destination image buffer or 2D graphics list.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the type of operation to perform																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_DRAW_BOX +	Draws a bounding box around each edge. Note that to perform this operation, the M_BOX feature in MedgeControl() must be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_CENTER_OF_GRAVITY +	Draws a cross at the center of gravity of each edge. You can modify the cross size by setting MedgeControl() with M_DRAW_CROSS_SIZE to an appropriate value. Note that to perform this operation, the M_CENTER_OF_GRAVITY feature in MedgeControl() must be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_CIRCLE_FIT +	Draws the circle fit of each edge. Note that to perform this operation, the M_CIRCLE_FIT feature in MedgeControl() must be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_EDGE +	Draws each edge.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_EDGELS +	Draws a cross at each edgel. You can modify the cross size by setting MedgeControl() with M_DRAW_CROSS_SIZE to an appropriate value.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_ELLIPSE_FIT +	Draws the ellipse fit of each edge. Note that to perform this operation, the M_ELLIPSE_FIT feature in MedgeControl() must be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_FERET_GENERAL +	Draws the edges' general Feret, using an H-type line (|-|). This line is drawn on contact with the edge's two extrema edgels, at the general Feret diameter's angle. Note that to perform this operation, the M_FERET_GENERAL feature in MedgeControl() must be calculated.	c		e	f	g	a		i		j	k	l	m	n	b

M_DRAW_FERET_MAX +	Draws the edges' maximum Feret, using an H-type line (|-|). This line is drawn on contact with the edge's two extrema edgels, at the maximum Feret diameter's angle. Note that to perform this operation, the M_FERET_MAX_DIAMETER and M_FERET_MAX_ANGLE features in MedgeControl() must be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_FERET_MIN +	Draws the edges' minimum Feret, using an H-type line (|-|). This line is drawn on contact with the edge's two extrema edgels, at the minimum Feret diameter's angle. Note that to perform this operation, the M_FERET_MIN_DIAMETER and M_FERET_MIN_ANGLE features in MedgeControl() must be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_INDEX +	Draws each edge's index value. Note that this operation will only work for included edges. Therefore the IndexOrLabel parameter cannot be set to M_EXCLUDED_EDGES and can only be set to M_ALL_EDGES if all edges are included.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_LABEL +	Draws each edge's label value. Note that to perform this operation, the M_LABEL_VALUE feature in MedgeControl() must be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_LINE_FIT +	Draws the line fit of each edge. Note that to perform this operation, the M_LINE_FIT feature in MedgeControl() must be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_POSITION +	Draws a cross at the center of each edge. You can modify the cross size by setting MedgeControl() with M_DRAW_CROSS_SIZE to an appropriate value. Note that to perform this operation, the M_POSITION feature in MedgeControl() must be calculated.	c		e	f	g	a		i		j	k	l	m	n	b

M_DRAW_SEGMENTS +	Draws the segments of the edge approximation. Note that to perform this operation, edge approximations must have been previously calculated by setting MedgeControl() with M_CHAIN_APPROXIMATION to M_LINE or M_WORLD_LINE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_VERTICES +	Draws the segment intersections (or vertices) of the edge approximation. Note that to perform this operation, edge approximations must have been previously calculated by setting MedgeControl() with M_CHAIN_APPROXIMATION to M_LINE or M_WORLD_LINE.	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for M_DRAW_CENTER_OF_GRAVITY; M_DRAW_FERET_GENERAL; M_DRAW_FERET_MAX; M_DRAW_FERET_MIN; M_DRAW_POSITION.

You can add the following value to the above-mentioned values to specify that the operation's numeric value is drawn.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For M_DRAW_CENTER_OF_GRAVITY, M_DRAW_POSITION, M_DRAW_FERET_MIN, M_DRAW_FERET_MAX, and M_DRAW_FERET_GENERAL																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DRAW_VALUE +	Specifies that the operation's numerical value is drawn. For example, if you add M_DRAW_VALUE to M_DRAW_CENTER_OF_GRAVITY, the center of gravity's coordinates are drawn within parenthesis, and centered above the drawing cross.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values that can be used alone or as combination values for the values listed in For specifying the type of operation to perform, For M_DRAW_CENTER_OF_GRAVITY, M_DRAW_POSITION, M_DRAW_FERET_MIN, M_DRAW_FERET_MAX, and M_DRAW_FERET_GENERAL.

You can use one of the following values on its own, or add it to the above-mentioned values, to specify an operation that cannot use a 2D graphics list as a destination.

The following operations can only draw in an image buffer.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying operations that cannot use a 2D graphics list as a destination																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)

M_DRAW_ANGLE	Draws the internal angle buffer. Note that to perform this operation, the angle buffer must have been previously saved in the result buffer by enabling M_SAVE_ANGLE in MedgeControl().	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_CROSS_DERIVATIVE	Draws the internal cross-derivative buffer of the source image. M_DRAW_CROSS_DERIVATIVE is only relevant for M_CREST Edge Finder contexts. Note that to perform this operation, the derivatives must have been previously saved in the result buffer by enabling M_SAVE_DERIVATIVES in MedgeControl().	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_FIRST_DERIVATIVE_X	Draws the internal first derivative buffer of the source image, in the X-direction. M_DRAW_FIRST_DERIVATIVE_X is only relevant for M_CONTOUR Edge Finder contexts. Note that to perform this operation, the derivatives must have been previously saved in the result buffer by enabling M_SAVE_DERIVATIVES in MedgeControl().	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_FIRST_DERIVATIVE_Y	Draws the internal first derivative buffer of the source image, in the Y-direction. M_DRAW_FIRST_DERIVATIVE_Y is only relevant for M_CONTOUR Edge Finder contexts. Note that to perform this operation, the derivatives must have been previously saved in the result buffer by enabling M_SAVE_DERIVATIVES in MedgeControl().	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_IMAGE	Draws the source image. Note that to perform this operation, the source image must have been previously saved in the result buffer by enabling M_SAVE_IMAGE in MedgeControl().	c		e	f	g	a		i		j	k	l	m	n	b

M_DRAW_MAGNITUDE	Draws the internal magnitude buffer. Note that to perform this operation, the magnitude buffer must have been previously saved in the result buffer by enabling M_SAVE_MAGNITUDE in MedgeControl().	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_MASK	Draws the mask buffer. Note that to perform this operation, the mask buffer must have been previously saved in the result buffer by enabling M_SAVE_MASK in MedgeControl().	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_SECOND_DERIVATIVE_X	Draws the internal second derivative buffer of the source image, in the X-direction. M_DRAW_SECOND_DERIVATIVE_X can only be used with M_CREST Edge Finder contexts. Note that to perform this operation, the derivatives must have been previously saved in the result buffer by enabling M_SAVE_DERIVATIVES in MedgeControl().	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_SECOND_DERIVATIVE_Y	Draws the internal second derivative buffer of the source image, in the Y-direction. M_DRAW_SECOND_DERIVATIVE_Y can only be used with M_CREST Edge Finder contexts. Note that to perform this operation, the derivatives must have been previously saved in the result buffer by enabling M_SAVE_DERIVATIVES in MedgeControl().	c		e	f	g	a		i		j	k	l	m	n	b

IndexOrLabel

Specifies the edge(s) to draw. This parameter must be set to one of the values below.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the edge(s) to draw																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_INCLUDED_EDGES.	c		e	f	g	a		i		j	k	l	m	n	b

M_ALL_EDGES +	Specifies all edges.	c		e	f	g	a		i		j	k	l	m	n	b
M_EXCLUDED_EDGES +	Specifies all currently excluded edges.	c		e	f	g	a		i		j	k	l	m	n	b
M_INCLUDED_EDGES +	Specifies all currently included edges.	c		e	f	g	a		i		j	k	l	m	n	b
Value +	Specifies either the edge's index or label. For more information, see combination values below. Valid index values must fall within the following range: 0 to the number of included edges in the Edge Finder result buffer - 1. You can retrieve valid label values using MedgeGetResult() with M_LABEL_VALUE.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for any of the possible values of the IndexOrLabel parameter.

You can add one of the following values to the above-mentioned values to specify whether you are providing an index or a label value.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For an index or a label value																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_TYPE_INDEX	Specifies an edge using its index value. Note that only included edges (M_INCLUDED_EDGES) can be drawn. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_LABEL	Specifies an edge using its label value.	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailedge11.lib.
DLL	Requires ail11.dll; ailedge11.dll.

MedgeFree

Synopsis

Free an Edge Finder context or an Edge Finder result buffer.

Syntax

```
void MedgeFree(
    AIL_ID ObjectId //in
)
```

Description

This function deletes the specified Edge Finder context or Edge Finder result buffer identifier, and releases any memory associated with it.

All Edge Finder contexts and all Edge Finder result buffers allocated on a particular system must be freed before the system can be freed, unless M_UNIQUE_ID was specified during allocation.

Parameter

ObjectId

Specifies the identifier of the Edge Finder context or Edge Finder result buffer to free. These must have been successfully allocated (with MedgeAlloc() or MedgeAllocResult()) prior to calling this function.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailedge11.lib.
DLL	Requires ail11.dll; ailedge11.dll.

MedgeGetNeighbors

Synopsis

Get edgels from an Edge Finder result buffer that are the closest neighbors to a list of user-specified point coordinates.

Syntax

```
void MedgeGetNeighbors(
    AIL_ID EdgeResultId,                //in
    AIL_INT SizeOfArray,                //in
    const AIL_DOUBLE *SrcArrayXPtr,     //in
    const AIL_DOUBLE *SrcArrayYPtr,     //in
    const AIL_DOUBLE *SrcArrayAnglePtr, //in
    AIL_DOUBLE *DstArrayXPtr,           //out
    AIL_DOUBLE *DstArrayYPtr,           //out
    AIL_INT *DstArrayIndexPtr,          //out
    AIL_INT *DstArrayLabelPtr,          //out
    AIL_INT64 ControlFlag               //in
)
```

Description

This function retrieves the coordinates of edgels, from an Edge Finder result buffer, that correspond to the closest neighbors from a list of user-specified source point coordinates. You can also set a series of constraints that potential results must adhere to before being returned. To do so, use the appropriate settings in MedgeControl().

If your target image was associated with a camera calibration context but you want to retrieve positional and dimensional results in pixel units, use MedgeControl() with the M_RESULT_OUTPUT_UNITS control type set to M_PIXEL. Note that if you set M_RESULT_OUTPUT_UNITS to M_WORLD and the results were not obtained from a calibrated image, **MedgeGetNeighbors**() will generate an error. Additionally, if results were obtained from a calibrated source image, and you specify to retrieve results in world units, user-specified coordinates and constraints must be specified in the same world units as the coordinate system used to calculate the results.

Parameters

EdgeResultId

Specifies the identifier of the Edge Finder result buffer from which to search for edgel candidates.

SizeOfArray

Specifies the number of points provided. This value represents how many user-specified point coordinates will be searched for in the Edge Finder result buffer.

> When using a standard vector (std::vector) overload function in C++, you can pass **M_DEFAULT** to this parameter; the size will be determined automatically based on the number of items in the vector passed to the SrcArrayXPtr, SrcArrayYPtr, SrcArrayAnglePtr, DstArrayXPtr, DstArrayYPtr, DstArrayIndexPtr, or DstArrayLabelPtr parameter.

SrcArrayXPtr

Accepts the address of one of the following:

- array of type AIL_DOUBLE [optionally, in C++: a reference to a constant std::vector<AIL_DOUBLE>] (and a maximum array rank for C# of 1)
  Required array size:(SizeOfArray)

Specifies the address of the array containing the X-coordinate(s) of the source point(s).

SrcArrayYPtr
Accepts the address of one of the following:

- array of type AIL_DOUBLE [optionally, in C++: a reference to a constant std::vector<AIL_DOUBLE>] (and a maximum array rank for C# of 1)
  Required array size:(SizeOfArray)

Specifies the address of the array containing the Y-coordinate(s) of the source point(s).

SrcArrayAnglePtr
Accepts the address of one of the following:

- array of type AIL_DOUBLE [optionally, in C++: a reference to a constant std::vector<AIL_DOUBLE>] (and a maximum array rank for C# of 1)
  Required array size:(SizeOfArray)

Specifies the address of the array containing the constraint angle(s) used to refine the search for edgel candidates. Note that angle values (0° to 360°) are measured counter-clockwise and must be mapped between either 0 to 255 (for contour contexts) or 0 to 127 (for crest contexts).

If no value is required, set this parameter to **M_NULL**.

DstArrayXPtr
Accepts the address of one of the following:

- array of type AIL_DOUBLE [optionally, in C++: a reference to a constant std::vector<AIL_DOUBLE>] (and a maximum array rank for C# of 1)
  Required array size:(SizeOfArray)

Specifies the address of the array in which the X-coordinate(s) of the edgel(s) found in the Edge Finder result buffer are to be written. If no edgels were found, ignore the information written. Note that if no edgels were found, the index (DstArrayIndexPtr) and label (DstArrayLabelPtr) of the edgels is **M_NULL**.

The size of the array must be equal to SizeOfArray multiplied by the maximum number of edgel candidates for each point (M_NEIGHBOR_MAXIMUM_NUMBER).

If no value is required, set this parameter to **M_NULL**.

DstArrayYPtr
Accepts the address of one of the following:

- array of type AIL_DOUBLE [optionally, in C++: a reference to a constant std::vector<AIL_DOUBLE>] (and a maximum array rank for C# of 1)
  Required array size:(SizeOfArray)

Specifies the address of the array in which the Y-coordinate(s) of the edgel(s) found in the Edge Finder result buffer is to be written. If no edgels were found, ignore the information written. Note that if no edgels were found, the index (DstArrayIndexPtr) and label (DstArrayLabelPtr) of the edgels is **M_NULL**.

The size of the array must be equal to SizeOfArray multiplied by the maximum number of edgel candidates for each point (M_NEIGHBOR_MAXIMUM_NUMBER).

If no value is required, set this parameter to **M_NULL**.

DstArrayIndexPtr
Accepts the address of one of the following:

- array of type AIL_DOUBLE [optionally, in C++: a reference to a constant std::vector<AIL_DOUBLE>] (and a maximum array rank for C# of 1)
  Required array size:(SizeOfArray)

Specifies the address of the array in which the index of the edgel(s) found in the Edge Finder result buffer is to be written. If no edgels were found, **M_NULL** is written.

The size of the array must be equal to SizeOfArray multiplied by the maximum number of edgel candidates for each point (M_NEIGHBOR_MAXIMUM_NUMBER).

If no value is required, set this parameter to **M_NULL**.

DstArrayLabelPtr
Accepts the address of one of the following:

- array of type AIL_DOUBLE [optionally, in C++: a reference to a constant std::vector<AIL_DOUBLE>] (and a maximum array rank for C# of 1)
  Required array size:(SizeOfArray)

Specifies the address of the array in which the label of the edgel's edge found in the Edge Finder result buffer is to be written. If no edgels were found, **M_NULL** is written.

The size of the array must be equal to SizeOfArray multiplied by the maximum number of edgel candidates for each point (M_NEIGHBOR_MAXIMUM_NUMBER).

If no value is required, set this parameter to **M_NULL**.

ControlFlag

Specifies the accuracy with which to return edgels.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the accuracy																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_GET_EDGELS.	c		e	f	g	a		i		j	k	l	m	n	b
M_GET_EDGELS	Specifies that edgels are returned with normal accuracy. In this case, only edgels explicitly located in the Edge Finder result buffer can be returned.	c		e	f	g	a		i		j	k	l	m	n	b

M_GET_SUBEDGELS	Specifies that edgels are returned with high accuracy. In this case, a point between two edgels can be returned. When using M_GET_SUBEDGELS, the M_NEIGHBOR_MAXIMUM_NUMBER constraint in MedgeControl() must be set to 1.	c		e	f	g	a		i		j	k	l	m	n	b

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailedge11.lib.
DLL	Requires ail11.dll; ailedge11.dll.

MedgeGetResult

Synopsis

Get results of the included edges from an Edge Finder result buffer.

Syntax

```
void MedgeGetResult(
    AIL_ID EdgeResultId,           //in
    AIL_INT EdgeIndexOrLabelValue, //in
    AIL_INT64 ResultType,          //in
    void *FirstResultArrayPtr,     //out
    void *SecondResultArrayPtr     //out
)
```

Description

This function retrieves the results of a specified type from an Edge Finder result buffer, after an MedgeCalculate() call.

The following result types can be retrieved from the Edge Finder result buffer:

- General results related to the edge extraction and edge processing.
- Edge chain results.
- Edge feature results.

If your target image was associated with a camera calibration context, positional and dimensional results are, by default, returned with respect to the relative coordinate system of the image. Otherwise, these results are returned in pixels, relative to the top-left pixel of the image.

> If your target image was associated with a camera calibration context but you want to retrieve positional and dimensional results in pixel units, use MedgeControl() with the M_RESULT_OUTPUT_UNITS control type set to M_PIXEL. However, note that if you set M_RESULT_OUTPUT_UNITS to M_WORLD without specifying a calibrated image in which to calculate the results, **MedgeGetResult**() will generate an error.

Note that in the presence of distortion, some results are meaningless when converted from real-world to pixel units (for example, the Feret angles). For example, if an edge appears warped in the source image, but the camera calibration context of the source image compensates for this during the extraction, the resulting Feret angles are meaningful in the real-world coordinate system, and meaningless in the pixel coordinate system.

Note that, unless otherwise specified, results are only returned to FirstResultArrayPtr; when this is the case, SecondResultArrayPtr must be set to **M_NULL**.

Parameters

EdgeResultId

Specifies the identifier of the Edge Finder result buffer from which to get results.

EdgeIndexOrLabelValue

Specifies the edge(s) from which to get results. This parameter can be set to one of the following values.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For edge results																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT	Same as M_ALL.	c		e	f	g	a		i		j	k	l	m	n	b
M_ALL +	Specifies all edges. That is, the target array(s) will be filled with the specified type of result for all the included edges. Note that target array values are ordered by edge indices.	c		e	f	g	a		i		j	k	l	m	n	b
Value >= 0 +	Specifies either the edge's index or label. Index values must fall within the following range: 0 to the number of included edges in the Edge Finder result buffer - 1. Label values can be returned with the result type M_LABEL_VALUE.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for the values listed in For edge results.

You can add one of the following values to the above-mentioned values to specify whether you are providing an index or a label value.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying an index or label value																
Combination value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_TYPE_INDEX	Specifies an edge using its index value. This is the default value.	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_LABEL	Specifies an edge using its label value.	c		e	f	g	a		i		j	k	l	m	n	b

ResultType

Specifies the type of result to retrieve.

See the Parameter associations section for possible values that can be specified.

FirstResultArrayPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]
- array of type AIL_ID [optionally, in C++: a reference to a std::vector<AIL_ID>]
- array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]
- array of type AIL_INT32 [optionally, in C++: a reference to a std::vector<AIL_INT32>]
- array of type AIL_INT64 [optionally, in C++: a reference to a std::vector<AIL_INT64>]

- AIL_DOUBLE
- AIL_ID
- AIL_INT
- AIL_INT32
- AIL_INT64

Specifies the address of the first array in which to write the requested information.

See the Parameter associations section for possible values that can be returned.

SecondResultArrayPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]
- array of type AIL_ID [optionally, in C++: a reference to a std::vector<AIL_ID>]
- array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]
- array of type AIL_INT32 [optionally, in C++: a reference to a std::vector<AIL_INT32>]
- array of type AIL_INT64 [optionally, in C++: a reference to a std::vector<AIL_INT64>]
- AIL_DOUBLE
- AIL_ID
- AIL_INT
- AIL_INT32
- AIL_INT64
- M_NULL

Specifies the address of the second array in which to write the requested information. If nothing is to be written, set SecondResultArrayPtr to **M_NULL**.

See the Parameter associations section for possible values that can be returned.

Parameter associations

The tables below list possible values for the **ResultType** parameter and possible values returned to the **FirstResultArrayPtr** and **SecondResultArrayPtr** parameters.

- For retrieving results related to edge extraction and edge processing
- For retrieving information about the internal processing buffers
- For retrieving edge chain results
- For retrieving edge feature results

To retrieve general results related to the edge extraction and edge processing, the ResultType parameter must be set to one of the values below.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving results related to edge extraction and edge processing																	
ResultType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
FirstResultArrayPtr - Possible values returned																

SecondResultArrayPtr - Possible values returned																
M_CONTEXT_TYPE +	Retrieves the type of the Edge Finder context that was used to extract edges.															
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* Returns one of the following.	c		e	f	g	a		i		j	k	l	m	n	b
M_CONTOUR	Specifies a contour context type, which is used to find object contours in images.	c		e	f	g	a		i		j	k	l	m	n	b
M_CREST	Specifies a crest context type, which is used to find thin line crests in images.	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_CHAINS +	Retrieves the number of included edges.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_X +	Retrieves the width of the source image and internal processing buffers, in pixels. Note that if you have used MedgePut() to add edge chains to the Edge Finder result buffer, the width returned can be bigger than the width of the source image buffer if the edge chains exceed the boundaries of the source image buffer.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_Y +	Retrieves the height of the source image and internal processing buffers, in pixels. Note that if you have used MedgePut() to add edge chains to the Edge Finder result buffer, the height returned can be bigger than the height of the source image buffer if the edge chains exceed the boundaries of the source image buffer.	c		e	f	g	a		i		j	k	l	m	n	b

FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_THRESHOLD_HIGH +	Retrieves the upper bound value, used during hysteresis thresholding to extract edges.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_THRESHOLD_LOW +	Retrieves the lower bound value, used during hysteresis thresholding to extract edges.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_THRESHOLDS +	Retrieves both the lower and upper bound values, used during hysteresis thresholding to extract edges.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* Returns the lower bound threshold value.	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* Returns the upper bound threshold value.	c		e	f	g	a		i		j	k	l	m	n	b
M_TIMEOUT_END +	Retrieves whether the timeout limit was reached. You can set the timeout limit using MedgeControl() with M_TIMEOUT. By default, there is no limit.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* Returns one of the following.	c		e	f	g	a		i		j	k	l	m	n	b
M_FALSE	Specifies that the timeout limit has not been reached.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRUE	Specifies that the timeout limit has been reached.	c		e	f	g	a		i		j	k	l	m	n	b

	SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b

To retrieve information about the internal processing buffers, you must specify the internal buffer about which to retrieve the information and the type of information to retrieve. To do so, set ResultType to a combination of two values, one from each of the following two tables. The information retrieved can be used to allocate a buffer with the same properties as the internal processing buffer. This newly allocated buffer can then be used, for example, as the destination buffer for MedgeDraw().

> The internal Edge Finder buffers cannot be accessed directly; however, they can be accessed by creating a buffer with the same characteristics as the internal buffer and using MedgeDraw() to draw the contents of the internal Edge Finder buffer into your new buffer.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving information about the internal processing buffers																		
ResultType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	FirstResultArrayPtr - Possible values returned																
	SecondResultArrayPtr - Possible values returned																
M_ANGLE_ID +		Retrieves information about the internal angle buffer. Note that information about the internal angle buffer is only available if it was saved using MedgeControl() with M_SAVE_ANGLE set to M_ENABLE. You must specify a combination value from the following table: • To specify the type of information to retrieve from the specified internal processing buffer															
	FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* Returns the identifier of the internal angle buffer.	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b

M_CROSS_DERIVATIVE_ID +	Retrieves information about the internal cross derivative buffer. Note that information about the internal cross derivative buffer is only available if it was saved using MedgeControl() with M_SAVE_DERIVATIVES set to M_ENABLE. You must specify a combination value from the following table: • To specify the type of information to retrieve from the specified internal processing buffer	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* Returns the identifier of the internal cross derivative buffer.	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_FIRST_DERIVATIVE_X_ID +	Retrieves information about the internal first derivative buffer in the X-direction. Note that information about the internal first derivative in the X-direction buffer is only available if it was saved using MedgeControl() with M_SAVE_DERIVATIVES set to M_ENABLE. You must specify a combination value from the following table: • To specify the type of information to retrieve from the specified internal processing buffer	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* Returns the identifier of the internal first derivative buffer in the X-direction.	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b

M_FIRST_DERIVATIVE_Y_ID +	Retrieves information about the internal first derivative buffer in the Y-direction. Note that information about the internal first derivative buffer in the Y-direction is only available if it was saved using MedgeControl() with M_SAVE_DERIVATIVES set to M_ENABLE. You must specify a combination value from the following table: • To specify the type of information to retrieve from the specified internal processing buffer	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* Returns the identifier of the internal first derivative buffer in the Y-direction.	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_FIRST_DERIVATIVES_ID +	Retrieves information about the internal first derivative buffers in both the X- and Y-directions. Note that information about the internal first derivative buffers in both the X- and Y-directions is only available if it was saved using MedgeControl() with M_SAVE_DERIVATIVES set to M_ENABLE. You must specify a combination value from the following table: • To specify the type of information to retrieve from the specified internal processing buffer	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* Returns the identifier of the internal first derivative buffer in the X-direction.	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* Returns the identifier of the internal first derivative buffer in the Y-direction.	c		e	f	g	a		i		j	k	l	m	n	b

M_IMAGE_ID +	Retrieves information about the source buffer. Note that information about the source buffer is only available if it was saved using MedgeControl() with M_SAVE_IMAGE set to M_ENABLE. You must specify a combination value from the following table: • To specify the type of information to retrieve from the specified internal processing buffer	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* Returns the identifier of the source buffer.	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_MAGNITUDE_ID +	Retrieves information about the internal magnitude buffer. Note that information about the internal magnitude buffer is only available if it was saved using MedgeControl() with M_SAVE_MAGNITUDE set to M_ENABLE. The magnitude buffer is calculated according to MedgeControl() with M_MAGNITUDE_TYPE. If M_MAGNITUDE_TYPE is set to M_SQR_NORM, magnitude values are truncated on 15.0 bits. If M_MAGNITUDE_TYPE is set to M_NORM, magnitude values are truncated on 8.7 bits, with fixed-point precision. 32-bit float buffers use the same conventions, but with floating-point precision. You must specify a combination value from the following table: • To specify the type of information to retrieve from the specified internal processing buffer	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* Returns the identifier of the internal magnitude buffer.	c		e	f	g	a		i		j	k	l	m	n	b

	SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_MASK_ID +		Retrieves information about the internal mask buffer. Note that information about the internal mask buffer is only available if it was saved using MedgeControl() with M_SAVE_MASK set to M_ENABLE. You must specify a combination value from the following table: • To specify the type of information to retrieve from the specified internal processing buffer	c		e	f	g	a		i		j	k	l	m	n	b
	FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* Returns the identifier of the internal mask buffer.	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_SECOND_DERIVATIVE_X_ID +		Retrieves information about the internal second derivative buffer in the X-direction. Note that information about the internal second derivative buffer in the X-direction is only available if it was saved using MedgeControl() with M_SAVE_DERIVATIVES set to M_ENABLE. You must specify a combination value from the following table: • To specify the type of information to retrieve from the specified internal processing buffer	c		e	f	g	a		i		j	k	l	m	n	b
	FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* Returns the identifier of the internal second derivative buffer in the X-direction.	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b

M_SECOND_DERIVATIVE_Y_ID +	Retrieves information about the internal second derivative buffer in the Y-direction. Note that information about the internal second derivative buffer in the Y-direction is only available if it was saved using MedgeControl() with M_SAVE_DERIVATIVES set to M_ENABLE. You must specify a combination value from the following table: • To specify the type of information to retrieve from the specified internal processing buffer	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* Returns the identifier of the internal second derivative buffer in the Y-direction.	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_SECOND_DERIVATIVES_ID +	Retrieves information about the internal second derivative buffers in both the X- and Y-directions. Note that information about the internal second derivative buffers in both the X- and Y-directions is only available if it was saved using MedgeControl() with M_SAVE_DERIVATIVES set to M_ENABLE. You must specify a combination value from the following table: • To specify the type of information to retrieve from the specified internal processing buffer	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* Returns the identifier of the internal second derivative buffer in the X-direction.	c		e	f	g	a		i		j	k	l	m	n	b

SecondResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* Returns the identifier of the internal second derivative buffer in the Y-direction.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for the values listed in For retrieving information about the internal processing buffers.

You must add one of the following values to the above-mentioned values to specify the type of information to retrieve from the specified internal processing buffer.

Note that to inquire the width or height of the internal processing buffers, use M_SIZE_X or M_SIZE_Y.

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For specifying the type of information to retrieve from an internal processing buffer																
ResultType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
FirstResultArrayPtr - Possible values returned																
SecondResultArrayPtr - Possible values returned																
M_SIGN	Retrieves the buffer range.															
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* Returns one of the following.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIGNED	Specifies that the buffer range is signed.	c		e	f	g	a		i		j	k	l	m	n	b
M_UNSIGNED	Specifies that the buffer range is unsigned.	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_BAND	Retrieves the number of buffer color bands.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE_BIT	Retrieves the depth per band, in bits.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE	Retrieves the buffer data type and depth. Depth is returned in bits.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* Returns one of the following.	c		e	f	g	a		i		j	k	l	m	n	b
M_FLOAT + 32	Specifies 32-bit float data.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIGNED + 8	Specifies 8-bit signed data.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIGNED + 16	Specifies 16-bit signed data.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIGNED + 32	Specifies 32-bit signed data.	c		e	f	g	a		i		j	k	l	m	n	b
M_UNSIGNED + 1	Specifies 1-bit unsigned data.	c		e	f	g	a		i		j	k	l	m	n	b
M_UNSIGNED + 8	Specifies 8-bit unsigned data.	c		e	f	g	a		i		j	k	l	m	n	b
M_UNSIGNED + 16	Specifies 16-bit unsigned data.	c		e	f	g	a		i		j	k	l	m	n	b
M_UNSIGNED + 32	Specifies 32-bit unsigned data.	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b

To retrieve chained edgel results, the ResultType parameter can be set to one of the values below. If EdgeIndexOrLabelValue is set to a specific edge, results are returned for that edge. If EdgeIndexOrLabelValue is set to M_ALL, results are returned for all the included edges.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving edge chain results																		
ResultType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
FirstResultArrayPtr - Possible values returned																
SecondResultArrayPtr - Possible values returned																
M_CHAIN +	Retrieves the coordinates of the edge(s)'s edgels.															

	FirstResultArrayPtr	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]* Returns the edgels' X-coordinates.	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]* Returns the edgels' Y-coordinates.	c		e	f	g	a		i		j	k	l	m	n	b
M_CHAIN_ANGLE +		Retrieves the angle values of the edge(s)'s edgels. The angle is always returned in uncalibrated pixel units and is measured counter-clockwise in the image, from the positive X-axis towards the negative Y-axis of the image. Possible values range from 0 to 360 degrees and mapped in the range of 0 to 255 (that is, an 8-bit range). If M_FLOAT_MODE is set to M_ENABLE, 0° corresponds to 0, and 360° corresponds to 256. If M_FLOAT_MODE is set to M_DISABLE, 360° corresponds to 0, and 0° corresponds to 0. Note that the angle value of the edge at each edgel position is only available if it was saved using MedgeControl() with M_SAVE_CHAIN_ANGLE set to M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	FirstResultArrayPtr	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b

M_CHAIN_CODE +	Retrieves the edge(s)'s chain code. The chain code describes how edgels are connected using the following neighbor code: $\begin{bmatrix} 3 & 2 & 1 \\ 4 & 8 & 0 \\ 5 & 6 & 7 \end{bmatrix}$	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_CHAIN_INDEX +	Retrieves the index of each edgel's edge.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_CHAIN_MAGNITUDE +	Retrieves the magnitude values of the edge(s)'s edgels. Note that the magnitude value of the edge at each edgel position is only available if it was saved using MedgeControl() with M_SAVE_CHAIN_MAGNITUDE set to M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b

M_CHAIN_MAGNITUDE + M_CHAIN_ANGLE +	Retrieves the magnitude values and the angle values of the edge(s)'s edgels. Angles are returned, counter-clockwise, from 0 degrees to 360 degrees and mapped in the range of 0 to 255 (that is, an 8-bit range). Note that the magnitude and angle values of the edge at each edgel position are only available if they were saved using MedgeControl() with M_SAVE_CHAIN_MAGNITUDE and M_SAVE_CHAIN_ANGLE set to M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]* Returns the magnitude values of the edge(s)'s edgels.	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]* Returns the angle values of the edge(s)'s edgels.	c		e	f	g	a		i		j	k	l	m	n	b
M_CHAIN_X +	Retrieves the X-coordinates of the edge(s)'s edgels.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_CHAIN_Y +	Retrieves the Y-coordinates of the edge(s)'s edgels.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

	SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_CHAINED_EDGELS +		Retrieves the number of edgels in the edge(s).	c		e	f	g	a		i		j	k	l	m	n	b
	FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_VERTICES +		Retrieves the number of vertices in the chain approximation. This result is not valid if MedgeControl() with M_CHAIN_APPROXIMATION is disabled.	c		e	f	g	a		i		j	k	l	m	n	b
	FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_VERTICES +		Retrieves the coordinates of the vertices in the chain approximation. This result is not valid if MedgeControl() with M_CHAIN_APPROXIMATION is disabled.	c		e	f	g	a		i		j	k	l	m	n	b
	FirstResultArrayPtr	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]* Returns the X-coordinates of the vertices in the chain approximation.	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]* Returns the Y-coordinates of the vertices in the chain approximation.	c		e	f	g	a		i		j	k	l	m	n	b

M_VERTICES_CHAIN_INDEX +		Retrieves the index of the vertices' corresponding edge, in a chain approximation. This result is not valid if MedgeControl() with M_CHAIN_APPROXIMATION is disabled.	c		e	f	g	a		i		j	k	l	m	n	b
	FirstResultArrayPtr	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_VERTICES_INDEX +		Retrieves the index of the vertices' corresponding edgels, in a chain approximation. This result is not valid if MedgeControl() with M_CHAIN_APPROXIMATION is disabled.	c		e	f	g	a		i		j	k	l	m	n	b
	FirstResultArrayPtr	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_VERTICES_X +		Retrieves the X-coordinates of the vertices in the chain approximation. This result is not valid if MedgeControl() with M_CHAIN_APPROXIMATION is disabled.	c		e	f	g	a		i		j	k	l	m	n	b
	FirstResultArrayPtr	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b

M_VERTICES_Y +		Retrieves the Y-coordinates of the vertices in the chain approximation. This result is not valid if MedgeControl() with M_CHAIN_APPROXIMATION is disabled.	c		e	f	g	a		i		j	k	l	m	n	b
	FirstResultArrayPtr	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b

To retrieve edge feature results, the ResultType parameter must be set to one of the values below. Note that the specified feature must have already been calculated with MedgeCalculate().

If EdgeIndexOrLabelValue is set to an index value or a label value, the edge feature for the specified edge is returned. If EdgeIndexOrLabelValue is set to M_ALL, the edge feature for all the included edges in the Edge Finder result buffer is returned.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving edge feature results																	
ResultType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	FirstResultArrayPtr - Possible values returned																
	SecondResultArrayPtr - Possible values returned																
M_AVERAGE_STRENGTH +		Retrieves the average strength value of each edge.															
	FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_X_MAX +		Retrieves the X-coordinate of each edge's extreme right edgel.	c		e	f	g	a		i		j	k	l	m	n	b
	FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b

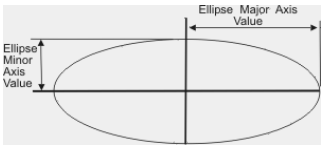
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_X_MIN +	Retrieves the X-coordinate of each edge's extreme left edgel.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_Y_MAX +	Retrieves the Y-coordinate of each edge's extreme bottom edgel.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_Y_MIN +	Retrieves the Y-coordinate of each edge's extreme top edgel.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_OF_GRAVITY +	Retrieves the coordinates of each edge's center of gravity.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]* Returns the X-coordinate of each edge's center of gravity.	c		e	f	g	a		i		j	k	l	m	n	b

SecondResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]* Returns the Y-coordinate of each edge's center of gravity.	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_OF_GRAVITY_X +	Retrieves the X-coordinate of each edge's center of gravity.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_OF_GRAVITY_Y +	Retrieves the Y-coordinate of each edge's center of gravity.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_CIRCLE_FIT_CENTER_X +	Retrieves the X-coordinate of the center of the circle that is the best fit for each edge.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_CIRCLE_FIT_CENTER_Y +	Retrieves the Y-coordinate of the center of the circle that is the best fit for each edge.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b

M_CIRCLE_FIT_COVERAGE +	Retrieves the coverage of the circle that is the best fit for each edge. The coverage describes the portion of the circle covered by the edge.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]* Returns the following.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 1.0	Specifies the amount of coverage of the circle. The lower the value, the lower the coverage. For example, 0 equals no coverage, 0.5 equals 50 percent coverage, and 1.0 equals 100 percent coverage.	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_CIRCLE_FIT_ERROR +	Retrieves the fit error of the circle that is the best fit for each edge. This is calculated as the average quadratic error.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_CIRCLE_FIT_RADIUS +	Retrieves the radius of the circle that is the best fit for each edge.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_CLOSURE +	Retrieves the closure status of each edge.	c		e	f	g	a		i		j	k	l	m	n	b

FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]* Returns one of the following.	c		e	f	g	a		i		j	k	l	m	n	b
M_FALSE	Specifies an open edge.	c		e	f	g	a		i		j	k	l	m	n	b
M_TRUE	Specifies a closed edge.	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_CONVEX_PERIMETER +	Retrieves the convex elongation of each edge.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_ELLIPSE_FIT_ANGLE +	Retrieves the angle, in degrees, of the ellipse that is the best fit for each edge, relative to the output coordinate system specified using MedgeControl() with M_RESULT_OUTPUT_UNITS. An angle interpreted with respect to the pixel coordinate system is always measured counter-clockwise. For information on the angle's direction of rotation when interpreting the angle with respect to the relative coordinate system, see the Angle convention in Aurora Imaging Library subsection of the Working with real-world units section of Chapter 28: Calibrating your camera setup.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b

M_ELLIPSE_FIT_CENTER_X +	Retrieves the X-coordinate of the center of the ellipse that is the best fit for each edge.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_ELLIPSE_FIT_CENTER_Y +	Retrieves the Y-coordinate of the center of the ellipse that is the best fit for each edge.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_ELLIPSE_FIT_COVERAGE +	Retrieves the coverage of the ellipse that is the best fit for each edge. The coverage describes the portion of the ellipse covered by the edge.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]* Returns the following.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 1.0	Specifies the amount of coverage of the ellipse. The lower the value, the lower the coverage. For example, 0.0 equals no coverage, 0.5 equals 50 percent coverage, and 1.0 equals 100 percent coverage.	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_ELLIPSE_FIT_ERROR +	Retrieves the fit error of the ellipse that is the best fit for each edge.	c		e	f	g	a		i		j	k	l	m	n	b

	FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_ELLIPSE_FIT_MAJOR_AXIS +		Retrieves the major axis of the ellipse that is the best fit for each edge. The major axis divides the ellipse across its long dimension into two equal halves. 	c		e	f	g	a		i		j	k	l	m	n	b
	FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_ELLIPSE_FIT_MINOR_AXIS +		Retrieves the minor axis of the ellipse that is the best fit for each edge. The minor axis divides the ellipse across its short dimension into two equal halves, perpendicular to the major axis.	c		e	f	g	a		i		j	k	l	m	n	b
	FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_FAST_LENGTH +		Retrieves the fast length of each edge.	c		e	f	g	a		i		j	k	l	m	n	b
	FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b

M_FERET_BOX +	Retrieves the X- and Y-Feret values of each edge. The X-Feret is the dimension of the minimum bounding box of an edge in the horizontal direction. The Y-Feret is the dimension of the minimum bounding box of an edge in the vertical direction.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]* Returns the X-Feret value of each edge.	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]* Returns the Y-Feret value of each edge.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_ELONGATION +	Retrieves the Feret elongation of each edge. It is accurate for reasonably compact objects, but becomes less accurate for very elongated objects.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_GENERAL +	Retrieves the general Feret of each edge. This is the Feret diameter calculated at M_FERET_GENERAL_ANGLE, which is set in MedgeControl().	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b

M_FERET_MAX_ANGLE +	Retrieves the angle, in degrees, at which the maximum Feret diameter is found, relative to the output coordinate system specified using MedgeControl() with M_RESULT_OUTPUT_UNITS. An angle interpreted with respect to the pixel coordinate system is always measured counter-clockwise. For information on the angle's direction of rotation when interpreting the angle with respect to the relative coordinate system, see the Angle convention in Aurora Imaging Library subsection of the Working with real-world units section of Chapter 28: Calibrating your camera setup.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_MAX_DIAMETER +	Retrieves the maximum Feret diameter of each edge.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_MEAN_DIAMETER +	Retrieves the average Feret diameter at all the angles checked (see M_NUMBER_OF_FERETS).	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b

M_FERET_MIN_ANGLE +	Retrieves the angle, in degrees, at which the minimum Feret diameter is found, relative to the output coordinate system specified using MedgeControl() with M_RESULT_OUTPUT_UNITS. An angle interpreted with respect to the pixel coordinate system is always measured counter-clockwise. For information on the angle's direction of rotation when interpreting the angle with respect to the relative coordinate system, see the Angle convention in Aurora Imaging Library subsection of the Working with real-world units section of Chapter 28: Calibrating your camera setup.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_MIN_DIAMETER +	Retrieves the minimum Feret diameter of each edge.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_X +	Retrieves the X-Feret value of each edge. This is the dimension of the minimum bounding box of an edge in the horizontal direction.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b

M_FERET_Y +	Retrieves the Y-Feret value of each edge. This is the dimension of the minimum bounding box of an edge in the vertical direction.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_FIRST_POINT +	Retrieves the coordinates of each edge's first point.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]* Returns the first point X-coordinate of each edge.	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]* Returns the first point Y-coordinate of each edge.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIRST_POINT_X +	Retrieves the X-coordinate of each edge's first point.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_FIRST_POINT_Y +	Retrieves the Y-coordinate of each edge's first point.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b

M_LABEL_VALUE +	Retrieves the label value of each edge in an image. The label value is a positive integer greater or equal to one; each edge in an image has a unique label value.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_LENGTH +	Retrieves the length of each edge. M_LENGTH gives a more accurate but slower approximation of the edge's length than M_FAST_LENGTH.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_LINE_FIT_A +	Retrieves the *A* variable of the line that is the best fit for each edge.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_LINE_FIT_B +	Retrieves the *B* variable of the line that is the best fit for each edge.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b

M_LINE_FIT_C +	Retrieves the *C* variable of the line that is the best fit for each edge.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_LINE_FIT_ERROR +	Retrieves the fit error of the line that is the best fit for each edge. This is calculated as the average quadratic error.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_MOMENT_ELONGATION +	Retrieves the moment elongation of each edge.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]* Returns the following.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 1.0	Specifies the moment elongation of each edge. Values closer to 1.0 are considered to have a low elongation, while values closer to 0.0 are considered to have a high elongation. For example, a straight edge has a null elongation value, while a circular edge has an elongation of 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b

M_MOMENT_ELONGATION_ANGLE +	Retrieves the angle, in degrees, of the principal axis along each edge's moment elongation, relative to the output coordinate system specified using MedgeControl() with M_RESULT_OUTPUT_UNITS. An angle interpreted with respect to the pixel coordinate system is always measured counter-clockwise. For information on the angle's direction of rotation when interpreting the angle with respect to the relative coordinate system, see the Angle convention in Aurora Imaging Library subsection of the Working with real-world units section of Chapter 28: Calibrating your camera setup.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION +	Retrieves the X- and Y-position of each edge. The position of the edge is defined by the middle edgel of the edge.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]* Returns the X-position of each edge.	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]* Returns the Y-position of each edge.	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION_X +	Retrieves the X-position of each edge. The position of the edge is defined by the middle edgel of the edge.	c		e	f	g	a		i		j	k	l	m	n	b

	FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION_Y +		Retrieves the Y-position of each edge. The position of the edge is defined by the middle edgel of the edge.	c		e	f	g	a		i		j	k	l	m	n	b
	FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE +		Retrieves the number of edgels in each edge.	c		e	f	g	a		i		j	k	l	m	n	b
	FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLEarray of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_STRENGTH +		Retrieves the strength value of each edge.	c		e	f	g	a		i		j	k	l	m	n	b
	FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_TORTUOSITY +		Retrieves the tortuosity measure of each edge.	c		e	f	g	a		i		j	k	l	m	n	b
	FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b

M_X_MAX_AT_Y_MAX +	Retrieves the maximum X-coordinate at the maximum Y-coordinate of each edge. Together with M_BOX_Y_MAX, this is one of four contact points on the convex perimeter of the edge.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_X_MIN_AT_Y_MIN +	Retrieves the minimum X-coordinate at the minimum Y-coordinate of each edge. Together with M_BOX_Y_MIN, this is one of four contact points on the convex perimeter of the edge.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_Y_MAX_AT_X_MIN +	Retrieves the maximum Y-coordinate at the minimum X-coordinate of each edge. Together with M_BOX_X_MIN, this is one of four contact points on the convex perimeter of the edge.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_Y_MIN_AT_X_MAX +	Retrieves the minimum Y-coordinate at the maximum X-coordinate of each edge. Together with M_BOX_X_MAX, this is one of four contact points on the convex perimeter of the edge.	c		e	f	g	a		i		j	k	l	m	n	b

	FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for M_FERET_GENERAL; M_FERET_MAX_DIAMETER; M_FERET_MIN_DIAMETER.

You can add one of the following values to the above-mentioned values to get the indices of the edgels from which the specified Feret was calculated.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For M_FERET_MAX_DIAMETER, M_FERET_MIN_DIAMETER, or M_FERET_GENERAL																		
ResultType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	FirstResultArrayPtr - Possible values returned																
	SecondResultArrayPtr - Possible values returned																
M_FIRST_FERET_INDEX		Retrieves the index of the first edgel from which the specified Feret was calculated.															
	FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]* Returns the index of the first edgel.	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_FIRST_FERET_INDEX + M_SECOND_FERET_INDEX +		Retrieves the indices of the first and second edgel from which the specified Feret was calculated.	c		e	f	g	a		i		j	k	l	m	n	b

	FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]* Returns the index of the first edgel.	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]* Returns the index of the second edgel.	c		e	f	g	a		i		j	k	l	m	n	b
M_SECOND_FERET_INDEX		Retrieves the index of the second edgel from which the specified Feret was calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* **|** *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]* Returns the index of the second edgel.	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for M_CHAIN; M_CHAIN_MAGNITUDE + M_CHAIN_ANGLE; M_VERTICES; M_CENTER_OF_GRAVITY; M_FERET_BOX; M_FIRST_POINT; M_POSITION; M_FIRST_FERET_INDEX + M_SECOND_FERET_INDEX.

You can add the following value to the above-mentioned values to get the results in a packed format. Alternatively, you can add a value from For retrieving statistics about results.

If you pack values, result types that return values in both FirstResultArrayPtr and SecondResultArrayPtr will be returned together, interlaced in FirstResultArrayPtr. For example, if you decide to pack **MedgeGetResult**() with M_CENTER_OF_GRAVITY, FirstResultArrayPtr will contain both the X- and Y-coordinates of the edge's center of gravity (XY XY XY...).

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For packing values																		
ResultType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	FirstResultArrayPtr - Possible values returned																	
	SecondResultArrayPtr - Possible values returned																	
M_PACKED +		Retrieves the specified values in a packed format. Note that only result types that return values in both FirstResultArrayPtr and SecondResultArrayPtr can be packed.																
	FirstResultArrayPtr	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]*	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for the values listed in For retrieving edge chain results, For retrieving edge feature results; and for the following values: M_POSITION; M_PACKED.

You can add the following value to the above-mentioned values to determine the required array size (number of elements) to store the returned values.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For determining the required array size (number of elements) to store the returned values																		
ResultType combination value		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	FirstResultArrayPtr - Possible values returned																	
	SecondResultArrayPtr - Possible values returned																	
M_NB_ELEMENTS		Retrieves the required array size (number of elements) to store the returned values.																
	FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for the values listed in For retrieving edge feature results.

You can add one of the following values to the above-mentioned values to get statistical result information. Alternatively, you can add a value from For packing values.

Note that results are only returned to FirstResultArrayPtr; therefore, SecondResultArrayPtr must be set to **M_NULL**.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving statistics about results

ResultType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
FirstResultArrayPtr - Possible values returned																
SecondResultArrayPtr - Possible values returned																
M_MAX	Retrieves the maximum value of the requested result type.															
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_MAX_ABS	Retrieves the maximum absolute value of the requested result type.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_MEAN	Retrieves the mean value of the requested result type.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN	Retrieves the minimum value of the requested result type.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b
M_MIN_ABS	Retrieves the minimum absolute value of the requested result type.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b

M_STANDARD_DEVIATION		Retrieves the standard deviation of the requested result type.	c		e	f	g	a		i		j	k	l	m	n	b
	FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for the values listed in For retrieving edge feature results.

You can add the following value to the above-mentioned values to determine if the edge feature result is available in the result buffer.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For determining whether edge feature results are available																	
ResultType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	FirstResultArrayPtr - Possible values returned																
	SecondResultArrayPtr - Possible values returned																
M_AVAILABLE		Retrieves whether the requested result type is available for retrieval.															
	FirstResultArrayPtr	Data type info: Data type: *address of an AIL_DOUBLE* Returns one of the following.	c		e	f	g	a		i		j	k	l	m	n	b
		M_FALSE	c		e	f	g	a		i		j	k	l	m	n	b
		M_TRUE	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *Set this parameter to M_NULL*	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for
Tables:

- For retrieving results related to edge extraction and edge processing
- For retrieving information about the internal processing buffers
- For retrieving edge chain results
- For retrieving edge feature results

the values listed in all the parameter association tables.

You can add one of the following values to the above-mentioned values to cast the requested results to a

required data type.

> Note: In C++, when using a standard vector (std::vector) instead of an array, the values listed in this combination table must not be used. If you need to use a specific data type, instantiate the vector with the necessary data type.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the data type																	
ResultType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
FirstResultArrayPtr - Possible values returned																	
SecondResultArrayPtr - Possible values returned																	
M_TYPE_AIL_DOUBLE	Casts the requested results to an *AIL_DOUBLE*.																
FirstResultArrayPtr	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]* **|** *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *array of type AIL_DOUBLE [optionally, in C++: a reference to a std::vector<AIL_DOUBLE>]* **|** *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_ID	Casts the requested results to an *AIL_ID*.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *array of type AIL_ID [optionally, in C++: a reference to a std::vector<AIL_ID>]* **|** *address of an AIL_ID*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *array of type AIL_ID [optionally, in C++: a reference to a std::vector<AIL_ID>]* **|** *address of an AIL_ID*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT	Casts the requested results to an *AIL_INT*.	c		e	f	g	a		i		j	k	l	m	n	b
FirstResultArrayPtr	Data type info: Data type: *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]* **|** *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
SecondResultArrayPtr	Data type info: Data type: *array of type AIL_INT [optionally, in C++: a reference to a std::vector<AIL_INT>]* **|** *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT32	Casts the requested results to an *AIL_INT32*.	c		e	f	g	a		i		j	k	l	m	n	b

	FirstResultArrayPtr	Data type info: Data type: *array of type AIL_INT32 [optionally, in C++: a reference to a std::vector<AIL_INT32>]* **|** *address of an AIL_INT32*	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *array of type AIL_INT32 [optionally, in C++: a reference to a std::vector<AIL_INT32>]* **|** *address of an AIL_INT32*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT64		Casts the requested results to an *AIL_INT64*.	c		e	f	g	a		i		j	k	l	m	n	b
	FirstResultArrayPtr	Data type info: Data type: *array of type AIL_INT64 [optionally, in C++: a reference to a std::vector<AIL_INT64>]* **|** *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b
	SecondResultArrayPtr	Data type info: Data type: *array of type AIL_INT64 [optionally, in C++: a reference to a std::vector<AIL_INT64>]* **|** *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailedge11.lib.
DLL	Requires ail11.dll; ailedge11.dll.

MedgeInquire

Synopsis

Inquire about an Edge Finder context or an Edge Finder result buffer.

Syntax

```
AIL_INT MedgeInquire(
    AIL_ID ContextOrResultId, //in
    AIL_INT64 InquireType,    //in
    void *UserVarPtr          //out
)
```

Description

This function inquires information about the specified Edge Finder context or Edge Finder result buffer.

Note that for an Edge Finder result buffer, this function only retrieves information about result buffer settings (set with MedgeControl(), for example). To retrieve results from the Edge Finder result buffer, use MedgeGetResult().

If the inquired setting is set to M_DEFAULT (for example, in MedgeControl()), **MedgeInquire**() will return M_DEFAULT. To inquire the actual default value, add M_DEFAULT to the InquireType parameter.

Parameters

ContextOrResultId

Specifies either the Edge Finder context or the Edge Finder result buffer about which to inquire information. Both the Edge Finder context and the Edge Finder result buffer must have been previously allocated on the system using MedgeAlloc() or MedgeAllocResult(), respectively.

InquireType

Specifies the type of setting about which to inquire.

See the Parameter associations section for possible values that can be specified.

UserVarPtr
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- AIL_DOUBLE
- AIL_ID
- AIL_INT
- AIL_INT32
- AIL_INT64

Specifies the address in which to write the requested information. Since the **MedgeInquire**() function also returns the requested information, you can set this parameter to **M_NULL**.

See the Parameter associations section for possible values that can be returned.

Parameter associations

The tables below list possible values for the **InquireType** parameter and possible values returned to the **UserVarPtr** parameter.

- For operation settings (both M_CONTOUR and M_CREST)
- For operation settings (M_CREST only)
- For internal buffers (both M_CONTOUR and M_CREST)
- For edge features (both M_CONTOUR and M_CREST)
- For sorting keys (both M_CONTOUR and M_CREST)
- For general settings (both M_CONTOUR and M_CREST)
- For Feret settings (both M_CONTOUR and M_CREST)
- For post-calculations (both M_CONTOUR and M_CREST)
- For result buffer settings (general)
- For result buffer settings (closest-edgel operations)
- For inquiring about the system

To inquire about the operation settings for both M_CONTOUR and M_CREST Edge Finder contexts, set the InquireType parameter to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For operation settings (both M_CONTOUR and M_CREST)																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_ACCURACY +		Inquires the edgel accuracy of the edge extraction. SET															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_HIGH.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that edgel accuracy will be disabled. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_HIGH	Specifies high accuracy. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_VERY_HIGH	Specifies very high accuracy. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_ANGLE_ACCURACY +		Inquires the edgel angle accuracy of the edge extraction. M_ANGLE_ACCURACY can only be used with M_CONTOUR Edge Finder contexts. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Same as M_HIGH.	c		e	f	g	a		i		j	k	l	m	n	b
	M_HIGH	Specifies high precision. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_LOW	Specifies low precision. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_CHAIN_ALL_NEIGHBORS +		Inquires whether edge chains are built using all the available neighboring edgels. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that edge chains are built with the least amount of edgel information possible. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that edge chains are built with as much edgel information as possible. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_EXTRACTION_SCALE +		Inquires the image scale with which to extract the edges. Returns the scale of the image at which to do the edge extraction. Once the extraction is complete, the results are scaled to the original scale of the image. A lower extraction scale speeds up the search but can result in a less reliable result, including, the loss of important details and/or a reduction in the accuracy of the search results. M_EXTRACTION_SCALE is for advanced users of the Edge Finder module. The default setting usually provides the most accurate search results. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.0.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0	Specifies the extraction scale.	c		e	f	g	a		i		j	k	l	m	n	b

M_FILTER_SMOOTHNESS +		Inquires the degree of smoothness (strength of denoising) of the edge extraction filter. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 50.0.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the smoothness value. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_FILTER_TYPE +		Inquires the type of filter used when performing the edge extraction. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_SHEN.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DERICHE	Specifies a Deriche infinite support filter. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FREI_CHEN	Specifies a Frei Chen filter. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_PREWITT	Specifies a Prewitt filter. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SHEN	Specifies a Shen-Castan infinite support exponential filter. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SOBEL	Specifies a Sobel filter. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_FLOAT_MODE +		Inquires whether the entire edge extraction process is forced to be performed using floating-point precision calculations. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that all edge extractions are not forced to be performed using floating-point precision calculations.	c		e	f	g	a		i		j	k	l	m	n	b

M_ENABLE	Specifies that all edge extractions are forced to be performed using floating-point precision calculations.	c		e	f	g	a		i		j	k	l	m	n	b
M_KERNEL_SIZE +	Inquires the actual X and Y size (same in X and Y) of the convolution kernel used for kernel filtering mode. Note that the kernel size can change according to the filter type, the smoothness factor, the specified maximum kernel width, the specified kernel depth, and whether you are using floating-point calculations.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the actual size, in X and Y, of the convolution kernel.	c		e	f	g	a		i		j	k	l	m	n	b
M_MAGNITUDE_TYPE +	Inquires the type of magnitude value used to calculate the magnitude of the edge at each edgel position. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_NORM	Specifies that the magnitude will be used.	c		e	f	g	a		i		j	k	l	m	n	b
M_SQR_NORM	Specifies that the square of the magnitude will be used. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_MODIFICATION_COUNT +	Inquires the current value of the modification counter. The modification counter is increased by one each time settings for the context are modified. Although you cannot identify the modification counter's contents, you can compare them throughout your application to know if the context has been altered. If the modification counter has changed you can, for example, prompt the user to save before closing the application.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

Value	Specifies the current value of the modification counter.	c		e	f	g	a		i		j	k	l	m	n	b
M_OVERSCAN +	Inquires the type of overscan used by the convolution filters when extracting edges for kernel filtering mode. Note that M_OVERSCAN is ignored if using an IIR filter (e.g. Shen and Deriche filter types). SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_MIRROR.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that no overscan will be used, unless processing the border pixels is faster than ignoring them; in the latter case, Aurora Imaging Library automatically selects the overscan to optimize speed according to the specified operation and the target system.	c		e	f	g	a		i		j	k	l	m	n	b
M_MIRROR	Specifies that the border pixels of a source image are processed using overscan pixel values that mirror the source buffer pixel values. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_REPLACE	Specifies that the border pixels of a source image are processed using overscan pixel values set to the overscan replacement value (MedgeControl() with M_OVERSCAN_REPLACE_VALUE).	c		e	f	g	a		i		j	k	l	m	n	b
M_TRANSPARENT	Specifies that the border pixels of a source image are processed using transparent overscan pixel values. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_OVERSCAN_REPLACE_VALUE +	Inquires the replacement value for the overscan pixel values when using replacement type overscan. Note that M_OVERSCAN_REPLACE_VALUE is ignored unless M_OVERSCAN with MedgeControl() is set to M_REPLACE. In addition, M_OVERSCAN_REPLACE_VALUE is ignored if using an IIR filter (e.g. Shen and Deriche filter types). SET	c		e	f	g	a		i		j	k	l	m	n	b

	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.	c		e	f	g	a		i		j	k	l	m	n	b
	M_REPLACE_MAX	Specifies that the overscan neighborhood pixel values will be set to the maximum value of the source image buffer.	c		e	f	g	a		i		j	k	l	m	n	b
	M_REPLACE_MIN	Specifies that the overscan neighborhood pixel values will be set to the minimum value of the source image buffer.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the value of the overscan neighborhood pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_THRESHOLD_HIGH +		Inquires the user-defined upper bound of the hysteresis threshold. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the upper bound of the hysteresis threshold.	c		e	f	g	a		i		j	k	l	m	n	b
M_THRESHOLD_LOW +		Inquires the user-defined lower bound of the hysteresis threshold. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 0.0	Specifies the lower bound of the hysteresis threshold.	c		e	f	g	a		i		j	k	l	m	n	b
M_THRESHOLD_MODE +		Inquires the threshold mode of the edge extraction. Note that lower threshold values result in a more sensitive edgel detection. SET	c		e	f	g	a		i		j	k	l	m	n	b

	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_HIGH.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies no threshold. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_HIGH	Specifies a high threshold. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_LOW	Specifies a low threshold. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_MEDIUM	Specifies a medium threshold. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_USER_DEFINED	Specifies that the threshold values will be user-defined. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_VERY_HIGH	Specifies a very high threshold. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_THRESHOLD_TYPE +		Inquires the type of the hysteresis threshold used when performing the edge extraction. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_HYSTERESIS.	c		e	f	g	a		i		j	k	l	m	n	b
	M_FULL_HYSTERESIS	Specifies that the lower bound threshold value is 0.	c		e	f	g	a		i		j	k	l	m	n	b
	M_HYSTERESIS	Specifies that both the lower bound threshold value and the upper bound threshold value will be used.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NO_HYSTERESIS	Specifies that the lower bound threshold value is equal to the upper bound threshold value.	c		e	f	g	a		i		j	k	l	m	n	b
M_TIMEOUT +		Inquires the maximum edge extraction and calculation time for MedgeCalculate(), in msec. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies an infinite amount of edge extraction and calculation time.	c		e	f	g	a		i		j	k	l	m	n	b
	Value > 0	Specifies the maximum edge extraction and calculation time, in msec.	c		e	f	g	a		i		j	k	l	m	n	b

To inquire about the operation settings for M_CREST Edge Finder contexts, set the InquireType parameter to the value below.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For operation settings (M_CREST only)																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_FOREGROUND_VALUE +		Inquires the color used to extract line crests from the image. SET															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_FOREGROUND_BLACK.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ANY	Specifies that the line crests are both lighter and darker than the image's background color. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FOREGROUND_BLACK	Specifies that the line crests are darker than the image's background color. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FOREGROUND_WHITE	Specifies that the line crests are lighter than the image's background color. INFO	c		e	f	g	a		i		j	k	l	m	n	b

To inquire whether internal buffers have been saved in the Edge Finder result buffer, set the InquireType parameter to one of the values below. These values can be specified for both M_CONTOUR and M_CREST Edge Finder contexts.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For internal buffers (both M_CONTOUR and M_CREST)																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																

M_SAVE_ANGLE +		Inquires whether the internal angle buffer used to extract edges is saved in the Edge Finder result buffer. SET														
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the angle buffer will not be saved.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the angle buffer will be saved.	c		e	f	g	a		i		j	k	l	m	n	b
M_SAVE_CHAIN_ANGLE +		Inquires whether the angle value at each edgel position is saved in the Edge Finder result buffer. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the angle values will not be saved.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the angle values will be saved.	c		e	f	g	a		i		j	k	l	m	n	b
M_SAVE_CHAIN_MAGNITUDE +		Inquires whether the magnitude value at each edgel position is saved in the Edge Finder result buffer. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the magnitude values will not be saved.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the magnitude values will be saved.	c		e	f	g	a		i		j	k	l	m	n	b

M_SAVE_DERIVATIVES +	Inquires whether the internal derivative buffers used to extract edges are saved in the Edge Finder result buffer. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that the derivative buffers will not be saved.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies that the derivative buffers will be saved.	c		e	f	g	a		i		j	k	l	m	n	b
M_SAVE_IMAGE +	Inquires whether the source image is saved in the Edge Finder result buffer. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that the image will not be saved.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies that the image will be saved.	c		e	f	g	a		i		j	k	l	m	n	b
M_SAVE_MAGNITUDE +	Inquires whether the internal magnitude buffer used to extract edges is saved in the Edge Finder result buffer. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that the magnitude buffer will not be saved.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies that the magnitude buffer will be saved.	c		e	f	g	a		i		j	k	l	m	n	b
M_SAVE_MASK +	Inquires whether the mask buffer is saved in the Edge Finder result buffer. SET	c		e	f	g	a		i		j	k	l	m	n	b

	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the mask will not be saved.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the mask will be saved.	c		e	f	g	a		i		j	k	l	m	n	b

To inquire about which edge features will be calculated for each edge, set the InquireType parameter to one of the values below. These values can be specified for both M_CONTOUR and M_CREST Edge Finder contexts. Note that, when an edge feature has been calculated, M_ENABLE is returned.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For edge features (both M_CONTOUR and M_CREST)																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_AVERAGE_STRENGTH +		Inquires whether the average strength value of each edge will be calculated. SET															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the average strength will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the average strength will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_X_MAX +		Inquires whether the extreme right edgel coordinate of each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies that the extreme right edgel coordinate will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the extreme right edgel coordinate will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_X_MIN +		Inquires whether the extreme left edgel coordinate of each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the extreme left edgel coordinate will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the extreme left edgel coordinate will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_Y_MAX +		Inquires whether the extreme bottom edgel coordinate of each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the extreme bottom edgel coordinate will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the extreme bottom edgel coordinate will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_BOX_Y_MIN +		Inquires whether the extreme top edgel coordinate of each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the extreme top edgel coordinate will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the extreme top edgel coordinate will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_OF_GRAVITY_X +		Inquires whether the X-position of each edge's center of gravity will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the X-position of the center of gravity will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the X-position of the center of gravity will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_CENTER_OF_GRAVITY_Y +		Inquires whether the Y-position of each edge's center of gravity will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the Y-position of the center of gravity will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the Y-position of the center of gravity will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_CIRCLE_FIT_CENTER_X +		Inquires whether the X-coordinate of the center of the circle that is the best fit for each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b

	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the X-coordinate will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the X-coordinate will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_CIRCLE_FIT_CENTER_Y +		Inquires whether the Y-coordinate of the center of the circle that is the best fit for each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the Y-coordinate will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the Y-coordinate will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_CIRCLE_FIT_COVERAGE +		Inquires whether the coverage of the circle that is the best fit for each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the coverage of the circle will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the coverage of the circle will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_CIRCLE_FIT_ERROR +		Inquires whether the fit error of the circle that is the best fit for each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the fit error of the circle will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the fit error of the circle will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_CIRCLE_FIT_RADIUS +		Inquires whether the radius of the circle that is the best fit for each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the radius will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the radius will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_CLOSURE +		Inquires whether the closure of each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the edge's closure state will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the edge's closure state will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_CONVEX_PERIMETER +		Inquires whether the convex elongation of each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies that the convex elongation will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the convex elongation will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_ELLIPSE_FIT_ANGLE +		Inquires whether the angle of the ellipse that is the best fit for each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the angle will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the angle will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_ELLIPSE_FIT_CENTER_X +		Inquires whether the X-coordinate of the center of the ellipse that is the best fit for each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the X-coordinate will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the X-coordinate will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_ELLIPSE_FIT_CENTER_Y +		Inquires whether the Y-coordinate of the center of the ellipse that is the best fit for each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b

M_DISABLE	Specifies that the Y-coordinate will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies that the Y-coordinate will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_ELLIPSE_FIT_COVERAGE +	Inquires whether the coverage of the ellipse that is the best fit for each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that the coverage of the ellipse will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies that the coverage of the ellipse will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_ELLIPSE_FIT_ERROR +	Inquires whether the fit error of the ellipse that is the best fit for each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that the fit error of the ellipse will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies that the fit error of the ellipse will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_ELLIPSE_FIT_MAJOR_AXIS +	Inquires whether the major axis of the ellipse that is the best fit for each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that the major axis will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b

	M_ENABLE	Specifies that the major axis will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_ELLIPSE_FIT_MINOR_AXIS +		Inquires whether the minor axis the ellipse that is the best fit for each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the minor axis will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the minor axis will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_FAST_LENGTH +		Inquires whether the length of each edge will be quickly calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that a fast length will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that a fast length will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_ELONGATION +		Inquires whether the Feret elongation of each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the Feret elongation will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the Feret elongation will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b

M_FERET_GENERAL +		Inquires whether the general Feret of each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the general Feret will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the general Feret will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_MAX_ANGLE +		Inquires whether the maximum Feret angle of each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the maximum Feret angle will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the maximum Feret angle will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_MAX_DIAMETER +		Inquires whether the maximum Feret diameter of each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the maximum Feret diameter will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the maximum Feret diameter will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b

M_FERET_MEAN_DIAMETER +		Inquires whether the average Feret diameter of each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the average Feret diameter will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the average Feret diameter will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_MIN_ANGLE +		Inquires whether the minimum Feret angle of each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the minimum Feret angle will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the minimum Feret angle will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_MIN_DIAMETER +		Inquires whether the minimum Feret diameter of each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the minimum Feret diameter will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the minimum Feret diameter will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b

M_FERET_X +		Inquires whether the X-Feret value of each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the X-Feret value will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the X-Feret value will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_Y +		Inquires whether the Y-Feret value of each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the Y-Feret value will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the Y-Feret value will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_FIRST_POINT_X +		Inquires whether the X-coordinate of each edge's first point (starting point) will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the first point's X-coordinate will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the first point's X-coordinate will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b

M_FIRST_POINT_Y +	Inquires whether the Y-coordinate of each edge's first point (starting point) will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that the first point's Y-coordinate will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies that the first point's Y-coordinate will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_LABEL_VALUE +	Inquires whether the label value of each edge in an image will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_ENABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that the label value will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies that the label value will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_LENGTH +	Inquires whether the length of each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that the length of each edge will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies that the length of each edge will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_LINE_FIT_A +	Inquires whether the coefficient *A* of the line that is the best fit for each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b

	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the coefficient *A* will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the coefficient *A* will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_LINE_FIT_B +		Inquires whether the coefficient *B* of the line that is the best fit for each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the coefficient *B* will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the coefficient *B* will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_LINE_FIT_C +		Inquires whether the coefficient *C* of the line that is the best fit for each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the coefficient *C* will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the coefficient *C* will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_LINE_FIT_ERROR +		Inquires whether the fit error of the line that is the best fit for each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the fit error will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the fit error will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOMENT_ELONGATION +		Inquires whether the moment elongation of each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the moment elongation will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the moment elongation will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_MOMENT_ELONGATION_ANGLE +		Inquires whether the angle of the principal axis along each edge's moment elongation will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the angle of the moment elongation will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the angle of the moment elongation will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION_X +		Inquires whether the X-position of each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the X-position will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the X-position will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_POSITION_Y +		Inquires whether the Y-position of each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the Y-position will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the Y-position will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_SIZE +		Inquires whether the number of edgels in each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the number of edgels will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the number of edgels will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_STRENGTH +		Inquires whether the strength value of each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b

	M_DISABLE	Specifies that the strength will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the strength will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_TORTUOSITY +		Inquires whether the tortuosity measure of each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the tortuosity measure will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the tortuosity measure will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_X_MAX_AT_Y_MAX +		Inquires whether the X-maximum at Y-maximum contact point of each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the X-maximum at Y-maximum will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the X-maximum at Y-maximum will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_X_MIN_AT_Y_MIN +		Inquires whether the X-minimum at Y-minimum contact point of each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the X-minimum at Y-minimum will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b

M_ENABLE	Specifies that the X-minimum at Y-minimum will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_Y_MAX_AT_X_MIN +	Inquires whether the Y-maximum at X-minimum contact point of each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that the Y-maximum at X-minimum will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies that the Y-maximum at X-minimum will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_Y_MIN_AT_X_MAX +	Inquires whether the Y-minimum at X-maximum contact point of each edge will be calculated. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
M_DISABLE	Specifies that the Y-minimum at X-maximum will not be calculated.	c		e	f	g	a		i		j	k	l	m	n	b
M_ENABLE	Specifies that the Y-minimum at X-maximum will be calculated.	c		e	f	g	a		i		j	k	l	m	n	b

To inquire about the corresponding edge feature associated with the specified sorting key, set the InquireType parameter to one of the values below. These values can be specified for both M_CONTOUR and M_CREST Edge Finder contexts.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For sorting keys (both M_CONTOUR and M_CREST)																
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																

M_SORT*n*_DOWN +		Inquires the feature associated with the *n* th sorting key (in descending order), where *n* stands for an integer between 1 and 3.														
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_NULL	Specifies that no feature has been associated with the specified sorting key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_AVERAGE_STRENGTH	Sorts based on the average strength of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_BOX_X_MAX	Sorts based on the extreme right edgel coordinate of each edge.	c		e	f	g	a		i		j	k	l	m	n	b
	M_BOX_X_MIN	Sorts based on the extreme left edgel coordinate of each edge.	c		e	f	g	a		i		j	k	l	m	n	b
	M_BOX_Y_MAX	Sorts based on the extreme bottom edgel coordinate of each edge.	c		e	f	g	a		i		j	k	l	m	n	b
	M_BOX_Y_MIN	Sorts based on the extreme top edgel coordinate of each edge.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CENTER_OF_GRAVITY_X	Sorts based on the X-position of each edge's center of gravity. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_CENTER_OF_GRAVITY_Y	Sorts based on the Y-position of each edge's center of gravity. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_CIRCLE_FIT_CENTER_X	Sorts based on the X-coordinate of the center of the circle that is the best fit for each edge.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CIRCLE_FIT_CENTER_Y	Sorts based on the Y-coordinate of the center of the circle that is the best fit for each edge.	c		e	f	g	a		i		j	k	l	m	n	b

	M_CIRCLE_FIT_COVERAGE	Sorts based on the coverage of the circle that is the best fit for each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_CIRCLE_FIT_ERROR	Sorts based on the fit error of the circle that is the best fit for each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_CIRCLE_FIT_RADIUS	Sorts based on the radius of the circle that is the best fit for each edge.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CLOSURE	Sorts based on the closure state of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_CONVEX_PERIMETER	Sorts based on the convex elongation of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_ELLIPSE_FIT_ANGLE	Sorts based on the angle of the ellipse that is the best fit for each edge.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ELLIPSE_FIT_CENTER_X	Sorts based on the X-coordinate of the center of the ellipse that is the best fit for each edge.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ELLIPSE_FIT_CENTER_Y	Sorts based on the Y-coordinate of the center of the ellipse that is the best fit for each edge.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ELLIPSE_FIT_COVERAGE	Sorts based on the coverage of the ellipse that is the best fit for each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_ELLIPSE_FIT_ERROR	Sorts based on the fit error of the ellipse that is the best fit for each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_ELLIPSE_FIT_MAJOR_AXIS	Sorts based on the major axis of the ellipse that is the best fit for each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_ELLIPSE_FIT_MINOR_AXIS	Sorts based on the minor axis of the ellipse that is the best fit for each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_FAST_LENGTH	Sorts based on a coarse approximation for the length of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FERET_ELONGATION	Sorts based on the Feret elongation of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FERET_GENERAL	Sorts based on the general Feret of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FERET_MAX_ANGLE	Sorts based on the maximum Feret angle of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FERET_MAX_DIAMETER	Sorts based on the maximum Feret diameter of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FERET_MEAN_DIAMETER	Sorts based on the average Feret diameter at all the angles checked. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FERET_MIN_ANGLE	Sorts based on the minimum Feret angle of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FERET_MIN_DIAMETER	Sorts based on the minimum Feret diameter of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FERET_X	Sorts based on the X-Feret value of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FERET_Y	Sorts based on the Y-Feret value of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIRST_POINT_X	Sorts based on the X-coordinate of each edge's first point (starting point). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIRST_POINT_Y	Sorts based on the Y-coordinate of each edge's first point (starting point). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_LABEL_VALUE	Sorts based on the label value of each edge in an image. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_LENGTH	Sorts based on the length of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_LINE_FIT_A	Sorts based on the coefficient *A* of the line that is the best fit for each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_LINE_FIT_B	Sorts based on the coefficient *B* of the line that is the best fit for each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_LINE_FIT_C	Sorts based on the coefficient *C* of the line that is the best fit for each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_LINE_FIT_ERROR	Sorts based on the fit error of the line that is the best fit for each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_MOMENT_ELONGATION	Sorts based on the moment elongation of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_MOMENT_ELONGATION_ANGLE	Sorts based on the angle of the principal axis along each edge's moment elongation (M_MOMENT_ELONGATION).	c		e	f	g	a		i		j	k	l	m	n	b
	M_POSITION_X	Sorts based on the X-position of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_POSITION_Y	Sorts based on the Y-position of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SIZE	Sorts based on the number of edgels of each edge.	c		e	f	g	a		i		j	k	l	m	n	b
	M_STRENGTH	Sorts based on the strength of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_TORTUOSITY	Sorts based on the tortuosity measure of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_X_MAX_AT_Y_MAX	Sorts based on the maximum X-coordinate at the maximum Y-coordinate of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_X_MIN_AT_Y_MIN	Sorts based on the minimum X-coordinate at the minimum Y-coordinate of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_Y_MAX_AT_X_MIN	Sorts based on the maximum Y-coordinate at the minimum X-coordinate of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_Y_MIN_AT_X_MAX	Sorts based on the minimum Y-coordinate at the maximum X-coordinate of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_SORT*n*_UP +		Inquires the feature associated with the *n* th sorting key (in ascending order), where *n* stands for an integer between 1 and 3.	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_NULL	Specifies that no feature has been associated with the specified sorting key.	c		e	f	g	a		i		j	k	l	m	n	b
	M_AVERAGE_STRENGTH	Sorts based on the average strength of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_BOX_X_MAX	Sorts based on the extreme right edgel coordinate of each edge.	c		e	f	g	a		i		j	k	l	m	n	b
	M_BOX_X_MIN	Sorts based on the extreme left edgel coordinate of each edge.	c		e	f	g	a		i		j	k	l	m	n	b
	M_BOX_Y_MAX	Sorts based on the extreme bottom edgel coordinate of each edge.	c		e	f	g	a		i		j	k	l	m	n	b
	M_BOX_Y_MIN	Sorts based on the extreme top edgel coordinate of each edge.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CENTER_OF_GRAVITY_X	Sorts based on the X-position of each edge's center of gravity. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_CENTER_OF_GRAVITY_Y	Sorts based on the Y-position of each edge's center of gravity. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_CIRCLE_FIT_CENTER_X	Sorts based on the X-coordinate of the center of the circle that is the best fit for each edge.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CIRCLE_FIT_CENTER_Y	Sorts based on the Y-coordinate of the center of the circle that is the best fit for each edge.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CIRCLE_FIT_COVERAGE	Sorts based on the coverage of the circle that is the best fit for each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_CIRCLE_FIT_ERROR	Sorts based on the fit error of the circle that is the best fit for each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_CIRCLE_FIT_RADIUS	Sorts based on the radius of the circle that is the best fit for each edge.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CLOSURE	Sorts based on the closure state of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_CONVEX_PERIMETER	Sorts based on the convex elongation of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_ELLIPSE_FIT_ANGLE	Sorts based on the angle of the ellipse that is the best fit for each edge.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ELLIPSE_FIT_CENTER_X	Sorts based on the X-coordinate of the center of the ellipse that is the best fit for each edge.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ELLIPSE_FIT_CENTER_Y	Sorts based on the Y-coordinate of the center of the ellipse that is the best fit for each edge.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ELLIPSE_FIT_COVERAGE	Sorts based on the coverage of the ellipse that is the best fit for each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_ELLIPSE_FIT_ERROR	Sorts based on the fit error of the ellipse that is the best fit for each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_ELLIPSE_FIT_MAJOR_AXIS	Sorts based on the major axis of the ellipse that is the best fit for each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_ELLIPSE_FIT_MINOR_AXIS	Sorts based on the minor axis of the ellipse that is the best fit for each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FAST_LENGTH	Sorts based on a coarse approximation for the length of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FERET_ELONGATION	Sorts based on the Feret elongation of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FERET_GENERAL	Sorts based on the general Feret of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FERET_MAX_ANGLE	Sorts based on the maximum Feret angle of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FERET_MAX_DIAMETER	Sorts based on the maximum Feret diameter of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FERET_MEAN_DIAMETER	Sorts based on the average Feret diameter at all the angles checked. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FERET_MIN_ANGLE	Sorts based on the minimum Feret angle of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FERET_MIN_DIAMETER	Sorts based on the minimum Feret diameter of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FERET_X	Sorts based on the X-Feret value of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FERET_Y	Sorts based on the Y-Feret value of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b

	M_FIRST_POINT_X	Sorts based on the X-coordinate of each edge's first point (starting point). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_FIRST_POINT_Y	Sorts based on the Y-coordinate of each edge's first point (starting point). INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_LABEL_VALUE	Sorts based on the label value of each edge in an image. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_LENGTH	Sorts based on the length of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_LINE_FIT_A	Sorts based on the coefficient *A* of the line that is the best fit for each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_LINE_FIT_B	Sorts based on the coefficient *B* of the line that is the best fit for each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_LINE_FIT_C	Sorts based on the coefficient *C* of the line that is the best fit for each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_LINE_FIT_ERROR	Sorts based on the fit error of the line that is the best fit for each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_MOMENT_ELONGATION	Sorts based on the moment elongation of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_MOMENT_ELONGATION_ANGLE	Sorts based on the angle of the principal axis along each edge's moment elongation (M_MOMENT_ELONGATION).	c		e	f	g	a		i		j	k	l	m	n	b
	M_POSITION_X	Sorts based on the X-position of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_POSITION_Y	Sorts based on the Y-position of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SIZE	Sorts based on the number of edgels of each edge.	c		e	f	g	a		i		j	k	l	m	n	b

	M_STRENGTH	Sorts based on the strength of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_TORTUOSITY	Sorts based on the tortuosity measure of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_X_MAX_AT_Y_MAX	Sorts based on the maximum X-coordinate at the maximum Y-coordinate of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_X_MIN_AT_Y_MIN	Sorts based on the minimum X-coordinate at the minimum Y-coordinate of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_Y_MAX_AT_X_MIN	Sorts based on the maximum Y-coordinate at the minimum X-coordinate of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_Y_MIN_AT_X_MAX	Sorts based on the minimum Y-coordinate at the maximum X-coordinate of each edge. INFO	c		e	f	g	a		i		j	k	l	m	n	b

To inquire about general Edge Finder context settings, set the InquireType parameter to one of the values below. These values can be specified for both M_CONTOUR and M_CREST Edge Finder contexts.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For general settings (both M_CONTOUR and M_CREST)																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_CONTEXT_TYPE +		Inquires the type of the Edge Finder context. SET															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_CONTOUR	Specifies a contour context type, which is used to find object contours in images.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CREST	Specifies a crest context type, which is used to find thin line crests in images.	c		e	f	g	a		i		j	k	l	m	n	b

M_FILTER_POWER +	Inquires the power of the filter used to extract edges. This indicates the decreasing factor of the noise variance. In the following formula, *h* represents the filter values. The sum is computed over the filter support region. This region can be finite or infinite; it includes the neighboring image elements that are taken into account when computing the output image element. $$Power = \frac{\sum h^2}{\left(\sum \lvert h \rvert\right)^2}$$	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the power of the filter used to extract edges.	c		e	f	g	a		i		j	k	l	m	n	b
M_MASK_SIZE_X +	Inquires the X-size of the mask.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_NULL	Specifies that no mask has been set using MedgeMask().	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the X-size of the mask, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_MASK_SIZE_Y +	Inquires the Y-size of the mask.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_NULL	Specifies that no mask has been set using MedgeMask().	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the Y-size of the mask, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b

To inquire about the values used to calculate Feret settings, set the InquireType parameter to one of the values below. These values can be specified for both M_CONTOUR and M_CREST Edge Finder contexts.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For Feret settings (both M_CONTOUR and M_CREST)																
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																

M_FERET_ANGLE_SEARCH_END +	Inquires the end of the angular range at which to search for Feret diameters. The angular range is used to calculate the following Feret features: M_FERET_MAX_DIAMETER, M_FERET_MIN_DIAMETER, M_FERET_MAX_ANGLE, and M_FERET_MIN_ANGLE. SET															
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 360.0 °.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 360.0	Specifies the end of the angular region.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_ANGLE_SEARCH_START +	Inquires the start of the angular range at which to search for Feret diameters. The angular range is used to calculate the following Feret features: M_FERET_MAX_DIAMETER, M_FERET_MIN_DIAMETER, M_FERET_MAX_ANGLE, and M_FERET_MIN_ANGLE. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 0.0 °.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 360.0	Specifies the start of the angular region.	c		e	f	g	a		i		j	k	l	m	n	b
M_FERET_GENERAL_ANGLE +	Inquires the angle at which to calculate M_FERET_GENERAL. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 0.0 °.	c		e	f	g	a		i		j	k	l	m	n	b

	M_MOMENT_ELONGATION_ANGLE + *n*	Specifies to use the moment elongation angle plus the offset *n*. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 360.0	Specifies the angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_NUMBER_OF_FERETS +		Inquires the number of Ferets used to calculate M_FERET_MAX_DIAMETER, M_FERET_MIN_DIAMETER, M_FERET_MAX_ANGLE, M_FERET_MIN_ANGLE, M_FERET_MEAN_DIAMETER, M_FERET_ELONGATION, and M_CONVEX_PERIMETER. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 8.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the number of Ferets.	c		e	f	g	a		i		j	k	l	m	n	b

To inquire about the settings used when performing post-calculations on extracted edges, set the InquireType parameter to one of the values below. These values can be specified for both M_CONTOUR and M_CREST Edge Finder contexts.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For post-calculations (both M_CONTOUR and M_CREST)																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_APPROXIMATION_TOLERANCE +		Inquires the resolution of the edge approximation. SET															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 50.0.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the resolution. INFO	c		e	f	g	a		i		j	k	l	m	n	b

M_CHAIN_APPROXIMATION +		Inquires the simple geometric features used to approximate edges. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the edge approximation will not be performed. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_LINE	Specifies that the edge approximation will be performed using a polygonal segmentation of each edge, in pixel units. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_WORLD_LINE	Specifies that the edge approximation will be performed using a polygonal segmentation of each edge, in world units. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_FILL_GAP_ANGLE +		Inquires the aperture angle where Edge Finder searches for edge extremity candidates when filling edge gaps. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 360.0 °.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 360.0	Specifies the aperture angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_FILL_GAP_CANDIDATE +		Inquires whether an edge extremity is filled with the other extremity of the same edge, or with an extremity of any edge. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_ANY.	c		e	f	g	a		i		j	k	l	m	n	b

	M_ANY	Specifies that the extremity of an edge can be connected with the extremity of any edge.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SAME	Specifies that the extremity of an edge can only be connected with the other extremity of the same edge.	c		e	f	g	a		i		j	k	l	m	n	b
M_FILL_GAP_CONTINUITY +		Inquires the continuity constraint used when performing edge gap filling, when more than one edge extremity candidate is present. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 50.0.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 100.0	Specifies the continuity constraint. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_FILL_GAP_DISTANCE +		Inquires the maximum distance radius where Edge Finder searches for edge extremity candidates when filling edge gaps. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0 pixels.	c		e	f	g	a		i		j	k	l	m	n	b
	M_INFINITE	Specifies an infinite maximum distance radius.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the maximum distance radius, in pixels. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_FILL_GAP_POLARITY +		Inquires the polarity constraint used when performing the filling of edge gaps. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_ANY.	c		e	f	g	a		i		j	k	l	m	n	b

	M_ANY	Specifies that gaps between edges will be filled, regardless of their polarity.	c		e	f	g	a		i		j	k	l	m	n	b
	M_REVERSE	Specifies that gaps between edges that have reverse polarity will be filled. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SAME	Specifies that gaps between edges that have the same polarity will be filled. INFO	c		e	f	g	a		i		j	k	l	m	n	b

To inquire about general Edge Finder result buffer settings, set the InquireType parameter to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For result buffer settings (general)																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_AGM_COMPATIBLE +		Inquires whether the Edge Finder result buffer is ready to be used with an AGM context. SET															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the Edge Finder result buffer cannot be used with an AGM context.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the Edge Finder result buffer can be used with an AGM context.	c		e	f	g	a		i		j	k	l	m	n	b
M_DRAW_CROSS_SIZE +		Inquires the size of the cross used for certain drawing operations. For example, M_DRAW_EDGELS, M_DRAW_POSITION, and M_DRAW_CENTER_OF_GRAVITY are drawn using this cross. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

	M_DEFAULT	Specifies the default value; the default value is 4.0 pixels.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the size of the cross, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_MODEL_FINDER_COMPATIBLE +		Inquires whether the Edge Finder result buffer is ready to be used with a Model Finder context. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_DISABLE.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DISABLE	Specifies that the Edge Finder result buffer cannot be used with a Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ENABLE	Specifies that the Edge Finder result buffer can be used with a Model Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEAREST_NEIGHBOR_RADIUS +		Inquires the radius distance used to select (MedgeSelect()) the closest edge or edges from a point. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 2.0 pixels.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the radius distance, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_RESULT_OUTPUT_UNITS +		Inquires whether results are returned in pixel or world units. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_ACCORDING_TO_CALIBRATION.	c		e	f	g	a		i		j	k	l	m	n	b

	M_ACCORDING_TO_CALIBRATION	Specifies that results are returned in world units if the result was calculated on an image associated with a camera calibration context; otherwise, specifies that results are returned in pixel units.	c		e	f	g	a		i		j	k	l	m	n	b
	M_PIXEL	Specifies that results are returned in pixel units, with respect to the pixel coordinate system.	c		e	f	g	a		i		j	k	l	m	n	b
	M_WORLD	Specifies that results are returned in world units, with respect to the relative coordinate system. INFO	c		e	f	g	a		i		j	k	l	m	n	b

To inquire about Edge Finder result buffer settings used when performing closest-edgel operations with MedgeGetNeighbors(), set the InquireType parameter to one of the values below.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For result buffer settings (closest-edgel operations)																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_NEIGHBOR_ANGLE +		Inquires the gradient angle that an edgel must have, before being considered a candidate, when finding the closest edgels to a list of points. SET															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_ANY.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ANY	Specifies that the edgel candidate can have any gradient angle. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_REVERSE	Specifies that the gradient angle of the edgel candidate must have the reverse angle (+ 180°) of the source point.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SAME	Specifies that the gradient angle of the edgel candidate must have the same angle as the source point.	c		e	f	g	a		i		j	k	l	m	n	b

	M_SAME_OR_REVERSE	Specifies that the gradient angle of the edgel candidate must either have the same, or the reverse angle of the source point.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEIGHBOR_ANGLE_TOLERANCE +		Inquires the angular tolerance used for the angle constraint (M_NEIGHBOR_ANGLE), when finding the closest edgels to a list of points. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 180.0 °.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 360.0	Specifies the angular tolerance, in degrees. INFO	c		e	f	g	a		i		j	k	l	m	n	b
M_NEIGHBOR_MAXIMUM_NUMBER +		Inquires the maximum number of closest edgel candidates that can be returned (for each source point), when finding the closest edgels to a list of points. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 1.	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the maximum number of edgels.	c		e	f	g	a		i		j	k	l	m	n	b
M_NEIGHBOR_MINIMUM_SPACING +		Inquires the minimum distance separating edgels within the same edge, in order for each to be considered potential candidates, when finding the closest edgels to a list of points. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_INFINITE.	c		e	f	g	a		i		j	k	l	m	n	b

	M_INFINITE	Specifies no minimum distance separating two edgel candidates.	c		e	f	g	a		i		j	k	l	m	n	b
	Value >= 1	Specifies the minimum distance, in edgels.	c		e	f	g	a		i		j	k	l	m	n	b
M_SEARCH_ANGLE +		Inquires the search angle constraint (applied the Edge Finder result buffer), when finding the closest edgels to a list of points. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value; the default value is 0.0 °.	c		e	f	g	a		i		j	k	l	m	n	b
	0.0 <= *Value* <= 360.0	Specifies the angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_SEARCH_ANGLE_SIGN +		Inquires the orientation to use for the search angle constraint, when finding the closest edgels to a list of points. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Same as M_SAME.	c		e	f	g	a		i		j	k	l	m	n	b
	M_REVERSE	Specifies that the orientation of the angle must be the reverse (+ 180°) of the source angle. INFO	c		e	f	g	a		i		j	k	l	m	n	b
	M_SAME	Specifies that the orientation of the angle must be the same as the source angle.	c		e	f	g	a		i		j	k	l	m	n	b
	M_SAME_OR_REVERSE	Specifies that the orientation of the angle can be the same as, or the reverse of, the source angle.	c		e	f	g	a		i		j	k	l	m	n	b
M_SEARCH_ANGLE_TOLERANCE +		Inquires the angular tolerance used for the search angle constraint, when finding the closest edgels to a list of points. SET	c		e	f	g	a		i		j	k	l	m	n	b
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

M_DEFAULT	Specifies the default value; the default value is 360.0 °.	c		e	f	g	a		i		j	k	l	m	n	b
0.0 <= *Value* <= 360.0	Specifies the angle, in degrees.	c		e	f	g	a		i		j	k	l	m	n	b
M_SEARCH_RADIUS_MAX +	Inquires the maximum radius distance used to search for the closest edgel candidates that match source edgels. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Same as M_INFINITE.	c		e	f	g	a		i		j	k	l	m	n	b
M_INFINITE	Specifies a maximum distance radius that spans all edgels in the Edge Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the maximum distance radius, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b
M_SEARCH_RADIUS_MIN +	Inquires the minimum radius distance used to search for the closest edgel candidates that match source edgels. SET	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the default value; the default value is 0 pixels.	c		e	f	g	a		i		j	k	l	m	n	b
Value	Specifies the minimum distance radius, in pixels.	c		e	f	g	a		i		j	k	l	m	n	b

To inquire about the system on which either the Edge Finder context or Edge Finder result buffer has been allocated, set the InquireType parameter to the value below.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring about the system																
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																

M_OWNER_SYSTEM +		Inquires the identifier of the system on which either the Edge Finder context or Edge Finder result buffer has been allocated. SET SET														
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
	System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for
Tables:

- For operation settings (both M_CONTOUR and M_CREST)
- For operation settings (M_CREST only)
- For internal buffers (both M_CONTOUR and M_CREST)
- For edge features (both M_CONTOUR and M_CREST)
- For sorting keys (both M_CONTOUR and M_CREST)
- For general settings (both M_CONTOUR and M_CREST)
- For Feret settings (both M_CONTOUR and M_CREST)
- For post-calculations (both M_CONTOUR and M_CREST)
- For result buffer settings (general)
- For result buffer settings (closest-edgel operations)
- For inquiring about the system

the values listed in all the parameter association tables.

You can add the following value to the above-mentioned values to get the default value of an inquire type, regardless of the current value of the inquire type.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring the default value																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_DEFAULT		Inquires the default value of the specified inquire type.															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b

Combination value for
Tables:

- For operation settings (both M_CONTOUR and M_CREST)
- For operation settings (M_CREST only)

- For internal buffers (both M_CONTOUR and M_CREST)
- For edge features (both M_CONTOUR and M_CREST)
- For sorting keys (both M_CONTOUR and M_CREST)
- For general settings (both M_CONTOUR and M_CREST)
- For Feret settings (both M_CONTOUR and M_CREST)
- For post-calculations (both M_CONTOUR and M_CREST)
- For result buffer settings (general)
- For result buffer settings (closest-edgel operations)
- For inquiring about the system

the values listed in all the parameter association tables.

You can add the following value to the above-mentioned values to determine whether an inquire type is sortable or supported for the Edge Finder context currently being inquired.

Note that to inquire if an inquire type is sortable, you must add M_SUPPORTED to the appropriate sort value (M_SORTn_UP or M_SORTn_DOWN) and the specified inquire type (for example, M_SIZE + M_SORTn_UP + M_SUPPORTED). If the inquire type is not supported for the Edge Finder context, or sortable, M_FALSE is returned.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For inquiring if an inquire type is sortable or is supported																	
InquireType		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	UserVarPtr - Possible values returned																
M_SUPPORTED		Inquires whether the specified inquire type is either sortable, or supported for the Edge Finder context.															
	UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
	M_FALSE	Specifies that the inquire type is not supported.	c		e	f	g	a		i		j	k	l	m	n	b
	M_TRUE	Specifies that the inquire type is supported.	c		e	f	g	a		i		j	k	l	m	n	b

Combination values for
Tables:

- For operation settings (both M_CONTOUR and M_CREST)
- For operation settings (M_CREST only)
- For internal buffers (both M_CONTOUR and M_CREST)
- For edge features (both M_CONTOUR and M_CREST)
- For sorting keys (both M_CONTOUR and M_CREST)
- For general settings (both M_CONTOUR and M_CREST)
- For Feret settings (both M_CONTOUR and M_CREST)
- For post-calculations (both M_CONTOUR and M_CREST)
- For result buffer settings (general)
- For result buffer settings (closest-edgel operations)

- For inquiring about the system

the values listed in all the parameter association tables.

You can add one of the following values to the above-mentioned values to cast the requested information to a required data type.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the data type																	
InquireType	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
UserVarPtr - Possible values returned																
M_TYPE_AIL_DOUBLE	Casts the requested information to an *AIL_DOUBLE*.															
UserVarPtr extra info	Data type info: Data type: *address of an AIL_DOUBLE*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_ID	Casts the requested information to an *AIL_ID*. Note that M_TYPE_AIL_ID should only be used with M_OWNER_SYSTEM.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_ID*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT	Casts the requested information to an *AIL_INT*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT32	Casts the requested information to an *AIL_INT32*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT32*	c		e	f	g	a		i		j	k	l	m	n	b
M_TYPE_AIL_INT64	Casts the requested information to an *AIL_INT64*.	c		e	f	g	a		i		j	k	l	m	n	b
UserVarPtr extra info	Data type info: Data type: *address of an AIL_INT64*	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the requested information, cast to an *AIL_INT*. If the requested information does not fit into an *AIL_INT*, this function will return **M_NULL** or truncate the information.

Compilation information

Header	Include AIL.h.

Library	Use ail11.lib; ailedge11.lib.
DLL	Requires ail11.dll; ailedge11.dll.

MedgeMask

Synopsis

Mask regions of the Edge Finder context.

Syntax

```
void MedgeMask(
    AIL_ID ContextId,     //out
    AIL_ID MaskImageId,   //in
    AIL_INT64 ControlFlag //in
)
```

Description

This function allows you to set a mask for an Edge Finder context. The mask is applied to the image that uses the context. Subsequent calls to MedgeCalculate() will therefore extract edges only in the context's unmasked regions.

You can also mask edges during post-calculation. In this case, masked edges are excluded from the result buffer and subsequent calculations. Note that partially masked edges are cropped.

Parameters

ContextId

Specifies the Edge Finder context in which to set the mask. The Edge Finder context must have been previously allocated on the required system using MedgeAlloc().

MaskImageId

Specifies the identifier of the image buffer used to identify the masked pixels in the Edge Finder context.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For the identifier of the image buffer																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_NULL	Specifies to ignore this parameter. This parameter must be set to M_NULL when removing the mask.	c		e	f	g	a		i		j	k	l	m	n	b
Image buffer identifier	Specifies the image buffer to use as a mask. If the size of the mask image is greater than the source image, the mask will be clipped. A masked pixel corresponds to a non-zero value in the mask buffer. Note that this buffer can be calibrated; however, this is not necessary, since the mask is applied on a pixel basis. This image buffer must be a 1-band buffer and must not have a region of interest (ROI) associated with it. Using an image buffer with an ROI will cause an error.	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailedge11.lib.
DLL	Requires ail11.dll; ailedge11.dll.

MedgePut

Synopsis

Put edge chains from user-supplied arrays into an Edge Finder result buffer.

Syntax

```
void MedgePut(
    AIL_ID ResultEdgeId,                //out
    AIL_INT NumEdgels,                  //in
    const AIL_INT *ChainIndexArrayPtr,  //in
    const AIL_DOUBLE *PosXArrayPtr,     //in
    const AIL_DOUBLE *PosYArrayPtr,     //in
    const AIL_DOUBLE *AngleArrayPtr,    //in
    const AIL_DOUBLE *MagnitudeArrayPtr, //in
    AIL_INT64 ControlFlag               //in
)
```

Description

This function copies edge chains from user-supplied arrays to a specified Edge Finder result buffer. This can be useful if you want to construct an Edge Finder result buffer that cannot be obtained from one image. You can therefore combine results from multiple Edge Finder result buffers to, for example, build a complete model to add to a Model Finder context.

You must only add edge chains that respect the following constraints, whereby pixels are considered connected based on an 8-connected lattice:

- Consecutive edgels must occupy separate connected pixels.
- No branches are allowed (they start or end a separate chain).
- There must not be an edge in the edge map that is more than 1 pixel wide. This means that the first and fourth edgel in the edge chain must not be in neighboring pixels.

All edge chains must be at the same scale and this scale must be the scale at which they were originally extracted. If edges were extracted at a different scale (MedgeControl() with M_EXTRACTION_SCALE), you must convert them to their original extraction scale.

If edgels are calculated with pixel accuracy, you must provide edgels with coordinates that are integer values; in this case, the distance between consecutive edgels should be one pixel. If edgels are calculated with subpixel accuracy, you can provide edgels with coordinates that are double values; in this case, the distance between consecutive edgels is such that, each pixel touched by the edge corresponds to one edgel. To change the accuracy edgels are calculated with, use MedgeControl() with M_ACCURACY.

To use an Edge Finder result buffer that contains edges set using **MedgePut**() with the Model Finder module, you must respect the following limitation. First, edges in the source image buffer must have been previously extracted and saved in the Edge Finder result buffer, using MedgeCalculate(). Second, the coordinates of the edge chains in the user-supplied arrays must not exceed the boundaries of the source image buffer. You can inquire the width and height of the source image buffer using MedgeGetResult() with M_SIZE_X and M_SIZE_Y.

The X- and Y-coordinates of the edgels you are adding to the Edge Finder result buffer must be provided

in pixel units (as opposed to real-world values). If your current results have been calibrated, the coordinates added to those results will be automatically transformed to their appropriate real-world values. For more information, see the Working with real-world units section of Chapter 28: Calibrating your camera setup.

Parameters

ResultEdgeId

Specifies the identifier of the Edge Finder result buffer in which to put user-supplied data.

NumEdgels

Specifies the number of edgels to add to the Edge Finder result buffer.

> When using a standard vector (std::vector) overload function in C++, you can pass **M_DEFAULT** to this parameter; the size will be determined automatically based on the number of items in the vector passed to the PosXArrayPtr or PosYArrayPtr parameter.

ChainIndexArrayPtr
Accepts the address of one of the following:

- array of type AIL_DOUBLE [optionally, in C++: a reference to a constant std::vector<AIL_DOUBLE>] (and a maximum array rank for C# of 1)
  Required array size:(NumEdgels)
- array of type AIL_INT [optionally, in C++: a reference to a constant std::vector<AIL_INT>] (and a maximum array rank for C# of 1)
  Required array size:(NumEdgels)

Specifies the address of the array containing the indices of the edgels' edges to add to the Edge Finder result buffer. If you do not want to provide this information, set this parameter to **M_NULL**. In this case, only one edge will be added.

PosXArrayPtr
Accepts the address of one of the following:

- array of type AIL_DOUBLE [optionally, in C++: a reference to a constant std::vector<AIL_DOUBLE>] (and a maximum array rank for C# of 1)
  Required array size:(NumEdgels)
- array of type AIL_INT [optionally, in C++: a reference to a constant std::vector<AIL_INT>] (and a maximum array rank for C# of 1)
  Required array size:(NumEdgels)

Specifies the address of the array containing the X-coordinate(s) of the edgel(s) to add to the Edge Finder result buffer.

PosYArrayPtr
Accepts the address of one of the following:

- array of type AIL_DOUBLE [optionally, in C++: a reference to a constant std::vector<AIL_DOUBLE>] (and a maximum array rank for C# of 1)
  Required array size:(NumEdgels)
- array of type AIL_INT [optionally, in C++: a reference to a constant std::vector<AIL_INT>] (and a maximum array rank for C# of 1)
  Required array size:(NumEdgels)

Specifies the address of the array containing the Y-coordinate(s) of the edgel(s) to add to the Edge Finder result buffer.

AngleArrayPtr
Accepts the address of one of the following:

- array of type AIL_DOUBLE [optionally, in C++: a reference to a constant std::vector<AIL_DOUBLE>] (and a maximum array rank for C# of 1)
  Required array size:(NumEdgels)
- array of type AIL_INT [optionally, in C++: a reference to a constant std::vector<AIL_INT>] (and a maximum array rank for C# of 1)
  Required array size:(NumEdgels)

Specifies the address of the array containing the orientation of the edgel(s) to add to the Edge Finder result buffer. If you do not want to provide this information, set this parameter to **M_NULL**.

Angle values must be mapped in the range of 0 to 255. That is, 0° corresponds to 0, and 360° corresponds to 256.

MagnitudeArrayPtr
Accepts the address of one of the following:

- array of type AIL_DOUBLE [optionally, in C++: a reference to a constant std::vector<AIL_DOUBLE>] (and a maximum array rank for C# of 1)
  Required array size:(NumEdgels)
- array of type AIL_INT [optionally, in C++: a reference to a constant std::vector<AIL_INT>] (and a maximum array rank for C# of 1)
  Required array size:(NumEdgels)

Specifies the address of the array containing the magnitude of the edgel(s) to add to the Edge Finder result buffer. If you do not want to provide this information, set this parameter to **M_NULL**.

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailedge11.lib.
DLL	Requires ail11.dll; ailedge11.dll.

MedgeRestore

Synopsis

Restore an Edge Finder context from disk.

Syntax

```
AIL_ID MedgeRestore(
    AIL_CONST_TEXT_PTR FileName, //in
    AIL_ID SysId,                //in
    AIL_INT64 ControlFlag,       //in
    AIL_ID *ContextIdPtr         //out
)
```

Description

This function restores an Edge Finder context that was previously saved to a file, using MedgeSave() or MedgeStream(). This function restores all the Edge Finder context's settings that were in effect when the Edge Finder context was saved.

When the restored Edge Finder context is no longer required, release it using MedgeFree() unless M_UNIQUE_ID was specified during allocation; if M_UNIQUE_ID was specified, the smart identifier manages the restored Edge Finder context's lifetime and you must not manually free it.

Parameters

FileName

Specifies the name and path of the file from which to restore the Edge Finder context. The function handles (internally) the opening and closing of the file.

This parameter can be set to one of the following:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the file name and path																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_INTERACTIVE	[This is only applicable to Windows] Opens the **File Open** dialog box from which you can interactively specify the drive, directory, and name of the file.	c		e	f	g	a		i		j	k	l	m	n	b
AIL_TEXT("*FileName*") 1	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"). Typically, Edge Finder context files have an MEF extension. To specify the file on the remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").	c		e	f	g	a		i		j	k	l	m	n	b

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

SysId

Specifies the system on which to restore the Edge Finder context.

This parameter should be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the system identifier																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

ControlFlag

Reserved for future expansion and must be set to **M_DEFAULT**.

ContextIdPtr

Specifies the address of the variable in which to write the Edge Finder context identifier or specifies the data type that the function should use to return the Edge Finder context identifier.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For retrieving the identifier or specifying how to return it																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_NULL	Specifies that you will use this function's return value to obtain the identifier of the allocated Edge Finder context ; in this case, a standard Aurora Imaging Library identifier of type *AIL_ID* is returned.	c		e	f	g	a		i		j	k	l	m	n	b

M_UNIQUE_ID	Specifies that you will use this function's return value to obtain the identifier of the allocated Edge Finder context ; in this case, an Aurora Imaging Library smart identifier of type *AIL_UNIQUE_EDGE_ID* is returned instead of a standard Aurora Imaging Library identifier. > This setting is only available when using C++11 (or later). An Aurora Imaging Library smart identifier manages the lifespan of the Aurora Imaging Library object it owns (similar to a *std::unique_ptr*). Note, you can use an Aurora Imaging Library smart identifier as though it were a standard Aurora Imaging Library identifier, except that you cannot use it to manually free the Edge Finder context (it is freed automatically). For more information, see the Aurora Imaging Library smart identifiers subsection of the Aurora Imaging Library custom data types, void pointers, extensions, and portability functions section of Chapter 2: Key information and building an application.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the contour Edge Finder context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the restored contour Edge Finder context. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b
Address in which to write the crest Edge Finder context identifier	Specifies the address of an *AIL_ID* in which to write the identifier of the restored crest Edge Finder context. If allocation fails, **M_NULL** is written as the identifier.	c		e	f	g	a		i		j	k	l	m	n	b

Return value

The returned value is the Edge Finder context identifier either as a standard identifier (*AIL_ID*) or a smart

identifier (*AIL_UNIQUE_EDGE_ID*). If allocation fails, **M_NULL** is returned (or *nullptr* if M_UNIQUE_ID was specified).

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailedge11.lib.
DLL	Requires ail11.dll; ailedge11.dll.

MedgeSave

Synopsis

Save an Edge Finder context to a file, or save edge chains and/or edge approximations from an Edge Finder result buffer to a CAD (Computer-Aided Design) file.

Syntax

```
void MedgeSave(
    AIL_CONST_TEXT_PTR FileName, //in
    AIL_ID ContextOrResultId,    //in
    AIL_INT64 ControlFlag        //in
)
```

Description

This function saves all the information about the previously allocated Edge Finder context to disk. This information can be reloaded, using MedgeRestore() or MedgeStream(). However, any associated camera calibration contexts are not saved.

This function also saves calculated edges (edge chains and/or edge approximations) included in a previously calculated Edge Finder result buffer to disk, in a standard DXF format CAD file. Note that edge chains and edge approximations are saved in the DXF file in separate layers.

Note that the Edge Finder context or the calculated edges are saved in real-world units if there is a camera calibration context associated with them. Otherwise, they are saved in pixel units.

Parameters

FileName

Specifies the name and path of the file in which to save the Edge Finder context, or the destination CAD file. For easier use with other Aurora Imaging Library software products, when saving an Edge Finder context to a file, use the MEF file extension, and when saving calculated edges from an Edge Result buffer to a CAD file, use the DXF file extension. The function internally handles the opening and closing of this file. If this file already exists, it will be overwritten.

This parameter can be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the file name and path																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_INTERACTIVE	[This is only applicable to Windows] Opens the **File Save As** dialog box from which you can interactively specify the drive, directory, and name of the file.	c		e	f	g	a		i		j	k	l	m	n	b

AIL_TEXT("*FileName*") [1]	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"). Typically, Edge Finder contexts have an MEF extension. Edge Finder result buffers have a DXF extension. To save the file on the remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").	c		e	f	g	a		i		j	k	l	m	n	b

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

ContextOrResultId

Specifies either the identifier of the Edge Finder context to save, or the identifier of the Edge Finder result buffer from which to save to a CAD file.

ControlFlag

Specifies whether to save edge chains or edge approximations to the DXF CAD file for Edge Finder result buffers. For Edge Finder contexts, this parameter must be set to **M_DEFAULT**.

For Edge Finder result buffers, this parameter must at least be set to one of the following values. Note that to save both edge chains and edge approximations, ControlFlag values can be combined (M_CHAIN + M_CHAIN_APPROXIMATION).

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For Edge Finder result buffers																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_CHAIN	Specifies that edge chains will be saved.	c		e	f	g	a		i		j	k	l	m	n	b
M_CHAIN_APPROXIMATION	Specifies that edge approximations will be saved.	c		e	f	g	a		i		j	k	l	m	n	b

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailedge11.lib.
DLL	Requires ail11.dll; ailedge11.dll.

MedgeSelect

Synopsis

Select edges for calculations and result retrieval.

Syntax

```
void MedgeSelect(
    AIL_ID EdgeResultId,          //out
    AIL_INT64 Operation,          //in
    AIL_INT64 SelectionCriterion, //in
    AIL_INT64 Condition,          //in
    AIL_DOUBLE Param1,            //in
    AIL_DOUBLE Param2             //in
)
```

Description

This function selects edges that meet a specified criterion. These edges will be included in or excluded from future operations (calculations and result retrieval), or deleted entirely from the Edge Finder result buffer. To call this function, MedgeCalculate() must have been called at least once.

If this function is not called at least once, all edges are included by default. If there is more than one call to this function, the effect of the calls is cumulative, unless M_INCLUDE_ONLY or M_EXCLUDE_ONLY is specified as the operation to perform.

Once an edge has been excluded, it can typically be re-included by specifying M_INCLUDE or M_INCLUDE_ONLY in a future call to this function (with the correct criterion). However, if you use the result buffer with different images (in a call to MedgeCalculate()), all results in the result buffer are discarded and all new edges are re-included.

> Unlike most other Aurora Imaging Library functions, if the edges were extracted from a calibrated source image, you must specify relevant values in world units, although you will have control over the units when retrieving results using MedgeGetResult().

You can use this function to select edges based on one of the following:

- A calculated edge feature (see MedgeControl()), where the edge selection depends on whether the specified edge feature meets the specified condition.
- The inter-relationship of edges, where the edge selection depends on whether edges meet the specified box or chain condition of the specified edge or group of edges.
- The current status of edges, where the edge selection depends on the status of a specific edge, all edges, all included edges, or all excluded edges. These will be included, included only, excluded, excluded only, or deleted. For example, you can include only (M_INCLUDE_ONLY) the excluded edges (M_EXCLUDED_EDGES), which will essentially swap the previously included and excluded edges.
- The proximity of an edge or edges to a specified point, where the edge selection depends on the specified radius, and the nearest neighbor condition.

You can also use this function to crop and select a portion of a specified edge.

For more information on the different uses of this function, see the Calculating and retrieving results section of Chapter 10: Edge Finder and the Advanced edge extraction section of Chapter 10: Edge Finder.

Parameters

EdgeResultId

Specifies the identifier of the Edge Finder result buffer to be used in the edge selection process.

See the Parameter associations section for possible values that can be returned.

Operation

Specifies the operation to perform on the specified edges. Set this parameter to one of the following values.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the operation																	
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_DELETE	Deletes edges that meet the specified condition. M_DELETE affects only included edges, unless otherwise stated by the SelectionCriterion parameter. M_DELETE removes edges permanently from the Edge Finder result buffer and, consequently, prevents these edges from being re-included.	c		e	f	g	a		i		j	k	l	m	n	b
M_EXCLUDE	Excludes all edges that meet the specified condition. M_EXCLUDE affects only the status of currently included edges.	c		e	f	g	a		i		j	k	l	m	n	b
M_EXCLUDE_ONLY	Excludes only those edges that meet the specified condition and includes all others. The exclusion does not consider the present status of edges (whether they are excluded), except for edges that have been deleted (M_DELETE), unless otherwise stated by the SelectionCriterion parameter.	c		e	f	g	a		i		j	k	l	m	n	b
M_INCLUDE	Includes all edges that meet the specified condition. M_INCLUDE affects only the status of currently excluded edges.	c		e	f	g	a		i		j	k	l	m	n	b

M_INCLUDE_ONLY	Includes only those edges that meet the specified condition and excludes all others. The inclusion does not consider the present status of edges (whether they are included), except for edges that have been deleted (M_DELETE), unless otherwise stated by the SelectionCriterion parameter.	c		e	f	g	a		i		j	k	l	m	n	b

SelectionCriterion

Specifies on what the selection criterion will be based. Note that for an edge selection based on the proximity of the edges to a point, this parameter must be set to **M_NULL**.

See the Parameter associations section for possible values that can be specified.

Condition

Specifies the condition for the edge selection. Note that for an edge selection based on the current status of edges, this parameter must be set to **M_NULL**.

See the Parameter associations section for possible values that can be specified.

Param1

Specifies a value that is dependent on the feature and condition chosen. If the edges have been calculated using a calibrated source image, you must specify the relevant values in the real world. If this parameter is unused, set it to **M_NULL**.

See the Parameter associations section for possible values that can be specified.

Param2

Specifies a value that is dependent on the feature and condition chosen. If the edges have been calculated using a calibrated source image, you must specify the relevant values in the real world. If this parameter is unused, set it to **M_NULL**.

See the Parameter associations section for possible values that can be specified.

Parameter associations

The table below lists possible values for the **SelectionCriterion**, **Condition**, **Param1**, and **Param2** parameters and possible values returned to the **EdgeResultId** parameter.

- For specifying the edge selection method

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For specifying the edge selection method																	
EdgeResultId - Possible values returned		Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
	SelectionCriterion																
	Condition																
	Param1																

Param2																
Edge Finder result buffer identifier for a chain cropping based edge selection	Specifies an Edge Finder result buffer on which to perform a chain cropping edge selection. This type of selection essentially splits off a portion of a single specified edge, assigns the portion a new edge label, and selects the portion for inclusion, exclusion, or deletion from future operations on the result buffer. The other portion keeps its current edge label value and status. The two portions are considered two touching edges after the split. Therefore, all their calculated edge features are updated after the split. The specified edgel index value is the dividing point.															
SelectionCriterion	Sets the selection criterion.	c		e	f	g	a		i		j	k	l	m	n	b
M_CROP_CHAIN	Specifies a chain cropping edge selection.	c		e	f	g	a		i		j	k	l	m	n	b
Condition	Sets the condition that an edgel index value must satisfy to be part of the new edge that is selected.	c		e	f	g	a		i		j	k	l	m	n	b
M_GREATER	Specifies that the edgels with an index value greater than the specified edgel index value are considered part of the new edge.	c		e	f	g	a		i		j	k	l	m	n	b
M_GREATER_OR_EQUAL	Specifies that the edgels with an index value greater than or equal to the specified edgel index value are considered part of the new edge.	c		e	f	g	a		i		j	k	l	m	n	b
M_LESS	Specifies that the edgels with an index value less than the specified edgel index value are considered part of the new edge.	c		e	f	g	a		i		j	k	l	m	n	b

		M_LESS_OR_EQUAL	Specifies that the edgels with an index value less than or equal to the specified edgel index value are considered part of the new edge.	c		e	f	g	a		i		j	k	l	m	n	b
	Param1		Sets the label value of the edge to crop.	c		e	f	g	a		i		j	k	l	m	n	b
	Param2		Sets the edgel index value indicating where to split the edge.	c		e	f	g	a		i		j	k	l	m	n	b
Edge Finder result buffer identifier for a feature based edge selection			Specifies an Edge Finder result buffer on which to perform an edge selection based on a calculated edge feature. Edges whose specified feature satisfies the condition are included, excluded, or deleted from future operations on the Edge Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
	SelectionCriterion		Sets the calculated edge feature to use as the selection criterion. The result buffer must already contain the results for the specified feature. Otherwise, an error wll occur.	c		e	f	g	a		i		j	k	l	m	n	b
		M_AVERAGE_STRENGTH	Uses the average strength . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_BOX_X_MAX	Uses the extreme right edgel coordinate .	c		e	f	g	a		i		j	k	l	m	n	b
		M_BOX_X_MIN	Uses the extreme left edgel coordinate .	c		e	f	g	a		i		j	k	l	m	n	b
		M_BOX_Y_MAX	Uses the extreme bottom edgel coordinate .	c		e	f	g	a		i		j	k	l	m	n	b
		M_BOX_Y_MIN	Uses the extreme top edgel coordinate .	c		e	f	g	a		i		j	k	l	m	n	b
		M_CENTER_OF_GRAVITY_X	Uses the X-position of the center of gravity. INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_CENTER_OF_GRAVITY_Y	Uses the Y-position of the center of gravity. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_CIRCLE_FIT_CENTER_X	Uses the X-coordinate of the center of the circle that is the best fit .	c		e	f	g	a		i		j	k	l	m	n	b
		M_CIRCLE_FIT_CENTER_Y	Uses the Y-coordinate of the center of the circle that is the best fit .	c		e	f	g	a		i		j	k	l	m	n	b
		M_CIRCLE_FIT_COVERAGE	Uses the coverage of the circle that is the best fit . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_CIRCLE_FIT_ERROR	Uses the fit error of the circle that is the best fit . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_CIRCLE_FIT_RADIUS	Uses the radius of the circle that is the best fit .	c		e	f	g	a		i		j	k	l	m	n	b
		M_CLOSURE	Uses the closure state . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_CONVEX_PERIMETER	Uses the convex elongation . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_ELLIPSE_FIT_ANGLE	Uses the angle of the ellipse that is the best fit .	c		e	f	g	a		i		j	k	l	m	n	b
		M_ELLIPSE_FIT_CENTER_X	Uses the X-coordinate of the center of the ellipse that is the best fit .	c		e	f	g	a		i		j	k	l	m	n	b
		M_ELLIPSE_FIT_CENTER_Y	Uses the Y-coordinate of the center of the ellipse that is the best fit .	c		e	f	g	a		i		j	k	l	m	n	b
		M_ELLIPSE_FIT_COVERAGE	Uses the coverage of the ellipse that is the best fit . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_ELLIPSE_FIT_ERROR	Uses the fit error of the ellipse that is the best fit . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_ELLIPSE_FIT_MAJOR_AXIS	Uses the major axis of the ellipse that is the best fit . INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_ELLIPSE_FIT_MINOR_AXIS	Uses the minor axis of the ellipse that is the best fit . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_FAST_LENGTH	Uses a coarse approximation for the length . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_FERET_ELONGATION	Uses the Feret elongation . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_FERET_GENERAL	Uses the general Feret . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_FERET_MAX_ANGLE	Uses the maximum Feret angle . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_FERET_MAX_DIAMETER	Uses the maximum Feret diameter . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_FERET_MEAN_DIAMETER	Uses the average Feret diameter at all the angles checked. INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_FERET_MIN_ANGLE	Uses the minimum Feret angle . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_FERET_MIN_DIAMETER	Uses the minimum Feret diameter . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_FERET_X	Uses the X-Feret value . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_FERET_Y	Uses the Y-Feret value . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_FIRST_POINT_X	Uses the X-coordinate of the first point . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_FIRST_POINT_Y	Uses the Y-coordinate of the first point . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_LABEL_VALUE	Uses the label value of each edge . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_LENGTH	Uses the length . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_LINE_FIT_A	Uses the coefficient *A* of the line that is the best fit . INFO	c		e	f	g	a		i		j	k	l	m	n	b

		M_LINE_FIT_B	Uses the coefficient *B* of the line that is the best fit . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_LINE_FIT_C	Uses the coefficient *C* of the line that is the best fit . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_LINE_FIT_ERROR	Uses the fit error of the line that is the best fit . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_MOMENT_ELONGATION	Uses the moment elongation . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_POSITION_X	Uses the X-position . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_POSITION_Y	Uses the Y-position . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_SIZE	Uses the number of edgels .	c		e	f	g	a		i		j	k	l	m	n	b
		M_STRENGTH	Uses the strength . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_TORTUOSITY	Uses the tortuosity measure . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_X_MAX_AT_Y_MAX	Uses the maximum X-coordinate at the maximum Y-coordinate . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_X_MIN_AT_Y_MIN	Uses the minimum X-coordinate at the minimum Y-coordinate . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_Y_MAX_AT_X_MIN	Uses the maximum Y-coordinate at the minimum X-coordinate . INFO	c		e	f	g	a		i		j	k	l	m	n	b
		M_Y_MIN_AT_X_MAX	Uses the minimum Y-coordinate at the maximum X-coordinate . INFO	c		e	f	g	a		i		j	k	l	m	n	b
	Condition		Sets the edge selection condition.	c		e	f	g	a		i		j	k	l	m	n	b

		M_EQUAL	Specifies that edges with values for the specified feature equal to Param1 are included, excluded, or deleted from future operations on the specified Edge Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GREATER	Specifies that edges with values for the specified feature greater than Param1 are included, excluded, or deleted from future operations on the specified Edge Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
		M_GREATER_OR_EQUAL	Specifies that edges with values for the specified feature greater than or equal to Param1 are included, excluded, or deleted from future operations on the specified Edge Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
		M_IN_RANGE	Specifies that edges with values for the specified feature in the range of Param1 to Param2, inclusive, are included, excluded, or deleted from future operations on the specified Edge Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
		M_LESS	Specifies that edges with values for the specified feature less than Param1 are included, excluded, or deleted from future operations on the specified Edge Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
		M_LESS_OR_EQUAL	Specifies that edges with values for the specified feature less than or equal to Param1 *are included, excluded, or deleted from future operations on the specified Edge Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
		M_NOT_EQUAL	Specifies that edges with values for the specified feature not equal to Param1 are included, excluded, or deleted from future operations on the specified Edge Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b

		M_OUT_RANGE	Specifies that edges with values for the specified feature less than Param1, or greater than Param2, are included, excluded, or deleted from future operations on the specified Edge Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
	Param1		Sets a condition limit. If the condition uses one limit, set this parameter to the required condition limit. If the condition uses two limits, set this parameter to the low condition limit.	c		e	f	g	a		i		j	k	l	m	n	b
	Param2		Sets the high condition limit. If the condition uses one limit, set this parameter to **M_NULL**. If the condition uses two limits, set this parameter to the high condition limit.	c		e	f	g	a		i		j	k	l	m	n	b
		M_NULL	Specifies that this parameter is unused.	c		e	f	g	a		i		j	k	l	m	n	b
		Value	Specifies the high condition limit.	c		e	f	g	a		i		j	k	l	m	n	b
Edge Finder result buffer identifier for an inter-relationship based edge selection			Specifies an Edge Finder result buffer on which to perform an edge selection based on the inter-relationship of edges. Edges that fall inside or outside of a boundary defined by other edges are included, excluded, or deleted from future operations on the Edge Finder result buffer. The specified condition establishes whether edges inside or outside are selected. You specify the delimiting edges; to define the boundary, you can specify to use the edges as is or to use their bounding box as a delimiter.	c		e	f	g	a		i		j	k	l	m	n	b
	SelectionCriterion		Sets which edges are used to delimit the boundary.	c		e	f	g	a		i		j	k	l	m	n	b
		M_ALL_EDGES	Specifies all edges regardless of their present status, except for edges that have been deleted.	c		e	f	g	a		i		j	k	l	m	n	b

		M_EXCLUDED_EDGES	Specifies all currently excluded edges, except for edges that have been deleted.	c		e	f	g	a		i		j	k	l	m	n	b
		M_INCLUDED_EDGES	Specifies all currently included edges.	c		e	f	g	a		i		j	k	l	m	n	b
		M_SPECIFIC_EDGE	Specifies a specific edge. Use Param1 to set the label value of the edge.	c		e	f	g	a		i		j	k	l	m	n	b
	Condition		Sets the edge selection condition based on the relationship to the specified delimiting edge(s).	c		e	f	g	a		i		j	k	l	m	n	b
		M_INSIDE_BOX	Specifies that edges inside the bounding box of the specified edge, or inside any individual bounding box of the specified group of edges, are included, excluded, or deleted from future operations on the specified Edge Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
		M_INSIDE_CHAIN	Specifies that edges inside the chain of the specified edge, or inside any individual chain of the specified group of edges, are included, excluded, or deleted from future operations on the specified Edge Finder result buffer. Note that no edges are selected from a delimiting edge that is not closed. For more information, see M_CLOSURE in MedgeControl(). Also, if an edge is not completely enclosed by a closed edge, it is considered as an outside or equal edge.	c		e	f	g	a		i		j	k	l	m	n	b
		M_INSIDE_OR_EQUAL_BOX	Specifies that edges inside or equal to the bounding box of the specified edge, or inside or equal to any individual bounding box of the specified group of edges, are included, excluded, or deleted from future operations on the specified Edge Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b

		M_INSIDE_OR_EQUAL_CHAIN	Specifies that edges inside or equal to the chain of the specified edge, or inside or equal to any individual chain of the specified group of edges, are included, excluded, or deleted from future operations on the specified Edge Finder result buffer. Note that no edges are selected from a delimiting edge that is not closed. For more information, see M_CLOSURE in MedgeControl(). Also, if an edge is not completely enclosed by a closed edge, it is considered as an outside or equal edge.	c		e	f	g	a		i		j	k	l	m	n	b
		M_OUTSIDE_BOX	Specifies that edges outside the bounding box of the specified edge, or outside any individual bounding box of the specified group of egdes, are included, excluded, or deleted from future operations on the specified Edge Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
		M_OUTSIDE_CHAIN	Specifies that edges outside the chain of the specified edge, or outside any individual chain of the specified group of edges, are included, excluded, or deleted from future operations on the specified Edge Finder result buffer. Note that no edges are selected from a delimiting edge that is not closed. For more information, see M_CLOSURE in MedgeControl(). Also, if an edge is not completely enclosed by a closed edge, it is considered as an outside or equal edge.	c		e	f	g	a		i		j	k	l	m	n	b
		M_OUTSIDE_OR_EQUAL_BOX	Specifies that edges outside or equal to the bounding box of the specified edge, or outside or equal to any individual bounding box of the specified group of edges, are included, excluded, or deleted from future operations on the specified Edge Finder result buffer.	c		e	f	g	a		i		j	k	l	m	n	b

	M_OUTSIDE_OR_EQUAL_CHAIN	Specifies that edges outside or equal to the chain of the specified edge, or outside or equal to any individual chain of the specified group of edges, are included, excluded, or deleted from future operations on the specified Edge Finder result buffer. Note that no edges are selected from a delimiting edge that is not closed. For more information, see M_CLOSURE in MedgeControl(). Also, if an edge is not completely enclosed by a closed edge, it is considered as an outside or equal edge.	c		e	f	g	a		i		j	k	l	m	n	b
Param1		Sets the label value of the edge to use to delimit the boundary, when delimiting the boundary using only one edge.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NULL	Specifies that this parameter is unused (when using a group of edges to delimit the boundary).	c		e	f	g	a		i		j	k	l	m	n	b
	Value	Specifies the label of the edge.	c		e	f	g	a		i		j	k	l	m	n	b
Param2		This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
Edge Finder result buffer identifier for a proximity based edge selection		Specifies an Edge Finder result buffer on which to perform an edge selection based on the proximity of an edge, or edges, to a specified point.	c		e	f	g	a		i		j	k	l	m	n	b
SelectionCriterion		This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
Condition		Sets the edge selection condition based on proximity.	c		e	f	g	a		i		j	k	l	m	n	b

	M_ALL_NEAREST_NEIGHBORS	Specifies that all edges within the specified radius from the stated point are included, excluded, or deleted from future operations on the specified Edge Finder result buffer. Define the search radius using MedgeControl() with M_NEAREST_NEIGHBOR_RADIUS.	c		e	f	g	a		i		j	k	l	m	n	b
	M_NEAREST_NEIGHBOR	Specifies that the closest edge from the specified point is included, excluded, or deleted from future operations on the specified Edge Finder result buffer. You can use MedgeControl() with M_NEAREST_NEIGHBOR_RADIUS to specify a circular area (around the specified point) in which to find the closest edge. If an edge is totally outside the search radius, it will not be considered as the closest edge. If some part of the edge chain lies within the search radius, it will be considered as a neighbor.	c		e	f	g	a		i		j	k	l	m	n	b
Param1		Sets the X-coordinate of the point (pixel or world units).	c		e	f	g	a		i		j	k	l	m	n	b
Param2		Sets the Y-coordinate of the point (pixel or world units).	c		e	f	g	a		i		j	k	l	m	n	b
Edge Finder result buffer identifier for a status based edge selection		Specifies an Edge Finder result buffer on which to perform an edge selection based on the current status (included or excluded) of edges.	c		e	f	g	a		i		j	k	l	m	n	b
SelectionCriterion		Sets the edge status to use as the selection criterion.	c		e	f	g	a		i		j	k	l	m	n	b
	M_ALL_EDGES	Specifies to select all edges regardless of their present status, except for edges that have been deleted.	c		e	f	g	a		i		j	k	l	m	n	b

		M_EXCLUDED_EDGES	Specifies to select all currently excluded edges, except for edges that have been deleted.	c		e	f	g	a		i		j	k	l	m	n	b
		M_INCLUDED_EDGES	Specifies to select all currently included edges.	c		e	f	g	a		i		j	k	l	m	n	b
	Condition		This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
	Param1		This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
	Param2		This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailedge11.lib.
DLL	Requires ail11.dll; ailedge11.dll.

MedgeStream

Synopsis

Load, restore, or save an Edge Finder context from/to a file or memory stream, or save calculated edges from an Edge Finder result buffer to a file or memory stream in CAD DXF format.

Syntax

```
void MedgeStream(
    AIL_TEXT_PTR MemPtrOrFileName,    //in-out
    AIL_ID SysId,                     //in
    AIL_INT64 Operation,              //in
    AIL_INT64 StreamType,             //in
    AIL_DOUBLE Version,               //in
    AIL_INT64 ControlFlag,            //in
    AIL_ID *ContextOrResultEdgeIdPtr, //in-out
    AIL_INT *SizeByteVarPtr           //out
)
```

Description

This function can load, restore, or save an Edge Finder context from/to a file or memory stream. Moreover, this function can save previously calculated edges (edge chains and/or edge approximations), stored in an Edge Finder result buffer, to a file or memory stream in a standard CAD DXF format. Edge chains and edge approximations are saved in the DXF file or memory stream in separate layers.

> Note that you cannot load or restore any CAD DXF file or memory stream using this function.

To inquire the number of bytes necessary to save an Edge Finder context or calculated edges to memory stream, you should first call this function (**MedgeStream**()) with M_INQUIRE_SIZE_BYTE.

The content saved to memory stream is equivalent to the content saved to file. This function is equivalent to a file saved using MedgeSave().

You can use this and other Aurora Imaging Library stream functions, for example, to save all required Aurora Imaging Library objects, as well as any other custom data, for your application to a memory stream. Once in a memory stream, you can write the stream to a single file or transfer it over a network. You are responsible for concatenating the streams and for saving the stream to file.

Using **MedgeStream**(), you can choose to save a backwards-compatible version of the Edge Finder context, which will work using a version of Aurora Imaging Library that is up to one major release older than the current version (depending on which version is specified). However, all settings and features unique to the higher version will be ignored when restored using the lower version. Besides saving backwards-compatible versions, you can also load or restore Edge Finder contexts saved using Aurora Imaging Library version 7.5 or above. Settings that do not exist in the lower version will be filled with default values when the Edge Finder context is loaded or restored.

Parameters

MemPtrOrFileName
Accepts the address of one of the following (see the Parameter associations section for specifics on which is expected):

- AIL_TEXT_PTR [optionally, in C++: a reference to an AIL_STRING]
- Optionally, in C++: a reference to a std::vector of type AIL_UINT8

Specifies the file or memory stream.

See the Parameter associations section for possible values that can be returned/specified.

SysId

Specifies the system on which to restore the Edge Finder context.

See the Parameter associations section for possible values that can be specified.

Operation

Specifies the operation to perform.

See the Parameter associations section for possible values that can be specified.

StreamType

Specifies the type of stream in which to store/from which to restore the Edge Finder context or calculated edges. This parameter must be set to one of the following values:

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For the type of stream																
Value	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
M_FILE	Specifies a file stream.	c		e	f	g	a		i		j	k	l	m	n	b
M_MEMORY	Specifies a memory stream. You are responsible for allocating a block of memory for the stream.	c		e	f	g	a		i		j	k	l	m	n	b

Version

Specifies the Aurora Imaging Library version of the Edge Finder context. When performing an M_LOAD or M_RESTORE operation, this parameter must be set to **M_DEFAULT**.

See the Parameter associations section for possible values that can be specified.

ControlFlag

Specifies the function's control flag.

See the Parameter associations section for possible values that can be specified.

ContextOrResultEdgeIdPtr

Specifies the address of the variable in which to write or from which to read the identifier of the Edge Finder context or the Edge Finder result buffer.

See the Parameter associations section for possible values that can be returned/specified.

SizeByteVarPtr

Specifies the address of the variable in which to write the size of the Edge Finder context or calculated

edges, in bytes. If the size is not required, you can set this parameter to **M_NULL**.

> Note that the size of Edge Finder context will vary depending on the Aurora Imaging Library version specified.

Parameter associations

The table below lists possible values for the **MemPtrOrFileName**, **SysId**, **Operation**, **Version**, **ControlFlag**, and **ContextOrResultEdgeIdPtr** parameters.

- For performing the stream operation.

Board legend: c – clarity uhd, d – concord poe with toe, e – gentl, f – geviq, g – gige vision, a – host system, h – indio, i – iris gtx, o – Rapixo CXP Pro Octo, j – radient ev-cl, k – rapixo cl, l – rapixo cof, m – rapixo cxp, n – usb3 vision, b – video4linux2

For performing the stream operation.																	
Operation	Description	(c)	(d)	(e)	(f)	(g)	(a)	(h)	(i)	(o)	(j)	(k)	(l)	(m)	(n)	(b)
MemPtrOrFileName - Possible values passed or returned																
SysId																
Version																
ControlFlag																
ContextOrResultEdgeIdPtr - Possible values passed or returned																
M_INQUIRE_SIZE_BYTE	Inquires the number of bytes required to save an Edge Finder context or result buffer to memory stream. This operation is not supported when the StreamType parameter is set to M_FILE.															
MemPtrOrFileName	This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
SysId	This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
Version	Specifies the Aurora Imaging Library version with which the Edge Finder context or result buffer must be compatible when saved. The size of an Edge Finder context or result buffer might differ from one Aurora Imaging Library version to another.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT	Specifies the current version of Aurora Imaging Library.	c		e	f	g	a		i		j	k	l	m	n	b

		M_PROC_VERSION_100_SP4	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 4.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PROC_VERSION_100_SP5	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 5.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PROC_VERSION_100_SP6	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 6.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PROC_VERSION_100_SP7	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 7.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PROC_VERSION_110	Specifies the version as being Aurora Imaging Library 11.0.	c		e	f	g	a		i		j	k	l	m	n	b
	ControlFlag		This parameter must be set to **M_DEFAULT**.	c		e	f	g	a		i		j	k	l	m	n	b
	ContextOrResultEdgeIdPtr		Specifies the address of the variable containing the identifier of the Edge Finder context or result buffer.	c		e	f	g	a		i		j	k	l	m	n	b
M_LOAD			Loads the content of a specified file or memory stream into a previously allocated Edge Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
	MemPtrOrFileName		Specifies the file or memory stream from which to load the context.	c		e	f	g	a		i		j	k	l	m	n	b
		M_INTERACTIVE	[This is only applicable to Windows] Opens a dialog box from which you can interactively specify the drive, directory, and name of the file, when the StreamType parameter is set to M_FILE.	c		e	f	g	a		i		j	k	l	m	n	b

	AIL_TEXT("*FileName*") 1	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"), when the StreamType parameter is set to M_FILE. Edge Finder contexts typically have an MEF file extension. The function handles (internally) the opening and closing of the file. To specify a file on a remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").	c		e	f	g	a		i		j	k	l	m	n	b
	MemPtr	Specifies the address of the block of memory, when the StreamType parameter is set to M_MEMORY. The block of memory should be of type *AIL_UINT8*. The specified address must correspond to the first memory address in which the object is located, and the block of memory must contain the entire object. Note that when using a C compiler (not a C++ or other compiler), you must cast the *AIL_UINT8* pointer to an *AIL_TEXT_PTR*.	c		e	f	g	a		i		j	k	l	m	n	b
SysId		This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
Version		This parameter must be set to **M_DEFAULT**.	c		e	f	g	a		i		j	k	l	m	n	b
ControlFlag		This parameter must be set to **M_DEFAULT**.	c		e	f	g	a		i		j	k	l	m	n	b
ContextOrResultEdgeIdPtr		Specifies the address of the variable containing the identifier of the Edge Finder context. All previous settings are overwritten with those of the loaded context, or are set to the default value if the settings didn't exist in the version of Aurora Imaging Library from which the context is being loaded.	c		e	f	g	a		i		j	k	l	m	n	b
M_RESTORE		Restores an Edge Finder context from a file or memory stream and assigns it an Aurora Imaging Library identifier.	c		e	f	g	a		i		j	k	l	m	n	b

MemPtrOrFileName	Specifies the file or memory stream from which to restore the context.	c		e	f	g	a		i		j	k	l	m	n	b
M_INTERACTIVE	[This is only applicable to Windows] Opens a dialog box from which you can interactively specify the drive, directory, and name of the file, when the StreamType parameter is set to M_FILE.	c		e	f	g	a		i		j	k	l	m	n	b
AIL_TEXT("*FileName*") 1	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"), when the StreamType parameter is set to M_FILE. Edge Finder contexts typically have an MEF file extension. The function handles (internally) the opening and closing of the file. To specify a file on a remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").	c		e	f	g	a		i		j	k	l	m	n	b
MemPtr	Specifies the address of the block of memory, when the StreamType parameter is set to M_MEMORY. The block of memory should be of type *AIL_UINT8*. The specified address must correspond to the first memory address in which the object is located, and the block of memory must contain the entire object. Note that when using a C compiler (not a C++ or other compiler), you must cast the *AIL_UINT8* pointer to an *AIL_TEXT_PTR*.	c		e	f	g	a		i		j	k	l	m	n	b
SysId	Specifies the system on which to allocate the Edge Finder context.	c		e	f	g	a		i		j	k	l	m	n	b
M_DEFAULT_HOST	Specifies the default Host system of the current Aurora Imaging Library application.	c		e	f	g	a		i		j	k	l	m	n	b
System identifier	Specifies a valid system identifier, previously allocated using MsysAlloc().	c		e	f	g	a		i		j	k	l	m	n	b

Version	This parameter must be set to **M_DEFAULT**.	c		e	f	g	a		i		j	k	l	m	n	b
ControlFlag	This parameter must be set to **M_DEFAULT**.	c		e	f	g	a		i		j	k	l	m	n	b
ContextOrResultEdgeIdPtr	Specifies the address of the variable in which to write the identifier of the Edge Finder context. If the operation is not successful, **M_NULL** is returned.	c		e	f	g	a		i		j	k	l	m	n	b
M_SAVE	Saves an Edge Finder context or result buffer to a specified file or memory stream; alternatively, if a result buffer is passed, this operation saves the calculated edges stored in the result buffer to the specified file or memory stream in a standard CAD DXF format.	c		e	f	g	a		i		j	k	l	m	n	b
MemPtrOrFileName	Specifies the file or memory stream.	c		e	f	g	a		i		j	k	l	m	n	b
M_INTERACTIVE	[This is only applicable to Windows] Opens a dialog box from which you can interactively specify the drive, directory, and name of the file, when the StreamType parameter is set to M_FILE.	c		e	f	g	a		i		j	k	l	m	n	b
AIL_TEXT("*FileName*") 1	Specifies the drive, directory, and name of the file (for example, "*C:\mydirectory\myfile*"), when the StreamType parameter is set to M_FILE. For easier use with other Aurora Imaging Library software products, when saving an Edge Finder context to a file, use the MEF file extension, and when saving calculated edges from an Edge Finder result buffer to a CAD file, use the DXF file extension. The function handles (internally) the opening and closing of the file. If the file already exists, it will be overwritten. To specify a file on a remote computer (under Distributed Aurora Imaging Library), prefix the specified file name string with "*remote:///*" (for example, "*remote:///C:\mydirectory\myfile*").	c		e	f	g	a		i		j	k	l	m	n	b

		MemPtr	Specifies the address of the block of memory, when the StreamType parameter is set to M_MEMORY. The block of memory should be of type *AIL_UINT8*. The specified address must correspond to the first memory address in which to write, and the block of memory must be large enough to stream the entire object. To determine the required size, call this function with M_INQUIRE_SIZE_BYTE. Note that when using a C compiler (not a C++ or other compiler), you must cast the *AIL_UINT8* pointer to an *AIL_TEXT_PTR*.	c		e	f	g	a		i		j	k	l	m	n	b
	SysId		This parameter must be set to **M_NULL**.	c		e	f	g	a		i		j	k	l	m	n	b
	Version		Specifies the Aurora Imaging Library version with which the Edge Finder context or result buffer must be compatible when saved. Note that the Edge Finder context or result buffer can only be up to one major release older than the current version.	c		e	f	g	a		i		j	k	l	m	n	b
		M_DEFAULT	Specifies the current version of Aurora Imaging Library.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PROC_VERSION_100_SP4	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 4.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PROC_VERSION_100_SP5	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 5.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PROC_VERSION_100_SP6	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 6.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PROC_VERSION_100_SP7	Specifies the version as being Aurora Imaging Library 10.0 Service Pack 7.	c		e	f	g	a		i		j	k	l	m	n	b
		M_PROC_VERSION_110	Specifies the version as being Aurora Imaging Library 11.0.	c		e	f	g	a		i		j	k	l	m	n	b

	ControlFlag	For any operation on Edge Finder contexts, this parameter must be set to M_DEFAULT. For calculated edges from an Edge Finder result buffer, this parameter stipulates whether to save edge chains and/or edge approximations to the CAD DXF file or memory stream. In this case, this parameter must be set to one or a combination (M_CHAIN + M_CHAIN_APPROXIMATION) of the following values.	c		e	f	g	a		i		j	k	l	m	n	b
	M_DEFAULT	Specifies the default value.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CHAIN	Specifies that edge chains will be saved.	c		e	f	g	a		i		j	k	l	m	n	b
	M_CHAIN_APPROXIMATION	Specifies that edge approximations will be saved.	c		e	f	g	a		i		j	k	l	m	n	b
	ContextOrResultEdgeIdPtr	Specifies the address of the variable containing the identifier of the Edge Finder context or result buffer to save.	c		e	f	g	a		i		j	k	l	m	n	b

[1] If you are passing the value in a variable, don't enclose it in AIL_TEXT().

Compilation information

Header	Include AIL.h.
Library	Use ail11.lib; ailedge11.lib.
DLL	Requires ail11.dll; ailedge11.dll.